

Navier–Stokes source propagation and the finite coupled bridge

Actual pressure defects, complete stress, and mean velocity correction

This cumulative reader preserves the complete preceding bridge and third-growth derivative, pulse-map and physical-residual proofs. The new parts compute the actual pressure-induced compatibility defects, the source’s five-row mean velocity correction and its remaining nonlinear terms, a compact symmetric-stress representation of the full averaged force, and the original-scale derivative costs of these operations.

The supplied source theorem remains attributed source input. This lane has not independently certified its entire proof. Its original positive swirl and exact negative reflection retain their corresponding forces. The distinct coupled candidate keeps every calculated residual; a finite stress representation is not asserted to be an added velocity or a completed infinite correction sequence. The companion 102-page reader contains the original finite coupled-stage proofs.

Contents

I	Complete preceding source bridge and original third-growth costs	4
1	Exact operator bridge to the released Navier–Stokes profiles	4
2	Scope and source equations	6
3	The coupled physical pair	7
4	Exact local residual map	7
5	Pressure reconstruction and full three-dimensional residual	8
6	Diffusion and energy comparison	8
7	Proven scope	9
8	Source objects and the finite calculation	9
9	Physical curl of the actual retained amplitude	10
10	Radial moments, support, and the original chart scales	11
11	The source covariance derivative and its signed right inverse	12

12	Assembly, the actual covariance change, and residual cancellation	13
13	Signs and the exact full-flow reflection	15
14	The released result and its actual signed path	15
15	Exact propagation at retained viscosities and orientations	16
16	The source positive branch and the exact negative branch	17
17	Propagation into the coupled calculation	18
18	The two moments of the actual compact curl increment	18
18.1	The actual harmonic curl has zero angular mean	19
18.2	Exact removal of the linear and mean-flow terms	20
18.3	The surviving axial fluxes and their signs	21
18.4	Bounds in the actual fields and original chart units	22
18.5	Exact relation to moments after phase evaluation	24
19	Time derivatives of the actual signed third growth	24
19.1	Original data, evolving parent, and proved interval	24
19.2	Signed amplitudes and first derivatives with the diffusion retained	25
19.3	Explicit coefficient rates	26
19.4	A closed all-orders derivative calculation	27
19.5	Derivative transport through the specified source clock	29
20	The actual third-growth pair in the source clock and frame	29
20.1	Exact physical differentiation of the source pulse clock	29
20.2	The actual frame, its inverse, and the direct candidate	30
20.3	The complete principal mismatch, including the cutoff	32
20.4	An explicit map of the actual seed to the actual source pulse	33
20.5	All scale costs of the finite intertwiner	34
21	Actual third-growth budgets in the physical curl and force	36
21.1	Two complete spatial shape fields for each retained harmonic	36
21.2	Quantitative radial moments with the original component powers	37
21.3	The full physical residual with all clock and diffusion costs	38
II	The actual source pressure and five-row correction	40
22	The actual wave increment and the source's three compatibility targets	40
22.1	The actual covariance increment before pressure reconstruction	40
22.2	Compact mean pressure and the corrected axial flux moment	41
22.3	New quantitative budgets for the three actual inputs	43
22.4	The exact five-row map for these targets	43
22.5	What the five-row operation actually leaves	45
22.6	Exact source operation dictionary and the bounds it requires	46

III	Compact completion of the full averaged force	50
23	Compact completion of the full averaged residual	50
23.1	The actual extended fields and all three averaged components	50
23.2	The compact fluxes and the original varying radial bumps	52
23.3	Corrected radial primitives and the complete symmetric tensor	53
23.4	Compact pressure, tracefree stress, and the exact gauge relation	54
23.5	All total force and torque components and the divergence image	54
23.6	Exact source chart, bump, stress, and pressure powers	56
IV	All-order actual moment and moving-scale costs	58
24	Original-scale derivative costs of the actual compatibility defects	58
24.1	Actual mixed derivatives of the third-growth curl	58
24.2	A contraction estimate that retains every product derivative	59
24.3	The actual three defects and the pressure-corrected axial flux	59
24.4	Exact derivatives of the original moving scale and bumps	61
V	Physical mean velocity and derivative costs	63
25	Physical derivative and full vector-force costs of the five-row mean operation	63
25.1	Exact inverse profiles and the five physical rows	63
25.2	Compact potential, induced radial velocity, and support	65
25.3	Every mixed physical derivative, with original scale powers	65
25.4	Explicit costs in full target derivative arrays	67
25.5	Actual pressure reconstruction and the surviving defects	68
25.6	Exact full cylindrical physical force increment	69
25.7	Finite all-order force costs with every inverse-radius term	70

Part I

Complete preceding source bridge and original third-growth costs

1 Exact operator bridge to the released Navier–Stokes profiles

This bounded comparison uses the locally saved released Navier–Stokes source (the source pages are 7–15 and 24–27 for the profile, stress and pulse construction). Its stated physical viscosity is denoted by $\nu_{\text{NS}} > 0$. The Boussinesq lane variables and physical viscosity remain d, μ and $\nu = \sqrt{A_0}$; none is identified with ν_{NS} .

Use right-handed cylindrical coordinates (r, ϑ, z) and define

$$y_1 = z, \quad y_2 = \frac{r^2}{2} > 0, \quad v = (u^z, ru^r) = J \nabla_y \Psi, \quad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

The meridional divergence identity is $\partial_{y_1} v_1 + \partial_{y_2} v_2 = 0$. Define the circulation and meridional vorticity variables

$$\Gamma = ru^\vartheta, \quad \xi = \frac{\omega^\vartheta}{r}, \quad h = \frac{\Gamma}{2y_2} = \frac{u^\vartheta}{r}, \quad Q = \partial_{y_1}^2 + 2y_2 \partial_{y_2}^2, \quad M = Q + 4\partial_{y_2}.$$

The azimuthal momentum and meridional vorticity rows of the cylindrical equations (with arbitrary smooth force components) become, on $y_2 > 0$,

$$D_v \Gamma = \nu_{\text{NS}} Q \Gamma + r f^\vartheta, \tag{1}$$

$$D_v \xi = \nu_{\text{NS}} M \xi + \partial_{y_1} (h^2) + \text{curl}_y (f^z, f^r/r), \tag{2}$$

where $D_v = \partial_t + v \cdot \nabla_y$ and $\text{curl}_y(a, b) = \partial_{y_1} b - \partial_{y_2} a$.

Indeed, $\partial_r = r \partial_{y_2}$ and $\partial_r^2 + r^{-1} \partial_r = 2y_2 \partial_{y_2}^2 + 2\partial_{y_2}$. For Γ , the azimuthal viscous operator is $\partial_r^2 - r^{-1} \partial_r + \partial_z^2 = Q$. For $r\xi = \omega^\vartheta$, the cylindrical operator $(\partial_r^2 + r^{-1} \partial_r - r^{-2} + \partial_z^2)(r\xi)$, divided by r , is $M\xi$; explicitly the radial terms are $2y_2 \xi_{y_2 y_2} + 4\xi_{y_2}$. The centrifugal term is $r^{-2} \partial_z((u^\vartheta)^2) = \partial_{y_1}(\Gamma^2/(4y_2^2)) = \partial_{y_1}(h^2)$. These calculations prove every coefficient in (1)–(2), including the r force conversion.

The velocity–vorticity relation is also retained. Since $u^z = -\Psi_{y_2}$ and $u^r = \Psi_{y_1}/r$,

$$\xi = \frac{\partial_z u^r - \partial_r u^z}{r} = \frac{\Psi_{y_1 y_1}}{2y_2} + \Psi_{y_2 y_2} = \frac{Q\Psi}{2y_2}.$$

Thus the two transport rows do not by themselves replace the elliptic velocity reconstruction. Conversely, on a simply connected meridional domain, a smooth Ψ, Γ satisfying this relation and both transport rows recovers u . Define its momentum residual without pressure as $a = \partial_t u + (u \cdot \nabla)u - \nu_{\text{NS}} \Delta u - f$. The circulation row is $a^\vartheta = 0$, while the vorticity row is $\partial_z a^r - \partial_r a^z = 0$. The one-form $a^r dr + a^z dz$ is therefore closed, and $p = -\int (a^r dr + a^z dz)$ along a path from a fixed base point is path independent. Its derivatives are $p_r = -a^r$, $p_z = -a^z$. This proves the local momentum reconstruction, with pressure unique up to a function of time; boundary data are not removed.

The change from circulation to swirl rate is invertible on the exact retained domain: $\Gamma = 2y_2 h$. Product differentiation gives

$$Q(2y_2 h) = 2y_2 M h, \quad D_v(2y_2 h) = 2y_2 D_v h + 2v_2 h.$$

Therefore (1) is equivalent to

$$D_v h = \nu_{\text{NS}} M h - \frac{v_2}{y_2} h + \frac{f^\vartheta}{r}. \quad (3)$$

The source profile uses precisely this relation: in its similarity variables, E is the azimuthal velocity profile and $H_{\text{src}} = \sqrt{2X}E$ is the scaled circulation profile (source equations (4.3), (4.8)). Since $r = \sqrt{2qX}$ and $u^\vartheta = q^{-A}E$, its physical circulation is $\Gamma = q^{1/2-A}H_{\text{src}} = q^{-h_{\text{src}}}H_{\text{src}}$, where the source exponents satisfy $A = 1/2 + h_{\text{src}}$. Here $q > 0$ is the geometric similarity coordinate and $h_{\text{src}} > 0$ is an exponent; neither is a signed amplitude scale. This is the exact scale dictionary; it does not identify H_{src} with the Boussinesq scalar amplitude.

The sign of the blowout amplitude must be kept separate from that positive geometric coordinate. The axisymmetric equations have the exact involution

$$(u^r, u^\vartheta, u^z, p; f^r, f^\vartheta, f^z) \mapsto (u^r, -u^\vartheta, u^z, p; f^r, -f^\vartheta, f^z).$$

It sends (Γ, h) to $(-\Gamma, -h)$, leaves ξ and the meridional flow unchanged, and preserves every displayed equation: the only quadratic swirl term is $\partial_{y_1}(h^2)$, while the azimuthal equation is linear in (Γ, f^ϑ) . Consequently a source-selected positive profile $E > 0$ (source pp. 7–9) does not justify assuming that every blowout branch is positive; its reflected branch has $E < 0$, $\Gamma < 0$ and the same blowup magnitude. The exponent $q^{-h_{\text{src}}}$ controls the magnitude in either branch. The source's positive covariance coefficients and viscosity are separate construction choices, not a sign claim about the velocity amplitude.

For comparison with the lane's controlled scalar equation, set $B = h^2$. The globally smooth square equation, including its geometric and viscous terms, is

$$D_v B = \nu_{\text{NS}} M B - 2\nu_{\text{NS}} |\nabla h|_Q^2 - \frac{2v_2}{y_2} B + \frac{2h}{r} f^\vartheta, \quad |\nabla h|_Q^2 = h_{y_1}^2 + 2y_2 h_{y_2}^2. \quad (4)$$

This follows by multiplying (3) by $2h$ and using $M(h^2) = 2hMh + 2|\nabla h|_Q^2$. At zeros of h equation (4) remains smooth; dividing by B is unnecessary and would lose that domain. On a component where $B > 0$, the equivalent expression is $D_v B = \nu_{\text{NS}} M B - \nu_{\text{NS}} |\nabla B|_Q^2 / (2B) - 2v_2 B / y_2 + 2h f^\vartheta / r$.

The exact relation to the Boussinesq lane is consequently an operator correspondence, not a scalar conjugacy. The lane's retained-viscosity scalar row has $D_v \Theta = d\Delta \Theta + f_\Theta$ in Cartesian material coordinates. A direct identification $\Theta = B$ would require, on the same physical domain, both $M = \Delta$ after the nonlinear coordinate pullback and $-2v_2 B / y_2 - 2\nu_{\text{NS}} |\nabla h|_Q^2 + 2h f^\vartheta / r$ to be absorbed into one prescribed force. The first condition fails as an operator identity: $M = \partial_{y_1}^2 + 2y_2 \partial_{y_2}^2 + 4\partial_{y_2}$ has variable metric and drift, whereas the lane's Δ is the Euclidean Cartesian Laplacian. The geometric, gradient and force terms must therefore be retained or calculated in the particular source state. Nonconstant profiles alone would not prove that such terms never cancel. The exact bridge is (1)–(4) with the full residual terms retained. It neither claims that the lane's controls are the source pulses nor changes the source's pressure reconstruction.

For the meridional velocity, the source's pulse construction supplies the following further exact local dictionary (source pp. 11–15 and 74–77). If w is a divergence-free pulse and π its pressure increment, then

$$R(u_B + w, p_B + \pi) = R(u_B, p_B) + L_{u_B}(w, \pi) + \nabla \cdot (w \otimes w),$$

$$L_{u_B}(w, \pi) = \partial_t w + (u_B \cdot \nabla) w + (w \cdot \nabla) u_B - \nu_{\text{NS}} \Delta w + \nabla \pi, \quad \nabla \cdot w = 0.$$

For the moving local phase with wavevector n and transverse amplitude t_m , the source equation is

$$t'_m + \mathcal{K} t_m + m^2 d_N t_m + i k m n \pi_m = -f_m, \quad n \cdot t_m = 0, \quad d_N = \varepsilon k^2 |n|^2,$$

with the source's moving shear matrix $\mathcal{K} = \begin{pmatrix} 0 & -2F & 0 \\ 2F + RF_R & 0 & 0 \\ G_R & 0 & 0 \end{pmatrix}$. The transverse projection uses

$$A_\Phi = -\mathcal{K} + \frac{n(n^T \mathcal{K} - n'^T)}{|n|^2}.$$

These are exact source pulse equations, including viscosity and pressure projection. Their quadratic covariance is chosen to cancel the annular stress; the source explicitly retains curl, cutoff, and correction interactions. The Boussinesq lane's two-dimensional pair and its full profile residuals have an exact algebraic form, but no identification with this three-dimensional pulse equation follows without an additional embedding that supplies the source's cylindrical geometry, moving plane, covariance cone and correction cycle. That embedding is not asserted here.

The current bridge therefore proves a precise morphism of variables, operators, forces and pulse residuals and records the exact obstruction to a direct scalar identification. It establishes the next programme datum while preserving the unproved status of any singular limit or third return.

2 Scope and source equations

This note records one local, exact comparison. It does not identify the coupled Boussinesq stages with the released Navier–Stokes construction and makes no claim about a global Navier–Stokes solution. The source is the released PDF `openai_navier_stokes.pdf`, SHA-256 `8c8a94ad9ac824c8b605b9827cadf7beaca48bd10b380de3cfc872a2c37afa81`. The source statements used here are Section 7, equations (7.2)–(7.8), (7.12)–(7.22), and Section 9, equations (9.1)–(9.2), on PDF pages 74–75, 77–81, and 101–103. In the source notation, on one lifted pulse rectangle,

$$\Phi = p\theta + p_z Z/\varepsilon + x_0 R - v(pF + p_z G), \quad n = \nabla_* \Phi, \quad (5)$$

with

$$n = (x_0 - v(pF_R + p_z G_R), p/R, p_z - \varepsilon v(pF_Z + p_z G_Z)). \quad (6)$$

For a nonzero integer harmonic m , the exact principal pulse equations are

$$n \cdot t_m = 0, \quad t'_m + K t_m + m^2 d_N t_m + i k m n \pi_m = -f_m, \quad d_N = \varepsilon k^2 |n|^2, \quad (7)$$

where a prime is the pulse-coordinate derivative and

$$K = \begin{pmatrix} 0 & -2F & 0 \\ 2F + RF_R & 0 & 0 \\ G_R & 0 & 0 \end{pmatrix}, \quad A_\Phi = -K + \frac{n(n^T K - (n')^T)}{|n|^2}. \quad (8)$$

The source pressure coefficient is exactly

$$\pi_m = \frac{i}{k m |n|^2} (n \cdot K t_m - (n') \cdot t_m + n \cdot f_m). \quad (9)$$

For $m = 1$, a real homogeneous pulse t_1 has the source energy identity (7.22):

$$\frac{1}{2}(|t_1|^2)' = -g \cdot (t_{1,r}(t_{1,\theta}, t_{1,z})) - \varepsilon k^2 |n|^2 |t_1|^2, \quad (10)$$

where $g = (RF_R, G_R)$ in the tangential coordinates. Thus the source pulse retains both shear transfer and viscous dissipation.

3 The coupled physical pair

For one activated layer of the coupled calculation, put

$$a_c = -\frac{J\zeta \cdot G_{<q}}{Kr^2}, \quad b_c = K\zeta_1, \quad \lambda_c^\Theta = \kappa K^2 r^2, \quad \lambda_c^\Omega = \varepsilon_c K^2 r^2. \quad (11)$$

Its exact retained-viscosity pair, including optional scalar and vorticity controls, is

$$\dot{y} = C_c y + c_c, \quad y = \begin{pmatrix} \Theta \\ \Omega \end{pmatrix}, \quad C_c = \begin{pmatrix} -\lambda_c^\Theta & a_c \\ b_c & -\lambda_c^\Omega \end{pmatrix}, \quad c_c = \begin{pmatrix} c_\theta \\ c_\omega \end{pmatrix}. \quad (12)$$

Here $G_{<q}$, ζ , and r evolve with the inherited background and phase ODEs; in particular no fixed-background substitution is made. Let v be a source pulse coordinate on an interval and let $q_v = dt/dv$ describe its physical-time parametrization. The forward source parametrization has $q_v > 0$; the algebra below also retains a negative orientation without changing the amplitude sign. Define $y'(v) = q_v(C_c y + c_c)$. Here K, r are the coupled frequency and phase length, not the source carrier k and cylindrical radius R . The original A_0, λ_0, μ and time scale $\nu = \sqrt{A_0}$ are unchanged. The source chart parameter ε is not identified with either physical diffusivity.

4 Exact local residual map

Let $B(v) \in \mathbb{R}^{3 \times 2}$ be a C^1 tangent frame with full column rank and $n(v)^T B(v) = 0$, so its range is exactly $n(v)^\perp$. Let $B_\ell(v)$ be any left inverse, $B_\ell B = I_2$; then BB_ℓ acts as the identity on $n^\perp$. Let $L(v) \in \mathbb{R}^{2 \times 2}$ be any C^1 coefficient map and embed the coupled pair into a transverse source amplitude by

$$t(v) = B(v)L(v)y(v). \quad (13)$$

For an embedding with a non-tangent frame the transversality condition would be

$$n(v)^T B(v)L(v)y(v) = 0. \quad (14)$$

For the tangent frame used here, the condition holds for every L, y . Set

$$H(v) = B_\ell(v)(A_\Phi(v)B(v) - B'(v)). \quad (15)$$

Then the exact projected source residual of the embedded coupled pair is

$$\mathcal{E}_{L,c} := B_\ell \text{proj}_{n^\perp} [t' - A_\Phi t + m^2 d_N t + f_m] = [L' + q_v L C_c - H L + m^2 d_N L] y + q_v L c_c \quad (16)$$

$$+ B_\ell \text{proj}_{n^\perp} f_m. \quad (17)$$

In particular, with $L = I$ and no added source, the defect is the explicit matrix expression

$$\mathcal{E}_{I,0} = (q_v C_c - H + m^2 d_N I_2) y. \quad (18)$$

This gives a local residual relation with every actual coefficient retained.

Proof. Differentiate (13): $t' = B'Ly + BL'y + BLy'$. Since $y' = q_v(C_c y + c_c)$, differentiate $n^T B = 0$ to obtain $n^T B' = -(n')^T B$. Also $n^T A_\Phi B = -(n')^T B$, directly from (8). Thus $A_\Phi B - B'$ has transverse range and equals BH . Applying B_ℓ to the transverse terms and $B_\ell \text{proj}_{n^\perp}$ to the force proves (17). More explicitly,

$$n^T(t' - A_\Phi t + m^2 d_N t) = 0, \quad n^T(t' - A_\Phi t + m^2 d_N t + f_m) = n^T f_m. \quad (19)$$

The unprojected force need not be transverse. The pressure below removes precisely this remaining normal component. $\square$

5 Pressure reconstruction and full three-dimensional residual

For any embedded transverse t and arbitrary f_m , define

$$\pi_{L,c} = \frac{i}{km|n|^2} (n \cdot \mathbf{K}t - (n') \cdot t + n \cdot f_m). \quad (20)$$

Then the full source residual satisfies the exact decomposition

$$\mathcal{R}_{L,c} := t' + \mathbf{K}t + m^2 d_N t + ikmn\pi_{L,c} + f_m \quad (21)$$

$$= \text{proj}_{n^\perp} (t' + \mathbf{K}t + m^2 d_N t + f_m) = B \mathcal{E}_{L,c}, \quad (22)$$

where the last equality uses the definition of H and the transverse condition. Thus the pressure removes precisely the normal component; it does not remove $\mathcal{E}_{L,c}$, the moving-frame, shear, diffusion, or coupled-background mismatch. If one elects to force the source pulse to obey the released equation, then the unique additional transverse forcing is

$$f_{m,\text{corr}} = -B \mathcal{E}_{L,c}, \quad (23)$$

added to the old f_m . The old $\mathcal{E}_{L,c}$ is evaluated before this addition, so the new projected residual is zero. Since $n^T f_{m,\text{corr}} = 0$, pressure is unchanged. This is an exact local forced relation, not an identification of the physical constructions.

6 Diffusion and energy comparison

The source diffusion multiplier is $m^2 d_N = m^2 \varepsilon k^2 |n|^2$ in (7). The coupled pair has the two retained physical losses λ_c^Θ and λ_c^Ω , which are generally unequal to $m^2 d_N$. Their signed discrepancy in (18) is explicitly

$$(m^2 d_N - q_v \lambda_c^\Theta) \Theta, \quad (m^2 d_N - q_v \lambda_c^\Omega) \Omega \quad (24)$$

when $H = 0$ and C_c is expanded entrywise; with general H the exact diagonal entries are those of $q_v C_c - H + m^2 d_N I_2$. No damping is dropped. If one compares nonnegative loss magnitudes instead, the corresponding cost gaps are $|m^2 d_N - q_v \lambda_c^\Theta|$ and $|m^2 d_N - q_v \lambda_c^\Omega|$. The required finite-interval coordinate map can be constructed explicitly. Let $I = [v_0, v_1]$, and let X_c, X_s be the identity-initial-value fundamental matrices for $D_c = q_v C_c$ and $D_s = H - m^2 d_N I_2$, respectively. For either continuous matrix D , define

$$X_D(v) = I_2 + \sum_{j \geq 1} \int_{v_0 \leq w_j \leq \dots \leq w_1 \leq v} D(w_1) \cdots D(w_j) dw_j \cdots dw_1.$$

With $M_D = \sup_I \|D\|$ and $T_I = v_1 - v_0$, the j th integral has norm at most $(M_D T_I)^j / j!$. Uniform convergence proves the integral equation $X_D = I_2 + \int_{v_0}^v D X_D$ and hence $X_D' = D X_D$. The same iteration for $Z' = -ZD$, $Z(v_0) = I_2$ gives $Z X_D = I_2$ by differentiation, proving invertibility on I . For a specified invertible L_0 , set

$$L = X_s L_0 X_c^{-1}, \quad L^{-1} = X_c L_0^{-1} X_s^{-1}.$$

This has norm at most $\|L_0\| e^{(M_c + M_s) T_I}$; its inverse has the same bound with L_0^{-1} . Differentiation proves

$$L' = H L - q_v L C_c - m^2 d_N L, \quad (25)$$

so the homogeneous embedded pair has zero projected residual. This proves an actual finite-interval conjugacy along the retained coefficient path. It does not make the source phase, pressure, cutoff, or nonlinear covariance equal to the coupled Boussinesq fields. For general controls c_c , the forced term is $q_v L c_c$ in (17).

For an arbitrary embedded field (possibly complex), the exact energy balance retains its full defect:

$$\begin{aligned} \frac{1}{2}(|t|^2)' = & -\operatorname{Re}(\bar{t}^T \mathbf{K} t) - m^2 d_N |t|^2 - \operatorname{Re}(\bar{t}^T f_m) \\ & + \operatorname{Re}(\bar{t}^T B \mathcal{E}_{L,c}). \end{aligned} \quad (26)$$

This follows by taking the real Hermitian product of (22) with t . Its pressure term vanishes because n is real and $n^T t = 0$. The $2F$ part of $\mathbf{K}$ is skew-symmetric, hence its real work is zero; the shear work equals $\operatorname{Re}(\bar{t}_r(g_\theta t_\theta + g_z t_z))$. Only for a zero residual, zero force, $m = 1$, and real t does this reduce to (10). The real coupled pair has

$$\frac{1}{2} \frac{d}{dt} |y|^2 = -\lambda_c^\Theta \Theta^2 - \lambda_c^\Omega \Omega^2 + (a_c + b_c) \Theta \Omega + c_\theta \Theta + c_\omega \Omega, \quad (27)$$

which is a two-component physical energy identity. Equations (17) and (27) exhibit exactly where the source's three-dimensional shear flux and the coupled pair's two-dimensional exchange differ.

7 Proven scope

The source uses a cylindrical, localized, divergence-free curl construction (Section 7.4 and Section 9, PDF pages 85–87 and 102–103); the displayed principal pulse equation is only one component of that construction. The comparison above proves an exact local map and its full pressure-projected residual for every finite interval on which the denominators and frame left inverse exist. It preserves source viscosity, moving phase normal, pressure, rotation/shear matrix, and all coupled physical diffusion coefficients. It does not prove that the coupled third-growth or steering stages produce the source's pulse phase Φ , covariance, cutoff tails, or smooth terminal force. Those remain separate bridge problems.

The sign conventions in this comparison are also explicit. The quantities $q_v > 0$, $\varepsilon > 0$, and $d_N = \varepsilon k^2 |n|^2 > 0$ record the chosen orientation of physical time and the positive viscous coefficient; they are not assumptions on the sign of a pulse or blowout amplitude. The equations (7)–(9) are linear in (t_m, f_m) , so the simultaneous replacement $(t_m, f_m, \pi_m) \mapsto (-t_m, -f_m, -\pi_m)$ is an exact reflected branch. Likewise, the local map (13) is sign-equivariant under $y \mapsto -y$ and $c_c \mapsto -c_c$, with the coefficients fixed at the actual path. This is a statement about the retained linear equations, not a sign symmetry of the entire coupled nonlinear system. The source covariance is quadratic, so reversing a wave leaves that covariance unchanged. Reversing the full Navier–Stokes velocity need not reverse its force, since $(u \cdot \nabla)u$ is even in u . The full-flow map that reverses axisymmetric swirl is a spatial reflection; its complete formula is proved in the accompanying covariance bridge. No sign of a blowout or endpoint is inferred from positive geometric scales.

8 Source objects and the finite calculation

This calculation uses the preserved 165-page supplied manuscript *Finite Time Blowup for Navier–Stokes*, source equations (7.23)–(7.42), (9.12)–(9.14). Its PDF SHA-256 is

The source pulse fields, auxiliary rectangles, phases, and leading covariance columns are the given objects; their global construction is source input. Every finite map asserted below is proved here. The existing coupled-stage proofs remain in the complete 102-page reader. The companion pulse note proves the exact frame residual and a constructed finite-interval intertwiner along its evolving coefficients. The source uses positive $q(z, t)$, band scales $Q_\beta = 2^{-\ell_\beta}$, $\varepsilon_\beta = Q_\beta^h$, $A = 1/2 + h$; none is set to one here or identified with the original coupled $A_0, \lambda_0, \mu, \nu = \sqrt{A_0}$ or its physical diffusivities.

All operations in this note are on a compact set inside $q > 0$ and the source active annulus. There are finitely many relevant bands and locally finite source labels. Supports of newly prescribed inputs are strictly inside the annulus, so no lower bound on a leading amplitude at a vanishing shell edge is assumed. Endpoint and shell-edge estimates are not inferred from compact-interior estimates.

9 Physical curl of the actual retained amplitude

Let $y = (\Theta, \Omega)^T$ be the actual smooth retained coupled solution, $y' = q_v(C_c y + c_c)$, with the coefficients of the companion pulse note. Use the smooth actual coefficients on the chosen slow/auxiliary patch; all harmonic coefficients and cutoffs are independent of θ in cylindrical components, and each carrier has nonzero integer angular frequency. For its source tangent frame B , smooth coefficient map L , and a fixed smooth real compact cutoff χ of this type, put

$$t_m = \chi B L y, \quad \mathcal{E}_\chi = \chi \{ [L' + q_v L C_c - H L + m^2 d_N L] y + q_v L c_c \} + \chi' L y, \quad H = B_\ell (A_\Phi B - B').$$

Here prime differentiates along the pulse coordinate; the other slow and auxiliary derivatives remain in the spatial formulas below. Smooth parameter dependence of the companion's constructed L follows directly from its integral equations on a fixed interval: for a nonzero parameter multi-index I ,

$$\partial^I X(v) = \int_{v_0}^v \left[D \partial^I X + \sum_{0 < J \leq I} \binom{I}{J} (\partial^J D) \partial^{I-J} X \right] dw.$$

The initial parameter derivatives vanish for identity initial data. Inductively the known forcing is smooth and bounded on the compact patch; the same convergent ordered-integral iteration solves this equation. This proves every finite mixed derivative of X, X^{-1}, L . Since $n^T B = 0$, $n^T t_m = 0$. Differentiation proves $t'_m - A_\Phi t_m + m^2 d_N t_m = B \mathcal{E}_\chi$. Thus even the pulse-cutoff term is explicit.

A physical version of the exact curl construction makes the intermediate map to momentum residual unambiguous. For each chosen harmonic let $\phi(x, t) = k_\beta m \Phi_\beta(x, t, Y(x, t))$ be its real evaluated phase, let $\xi = \nabla_x \phi \neq 0$, and let $a(x, t) = Q_\beta^{-A} t_m$ be the physical transverse amplitude. The source coordinate change is $r = \sqrt{Q_\beta} R$, $z = Q_\beta^D Z$, $1 - t = Q_\beta T$, with $D = 1/2 - h$ and normalized axial derivative $D_z = \varepsilon_\beta \partial_Z$. It gives $\xi = (k_\beta m / \sqrt{Q_\beta}) n_\Phi$; all auxiliary chain derivatives are included in n_Φ . Hence $\xi \cdot a = 0$. Define in Cartesian components

$$C = \frac{i \xi \times a}{|\xi|^2}, \quad \mathcal{A} = C e^{i\phi}, \quad \nabla \times \mathcal{A} = (a + r_a) e^{i\phi}, \quad r_a = \nabla \times C. \quad (28)$$

The vector triple-product identity gives $i \xi \times C = a$, proving the last equality. It also shows that the physical potential has exactly the source factor $Q_\beta^{1/2-A}$ relative to its chart coefficient $i n_\Phi \times t_m / (k_\beta m |n_\Phi|^2)$. Cartesian differentiation of C includes derivatives of the cylindrical basis;

the same terms are the R^{-1} terms of source (7.37)–(7.38). Take real parts and sum the finite retained harmonics. This constructs a smooth compact velocity w_y with $\nabla \cdot w_y = 0$.

Retain a finite smooth source state (u_*, p_*) , including its background, actual leading waves and their curl corrections, and the real physical pressure increments obtained from the companion pulse pressure formula with $f_m = 0$:

$$\pi_{\chi, m} = \frac{i}{k_\beta m |n|^2} (n^T \mathbf{K} t_m - (n')^T t_m), \quad \pi_y = \sum_{\beta, m} \operatorname{Re}(Q_\beta^{-2A} \pi_{\chi, m} e^{i\phi_{\beta, m}}).$$

These are smooth and compact wherever t_m is. This definition fixes the pressure map without assuming the pulse residual vanishes. Define the actual candidate residual

$$\begin{aligned} R_y &= R_{\nu_{\text{NS}}}(u_* + w_y, p_* + \pi_y), \\ R_{\nu_{\text{NS}}}(u, p) &:= \partial_t u + (u \cdot \nabla)u - \nu_{\text{NS}} \Delta u + \nabla p. \end{aligned} \quad (29)$$

This is a definition by explicit differentiated fields, not an assumed source term. Its exact expansion is

$$R_y = R_{\nu_{\text{NS}}}(u_*, p_*) + \partial_t w_y + (u_* \cdot \nabla)w_y + (w_y \cdot \nabla)u_* - \nu_{\text{NS}} \Delta w_y + \nabla \pi_y + (w_y \cdot \nabla)w_y.$$

Every cutoff, parent, phase, and curl derivative occurs in this formula. Physical ν_{NS} is retained and is distinct from every source time-scaling convention. Define $\Delta R_y = R_y - R_{\nu_{\text{NS}}}(u_*, p_*)$. We correct the new compact increment; this difference has compact support since every added term does. The baseline residual remains explicitly.

The averaging needed next is the source extended-field operation, not a spatial average of the evaluated field. Evaluate derivatives on the extended coefficients first by the complete chain rule. In a fixed chart set $\mathcal{D}_i = \partial_{x_i} + \sum_\alpha (\partial_{x_i} Y_\alpha) \partial_{Y_\alpha}$ and similarly $\mathcal{D}_t$. These are the pullbacks of commuting physical derivatives, so $[\mathcal{D}_i, \mathcal{D}_j] = 0$ on their lifted domain. For smooth periodic coefficients their auxiliary derivative terms have zero Haar average. In cylindrical components angular differentiation also differentiates the basis. This defines an extended residual $\tilde{R}_y$ whose evaluation is exactly (29). Average the tangential components of the increment at fixed slow variables:

$$F_2 = \langle \Delta \tilde{R}_{y, \theta} \rangle, \quad F_1 = \langle \Delta \tilde{R}_{y, z} \rangle, \quad \langle g \rangle = \int_{\mathbb{T}^2} \frac{1}{2\pi} \int_0^{2\pi} g \, d\theta \, dY. \quad (30)$$

These F_e are smooth and compact in the chosen physical annulus. Equations (28)–(30) are the explicit intermediate map from the retained coupled pair to a signed momentum residual. They do not identify it with the original two-dimensional Boussinesq force.

10 Radial moments, support, and the original chart scales

For each fixed (z, t) , choose $0 < r_- < r_+$ containing the compact supports of both F_e and the chosen bumps in the active annulus. The source bump is

$$b_e(r, z, t) = q^{-(e+1)/2} \hat{b}_e(r/\sqrt{q}), \quad \int_0^\infty x^e \hat{b}_e(x) \, dx = 1, \quad e = 1, 2,$$

with interior support. The change $r = \sqrt{q} x$ proves $\int r^e b_e \, dr = 1$ exactly. Define

$$M_e = \int_0^\infty r^e F_e(r) \, dr, \quad \sigma_e(r) = -r^{-e} \int_0^r s^e (F_e(s) - b_e(s) M_e) \, ds. \quad (31)$$

Then

$$(\partial_r + e/r)\sigma_e = -F_e + b_e M_e, \quad \int_0^\infty r^e (F_e - b_e M_e) dr = 0. \quad (32)$$

Differentiate $r^e \sigma_e$ to prove the first identity; integrate and use the bump normalization for the second. The integral is zero below r_- and above r_+ , so σ_e has compact smooth support in the same enclosing interval. A homogeneous solution is cr^{-e} , and a compact homogeneous solution has $c = 0$; thus the primitive is unique among compact solutions of its displayed equation. Moreover a compact solution of $(\partial_r + e/r)\sigma = -F_e$ exists if and only if $M_e = 0$: necessity follows by integrating the weighted derivative, and sufficiency follows from (31). The two moments are therefore the exact surviving finite-dimensional terms.

For $C_e = \int_{r_-}^{r_+} s^e ds$, $|M_e| \leq C_e \|F_e\|_\infty$ and

$$\|\sigma_e\|_\infty \leq r_-^{-e} C_e (\|F_e\|_\infty + \|b_e\|_\infty |M_e|).$$

If $G_e = F_e - b_e M_e$, successive radial derivatives obey the exact recurrence

$$\partial_r^{j+1} \sigma_e = -\partial_r^j G_e - e \sum_{i=0}^j \binom{j}{i} (-1)^i i! r^{-i-1} \partial_r^{j-i} \sigma_e.$$

This proves every finite radial derivative bound inductively. Parameter derivatives follow by differentiating the compact integral; all derivatives of q, b_e and the moving physical data are retained. On the present compact set q is bounded away from zero. No terminal uniform estimate is implied.

The source (9.12)–(9.14) uses physical $F_e = Q^{-2A-1/2} E_e^*$, where $E_2^* = \langle E_\theta \rangle_Y$ and $E_1^* = \langle E_z \rangle_Y$. For our fields define $E_e^* = Q^{2A+1/2} F_e$ in that same convention. With $r = \sqrt{Q}R$ and Q fixed in a chart, the exact conversions are

$$M_e = Q^{e/2-2A} M_e^*, \quad M_e^* = \int R^e E_e^* dR, \quad b_e^*(R) = Q^{(e+1)/2} b_e(\sqrt{Q}R), \\ \Sigma_e(R) = Q^{2A} \sigma_e(\sqrt{Q}R), \quad (\partial_R + e/R) \Sigma_e = -E_e^* + b_e^* M_e^*. \quad (33)$$

Substitute $r = \sqrt{Q}R$ including its measure in the moment, and differentiate Σ_e to prove each factor. The input F_e is already auxiliary independent. Taking only an averaged moment of an unaveraged $F(R, Y)$ would not suffice: $F = b_e(R) \cos(2\pi Y_1)$ has zero averaged moment but a nonzero exterior primitive at each generic Y .

11 The source covariance derivative and its signed right inverse

At each slow point let $w_{\text{tan}} = (w_\theta, w_z)$ and define

$$\mathcal{C}(w) = \langle w_r w_{\text{tan}} \rangle, \quad \mathcal{B}(w, v) = \langle w_r v_{\text{tan}} + v_r w_{\text{tan}} \rangle.$$

Direct expansion proves $\mathcal{C}(w + v) = \mathcal{C}(w) + \mathcal{B}(w, v) + \mathcal{C}(v)$ and $D\mathcal{C}(w)[v] = \mathcal{B}(w, v)$.

Use the exact source real pulses $b_\sigma = \chi_g(\xi_g) \psi(v) t_\sigma^h \cos(k\Phi_\sigma)$, $\sigma \in \{+, -\}$. They exclude the slow cutoff η_β and scalar amplitudes; χ_g is an auxiliary transverse cutoff. The two signs have disjoint auxiliary supports. Put $H_{\text{cov}} = [\mathcal{C}(b_+) \mid \mathcal{C}(b_-)]$. For slow scalar coefficients independent of both averages,

$$\mathcal{C}(a_+ b_+ + a_- b_-) = H_{\text{cov}} (a_+^2, a_-^2)^T.$$

All cross products vanish by disjointness. The factor $1/2$ from $\langle \cos^2(k\Phi_\sigma) \rangle_\theta$ is already inside the columns.

On the source interior cone the fixed primary amplitudes satisfy $a_\sigma = \sqrt{(H_{\text{cov}}^{-1}T_{0,*})_\sigma} > 0$, $T_{0,*} = (Q/q)^{A+1/2}T_0$, and $W_0 = \sqrt{\varepsilon} \sum_\sigma a_\sigma b_\sigma$. For any signed slow real vector Σ , define

$$d_\Sigma = H_{\text{cov}}^{-1}\Sigma/\varepsilon, \quad \delta a_\sigma = (d_\Sigma)_\sigma/(2a_\sigma), \quad L_\Sigma = \sqrt{\varepsilon} \sum_\sigma \delta a_\sigma b_\sigma. \quad (34)$$

It follows exactly that

$$\mathcal{B}(W_0, L_\Sigma) = \varepsilon H_{\text{cov}}(2a \odot \delta a) = \Sigma, \quad (35)$$

$$\mathcal{C}(W_0 + L_\Sigma) = \varepsilon T_{0,*} + \Sigma + \varepsilon H_{\text{cov}}(\delta a_+^2, \delta a_-^2)^T. \quad (36)$$

This proves a linear right inverse of the derivative. It is not an exact inverse of the quadratic covariance map. There is no sign or size restriction on Σ in these identities; no square root of $H_{\text{cov}}^{-1}(T_{0,*} + \Sigma/\varepsilon)$ is taken.

With Euclidean and induced operator norms and $a_{\min} = \min_\sigma |a_\sigma| > 0$, they give the explicit bound

$$\|\mathcal{C}(L_\Sigma)\| \leq \frac{\|H_{\text{cov}}\| \|H_{\text{cov}}^{-1}\|^2}{4\varepsilon a_{\min}^2} \|\Sigma\|^2. \quad (37)$$

Indeed $\|\delta a\| \leq \|H_{\text{cov}}^{-1}\| \|\Sigma\|/(2\varepsilon a_{\min})$ and $\|(\delta a_\sigma^2)_\sigma\|_2 \leq \|\delta a\|_2^2$. All higher derivative costs are explicit from Leibniz and

$$\partial^I H_{\text{cov}}^{-1} = -H_{\text{cov}}^{-1} \sum_{0 < J \leq I} \binom{I}{J} (\partial^J H_{\text{cov}}) \partial^{I-J} H_{\text{cov}}^{-1},$$

and, for any nonzero scalar a , $\partial^I(a^{-1}) = -a^{-1} \sum_{0 < J \leq I} \binom{I}{J} (\partial^J a) \partial^{I-J}(a^{-1})$. Together with (34) these recurrences give finite bounds for each order without assuming constant amplitudes or constant columns.

12 Assembly, the actual covariance change, and residual cancellation

For the physical target $\sigma = (\sigma_2, \sigma_1)$ from (31), use the source squared slow partition $\sum_\beta \eta_\beta^2 = 1$ on its support. Set

$$\begin{aligned} W_{0,\text{phys}} &= \sum_\beta Q_\beta^{-A} \eta_\beta W_{0,\beta}, \\ v_{\text{pr}} &= \sum_\beta Q_\beta^{-A} \eta_\beta L_\beta(Q_\beta^{2A} \sigma). \end{aligned} \quad (38)$$

Overlapping slow labels have disjoint auxiliary supports. Taking cross covariance and using (35) gives

$$\mathcal{B}(W_{0,\text{phys}}, v_{\text{pr}}) = \sum_\beta Q_\beta^{-2A} \eta_\beta^2 Q_\beta^{2A} \sigma = \sigma.$$

Every band scale remains distinct. Likewise the leading physical factor is $Q^{-2A} \varepsilon (Q/q)^{A+1/2} = Q^{-A+h+1/2} q^{-A-1/2} = q^{-A-1/2}$ because $A = 1/2 + h$.

Apply the exact curl (28) to each coefficient of v_{pr} , including the slow cutoffs before curling. Write the resulting divergence-free increment as $v_s = v_{\text{pr}} + r_s$. Let $t_{s,m}$ denote the chart transverse harmonic coefficients of $\eta_\beta L_\beta(Q_\beta^{2A} \sigma)$, before multiplication by Q_β^{-A} . For each such coefficient

choose $\pi_{s,m} = i(n^T K t_{s,m} - (n')^T t_{s,m}) / (k_\beta m |n|^2)$ and sum $\text{Re}(Q_\beta^{-2A} \pi_{s,m} e^{i\phi_{\beta,m}})$ to define the smooth compact π_s . Its full gradients remain below. Let the actual old wave w consist of the wave part of u_* plus the candidate w_y , including all existing curl corrections, and write $w = W_{0,\text{phys}} + e_w$. Define tensor covariance $\mathcal{T}(w) = \langle w \otimes w \rangle$ and $\mathcal{B}_T(u, v) = \langle u \otimes v + v \otimes u \rangle$. Expansion, with $e_w = w - W_{0,\text{phys}}$, proves

$$\Delta \mathcal{T} = \mathcal{B}_T(W_{0,\text{phys}}, v_{\text{pr}}) + \mathcal{B}_T(e_w, v_{\text{pr}}) + \mathcal{B}_T(w, r_s) + \mathcal{T}(v_s). \quad (39)$$

Its first radial–tangential pair is σ . All other entries and the remaining three tensors are retained. For example their radial–tangential pair has bound

$$2\|e_w\|_{L_{\theta,Y}^2} \|v_{\text{pr}}\|_{L_{\theta,Y}^2} + 2\|w\|_{L_{\theta,Y}^2} \|r_s\|_{L_{\theta,Y}^2} + \|v_s\|_{L_{\theta,Y}^2}^2,$$

by Cauchy–Schwarz. The compact-set estimates above and the explicit curl differentiate to bound all these terms at each finite order.

For clarity the complete residual after adding v_s and its selected pressure π_s is, without any averaging,

$$\begin{aligned} R_{\nu_{\text{NS}}}(u + v_s, p + \pi_s) &= R_{\nu_{\text{NS}}}(u, p) + \partial_t v_s + (u \cdot \nabla) v_s + (v_s \cdot \nabla) u \\ &\quad - \nu_{\text{NS}} \Delta v_s + \nabla \pi_s + (v_s \cdot \nabla) v_s. \end{aligned} \quad (40)$$

This follows by expanding the product and retains viscosity and every pressure term. For a symmetric tensor S the averaged tangential divergence in physical cylindrical coordinates is

$$(\langle \nabla \cdot S \rangle)_\theta = (\partial_r + 2/r) \langle S_{r\theta} \rangle + \partial_z \langle S_{z\theta} \rangle, \quad (\langle \nabla \cdot S \rangle)_z = (\partial_r + 1/r) \langle S_{rz} \rangle + \partial_z \langle S_{zz} \rangle.$$

To verify, write the Cartesian divergence in the orthonormal cylindrical basis, using $\partial_\theta e_r = e_\theta$, $\partial_\theta e_\theta = -e_r$; periodic angular derivatives average to zero and the two basis terms give $2/r$ and $1/r$ respectively. The auxiliary chain terms average to zero by periodicity, but the physical slow derivatives remain.

Let $U = u_* + w_y$, $P_y = p_* + \pi_y$, and set

$$\mathcal{L} = \partial_t v_s + (U \cdot \nabla) v_s + (v_s \cdot \nabla) U - \nu_{\text{NS}} \Delta v_s + \nabla \pi_s.$$

For a fully explicit averaged remainder, write $U = u_{\text{mean}} + w$, where u_{mean} is its angular mean, and put $\mathcal{V} = \mathcal{L} - (w \cdot \nabla) v_s - (v_s \cdot \nabla) w$. Then the average of (40) is $\langle R_y \rangle + \langle \mathcal{V} \rangle + \nabla \cdot \Delta \mathcal{T}$. Let $S_0 = \mathcal{B}_T(W_{0,\text{phys}}, v_{\text{pr}})$ and $S_{\text{rem}} = \mathcal{B}_T(e_w, v_{\text{pr}}) + \mathcal{B}_T(w, r_s) + \mathcal{T}(v_s)$. Consequently the radial divergence of the first pair in (39) changes the prescribed averaged residual F_e into $b_e M_e$ by (32). The axial tensor divergences, the other three tensors, the exact linear terms and the pressure terms of (40) remain in the new residual. Equivalently the new averaged tangential residual is exactly

$$\begin{aligned} \langle R_{\text{new}} \rangle_{\text{tan}} &= \langle R_{\nu_{\text{NS}}}(u_*, p_*) \rangle_{\text{tan}} + (b_2 M_2, b_1 M_1) + \langle \mathcal{V} \rangle_{\text{tan}} \\ &\quad + \partial_z ((S_0)_{z\theta}, (S_0)_{zz}) + (\nabla \cdot S_{\text{rem}})_{\text{tan}}. \end{aligned}$$

Every term here is defined by the displayed differentiated fields and tensors. None is assumed to vanish. The map from y through (30), (31), (34), and (38) is now an actual composite construction. It retains the signed residual produced by the coupled candidate and the source’s original physical and chart units.

13 Signs and the exact full-flow reflection

The maps $F_e \mapsto M_e \mapsto \sigma_e \mapsto L_\Sigma$ are linear and reverse sign with their inputs; $\mathcal{C}(L_\Sigma)$ is even. The two primary amplitudes may also be replaced independently by $s_\sigma a_\sigma$, $s_\sigma \in \{-1, 1\}$, provided the same fixed choice is used in each denominator: the cross identity is unchanged. This does not negate the nonlinear Navier–Stokes equation.

Here is the exact full-flow map. In Cartesian coordinates let $O = \text{diag}(1, -1, 1)$ and define

$$u^\sharp(x, t) = Ou(Ox, t), \quad p^\sharp(x, t) = p(Ox, t), \quad f^\sharp(x, t) = Of(Ox, t).$$

The chain rule gives $\nabla u^\sharp = O(\nabla u)(Ox, t)O$, $\Delta u^\sharp = O(\Delta u)(Ox, t)$, and $(u^\sharp \cdot \nabla)u^\sharp = O((u \cdot \nabla)u)(Ox, t)$. Thus $\nabla \cdot u^\sharp = 0$ and $R_{\nu_{\text{NS}}}(u^\sharp, p^\sharp) = OR_{\nu_{\text{NS}}}(u, p)(Ox, t)$ with the same positive viscosity. In cylindrical components this is

$$(u_r^\sharp, u_\theta^\sharp, u_z^\sharp)(r, \theta, z, t) = (u_r, -u_\theta, u_z)(r, -\theta, z, t),$$

and the force transforms identically. Pressure reflects in its argument. For an axisymmetric field this reverses exactly its swirl; meridional components remain fixed. The entire momentum tensor is mapped to OTO^T , so its radial–tangential pair transforms by $(T_{r\theta}, T_{rz}) \mapsto (-T_{r\theta}, T_{rz})$ after angular averaging. Vorticity transforms as $\omega^\sharp(x, t) = \det(O)O\omega(Ox, t)$, obtained by applying the same derivative identities to the Levi-Civita formula for curl. The map is an involution, preserves the kinetic-energy integral (by $|\det O| = 1$), compact support, smoothness, and blowup magnitude along the reflected path. It preserves zero initial data when present. It proves transport of any established axisymmetric signed swirl divergence to the opposite sign; it does not itself prove an endpoint.

The source five-moment correction, auxiliary mean inverse, and infinite summation are subsequent source operations. Their full estimates are not re-proved here. The earlier draft formulas for them are withdrawn: in particular the fifth moment row is $\int (2RG\gamma_d - RV\Delta v) \, dR = -J_z$, without an extra outer R , and its reserved profile in a Q chart includes $(Q/q)^A$. The finite map proved above retains the two moments and all other residual terms, so it establishes neither a third shooting return nor an infinite modified viscous sequence.

14 The released result and its actual signed path

The source in this section is the supplied 166-page edition of *Finite Time Blowup for Navier–Stokes*, whose SHA-256 is

0e779481c4da40bd28d1e642e1d8ca57447d129610df28dfa5a11e9af8ae228f.

Its Theorem 1.1, on page 1, states the following result for the standard three-dimensional incompressible Navier–Stokes equation. For every fixed physical viscosity $\nu_{\text{NS}} > 0$, the source constructs

$$f \in C_c^\infty(\mathbb{R}^3 \times (0, \infty); \mathbb{R}^3), \quad u, p \in C^\infty(\mathbb{R}^3 \times [0, 1)),$$

with a fixed compact spatial support K for u and p , such that

$$\begin{aligned} \partial_t u + (u \cdot \nabla)u - \nu_{\text{NS}}\Delta u + \nabla p &= f, \quad \nabla \cdot u = 0, \quad u(\cdot, 0) = 0, \\ \sup_{t < 1} \|u(t)\|_2 &< \infty, \quad \limsup_{t \uparrow 1} \|u(t)\|_\infty = \infty. \end{aligned} \tag{41}$$

It also excludes a global smooth solution with the same force and datum and uniformly bounded kinetic energy. Its Corollary 10.6, pages 125–126, gives a periodic construction on $\mathbb{R}^3/\mathbb{Z}^3$ with

smooth time-compact force, zero initial datum and no global smooth continuation. The source identifies these with alternatives (C) and (D) in the referenced Millennium problem statement. The force is nonzero, as the source explicitly notes on page 124. These are statements of the cited source theorem; the finite algebra checks in this lane do not constitute an independent verification of the complete 166-page proof.

The source proves more than an unsigned norm divergence. In its viscosity-one construction, equations (10.20)–(10.21), page 124, give

$$\begin{aligned} A &= \frac{1}{2} + h, & 0 < h < 1/100, & & X_{\text{in}} > 0, & & e_0 > 0, \\ x_\tau &= (\sqrt{2X_{\text{in}}\tau}, 0, 0), & t_\tau &= 1 - \tau, \\ u_\theta(x_\tau, t_\tau) &= \tau^{-A}(e_0 + \epsilon(\tau)), & |\epsilon(\tau)| &\leq C\tau^{2h} \quad (0 < \tau < \tau_0). \end{aligned} \tag{42}$$

Here C, τ_0 are finite constants supplied by the source asymptotic; e_0 and X_{in} retain their source values. If $C > 0$, choose $0 < \tau_1 \leq \tau_0$ so that $C\tau_1^{2h} \leq e_0/2$; if $C = 0$, take $\tau_1 = \tau_0$. It follows directly that

$$u_\theta(x_\tau, t_\tau) \geq \frac{e_0}{2}\tau^{-A} \longrightarrow +\infty \quad (0 < \tau < \tau_1).$$

The observation point moves to the origin. This assertion does not replace that path by the fixed point $x = 0$. On the path, $q = \tau > 0$, and at its cylindrical angle $\theta = 0$ one has $e_\theta = e_2$. The geometric domain $q > 0$, the positive exponent h , and the signed velocity value are three retained parts of the same explicit formula.

15 Exact propagation at retained viscosities and orientations

For any smooth fields u, p define the physical momentum residual

$$R_{\nu_b}(u, p) = \partial_t u + (u \cdot \nabla)u - \nu_b \Delta u + \nabla p.$$

Let $\nu_b > 0$ and $\nu_t > 0$ be the base and target physical viscosities, and retain

$$\rho = \sqrt{\nu_t/\nu_b}, \quad O^T O = O O^T = I, \quad a \in \mathbb{R}^3.$$

Define the affine spatial coordinate $y = O^T(x - a)/\rho$ and the fields

$$u^\sharp(x, t) = \rho O u(y, t), \quad p^\sharp(x, t) = \rho^2 p(y, t), \quad f^\sharp(x, t) = \rho O f(y, t). \tag{43}$$

No time coordinate is changed. These formulas give a bijection on smooth field triples, with inverse

$$u(y, t) = \rho^{-1} O^T u^\sharp(a + \rho O y, t), \quad p(y, t) = \rho^{-2} p^\sharp(a + \rho O y, t), \quad f(y, t) = \rho^{-1} O^T f^\sharp(a + \rho O y, t).$$

For the Jacobian convention $(Du)_{ij} = \partial_j u_i$, differentiation gives, without dropping an orientation or viscosity,

$$\begin{aligned} Du^\sharp &= O(Du)(y, t)O^T, & \nabla \cdot u^\sharp &= (\nabla \cdot u)(y, t), \\ \partial_t u^\sharp &= \rho O \partial_t u(y, t), & (u^\sharp \cdot \nabla_x)u^\sharp &= \rho O((u \cdot \nabla_y)u)(y, t), \\ \Delta_x u^\sharp &= \rho^{-1} O \Delta_y u(y, t), & \nabla_x p^\sharp &= \rho O \nabla_y p(y, t). \end{aligned}$$

The nonlinear identity follows by multiplying $Du^\sharp$ by $u^\sharp$ and using $O^T O = I$. The Laplacian identity follows by contracting the two spatial derivative indices and using orthogonality. Since $\nu_t = \rho^2 \nu_b$, these identities prove

$$R_{\nu_t}(u^\sharp, p^\sharp) - f^\sharp = \rho O(R_{\nu_b}(u, p) - f)(y, t). \tag{44}$$

In particular the map preserves the exact equation as well as each nonzero residual with its exact factor.

For vorticity, the cross-product identity $(Ov) \times (Ow) = \det(O)O(v \times w)$ gives

$$\omega^\sharp(x, t) = \det(O)O\omega(y, t). \quad (45)$$

One proof of this identity takes its scalar product with Oz : both sides give $\det(O)\det[v, w, z]$ for every z . Applying it to the contracted first derivatives proves the vorticity formula; the factor ρ in velocity cancels the inverse factor in a derivative.

Spatial support becomes $a + \rho OK$. The Jacobian has absolute determinant ρ^3 , for both signs of $\det O$. Hence, for $1 \leq p < \infty$,

$$\|u^\sharp(t)\|_p = \rho^{1+3/p}\|u(t)\|_p, \quad \|u^\sharp(t)\|_\infty = \rho\|u(t)\|_\infty, \quad \|u^\sharp(t)\|_2^2 = \rho^5\|u(t)\|_2^2. \quad (46)$$

The Frobenius norm of $O(Du)O^T$ equals that of Du . Consequently, with integrals over the same time interval,

$$\nu_t \int \|\nabla u^\sharp\|_2^2 dt = \rho^5 \nu_b \int \|\nabla u\|_2^2 dt.$$

For arbitrary directions $v_1, \dots, v_j$, the exact force derivative is

$$D_x^j \partial_t^m f^\sharp[v_1, \dots, v_j](x, t) = \rho^{1-j} O(D_y^j \partial_t^m f)[O^T v_1, \dots, O^T v_j](y, t).$$

Thus smoothness, compact space-time support and vanishing of every derivative at a terminal point are preserved. The initial datum remains zero. A global smooth bounded-energy competitor for the transformed datum and force would, under the displayed inverse, be a competitor for the base datum and force, because its energy is multiplied by the fixed factor ρ^{-5} . This proves propagation of the source noncontinuation conclusion through the map.

16 The source positive branch and the exact negative branch

Apply the map to the source field in (42). Here $\nu_b = 1$ specifies the source object being used; physical target viscosity ν_t remains arbitrary and $\rho = \sqrt{\nu_t}$ is retained. Take $a = 0$ and the two particular orthogonal matrices

$$O_+ = I, \quad O_- = \text{diag}(1, -1, 1).$$

Both fix e_1 , so they give the same spatial observation path $x_\tau^\sharp = \rho x_\tau$ and the same time $1 - \tau$. Since $O_+ e_2 = e_2$ and $O_- e_2 = -e_2$, their exact signed values are

$$u_{\pm, \theta}^\sharp(\rho x_\tau, 1 - \tau) = \pm \rho \tau^{-A}(e_0 + \epsilon(\tau)) \longrightarrow \begin{cases} +\infty, & +, \\ -\infty, & -. \end{cases} \quad (47)$$

All support, energy, force and viscosity statements in the preceding section apply to both triples. For the negative branch the force is $\rho O_- f(O_- x / \rho, t)$. Equality of this force with the positive branch force is not needed and is not asserted.

For a general, possibly nonaxisymmetric field the reflection alone has the exact cylindrical component formula

$$(u^r, u^\theta, u^z)^\sharp(r, \theta, z, t) = (u^r, -u^\theta, u^z)(r, -\theta, z, t).$$

Indeed $O_- e_r(-\theta) = e_r(\theta)$, $O_- e_\theta(-\theta) = -e_\theta(\theta)$ and $O_- e_z = e_z$. This proves the formula even in the presence of all oscillatory corrections. For the axisymmetric background the argument $-\theta$

can be dropped because its components are independent of θ . For a velocity covariance tensor the exact Cartesian transformation is $T^\sharp(x, t) = \rho^2 OT(y, t)O^T$: expand the outer product of ρOw with itself and use linearity of the retained average. For its angularly averaged symmetric tensor, the reflection basis calculation at $\rho = 1$ gives

$$(T_{r\theta}, T_{rz})^\sharp = (-T_{r\theta}, T_{rz}).$$

Under the additional viscosity dilation its components have factor ρ^2 at the corresponding points. Thus the two covariance columns and their target transform together. Positive coefficients in a source representation $T = c_1 v_1 + c_2 v_2$, $c_i > 0$, remain the same coefficients of the transformed columns; no change of coefficient sign is required to realize the negative-swirl branch.

The source's periodic construction also has this reflection: O_- preserves $\mathbb{Z}^3$, so $[x] \mapsto [O_-x]$ is well defined on $\mathbb{R}^3/\mathbb{Z}^3$. The derivative identities apply to periodic lifts. This gives both signed orientations of the source Corollary 10.6. Arbitrary $O \in O(3)$ need not preserve this particular lattice; only the lattice-preserving reflection just specified is used for this periodic assertion.

17 Propagation into the coupled calculation

The preceding sections use the released whole-space and periodic constructions at the precise places where those source results apply. The original coupled Boussinesq work is due to Levent Alpöge and Tristan Buckmaster. This lane's retained coupled system, its finite controlled stages and its residual maps remain the separate objects defined in the preceding parts of this reader.

For every compact candidate $(u_* + w_y, p_* + \pi_y)$ in the covariance part, (44) propagates its full computed residual. Linearity of the transformation and the exact nonlinear identity give in particular

$$\Delta R_y^\sharp(x, t) = \rho O \Delta R_y(y, t).$$

Thus the finite candidate's cutoff forces, curl errors and covariance remainders all have an exact transformed counterpart. A nonzero remainder is carried to a nonzero remainder, because the map is invertible. The source's residual flatness at the singular point belongs to its completed local source construction; its extended smooth force need not vanish on the entire time-one slice. The next section calculates the surviving moments of the actual finite candidate, so that propagation continues through its computed terms rather than assuming that its different coupled coefficients inherit the source's infinite estimates.

The earlier suggestion of a negative blowout imposes no condition on the programme. Formula (42) is the source-selected sign; formula (47) proves the exact relation of its two orientations. Both formulas retain a positive physical viscosity and the source coordinate domain $q > 0$.

18 The two moments of the actual compact curl increment

The objects in this calculation are the finite smooth source state (u_*, p_*) and the actual added curl field $v = w_y$ with its selected pressure $\pi = \pi_y$ from the signed covariance bridge. Their physical residual increment is

$$\Delta R = \partial_t v + (u_* \cdot \nabla)v + (v \cdot \nabla)u_* + (v \cdot \nabla)v - \nu_{\text{NS}} \Delta v + \nabla \pi. \quad (48)$$

The physical viscosity ν_{NS} is unchanged. Both velocities are divergence free: the source state uses its actual divergence-free background and curl increments, and v is the curl of its specified compact potential. No residual equation for u_* is required in (48); its residual has been subtracted exactly.

All radial supports involving v or π lie in a fixed compact subinterval of $(0, \infty)$ on the compact space–time patch at issue.

Here the source extended field is indispensable. Its absolute auxiliary coordinate is $Y \in \mathbb{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$, and physical evaluation uses precisely source (6.2)–(6.4):

$$\begin{aligned} J_g &= \begin{pmatrix} 3 & 1 \\ 1 & 5 \end{pmatrix}, \quad \Lambda_g = 4 - \sqrt{2}, \quad T_g = 4 + \sqrt{2}, \quad b_g = \sqrt{2} - 1, \\ \mathbf{v}_r &= (1, -b_g), \quad \mathbf{v}_t = (b_g, 1), \quad \rho_g = \frac{\log \Lambda_g}{\log T_g}, \quad \kappa_s = 10^{-5}, \quad d_r = 2((1+h)\rho_g - h\kappa_s), \\ Y(r, t) &= \mathbf{v}_r r^{d_r} + \mathbf{v}_t t \pmod{\mathbb{Z}^2}, \quad \mathcal{D}_r = \partial_r + d_r r^{d_r-1} \mathbf{v}_r \cdot \partial_Y, \quad \mathcal{D}_t = \partial_t + \mathbf{v}_t \cdot \partial_Y. \end{aligned} \quad (49)$$

These are the source’s original quantities, not the coupled physical frequency or diffusivity. Angular and axial physical derivatives are ∂_θ and ∂_z , because $Y(r, t)$ depends on neither variable. Every field below is first differentiated using these lifted physical derivatives. A tilde is suppressed for this extended representation. Evaluation at (49) recovers the physical field and its physical derivatives by the chain rule.

The incompressibility needed below holds at every independent Y . Indeed the actual finite source construction retains the form $u_* = \text{curl}_{\mathcal{D}} \mathcal{A}_* + B_* e_\theta$ with $\partial_\theta B_* = 0$, and the added field is $v = \text{curl}_{\mathcal{D}} \mathcal{A}_y$. The retained source potentials here include its finite mean and wave curl increments. The lifted derivatives in (49), together with ∂_θ and ∂_z , commute, and $\mathcal{D}_r r = 1$. Substitution in the cylindrical curl and divergence formulas therefore gives $\text{div}_{\mathcal{D}} \text{curl}_{\mathcal{D}} \mathcal{A} = 0$ identically in (r, θ, z, t, Y) : each mixed derivative cancels its opposite, including the derivative of r^{-1} . Also $\text{div}_{\mathcal{D}}(B_* e_\theta) = r^{-1} \partial_\theta B_* = 0$. Thus extended incompressibility is proved from the actual retained potential representation, rather than inferred from its restriction to the physical phase map.

Use the normalized averages

$$\langle a \rangle_\theta = \frac{1}{2\pi} \int_0^{2\pi} a \, d\theta, \quad \langle a \rangle_Y = \int_{\mathbb{T}^2} a \, dY, \quad \langle a \rangle = \langle \langle a \rangle_\theta \rangle_Y, \quad \int_{\mathbb{T}^2} 1 \, dY = 1.$$

They act on cylindrical components before physical evaluation, exactly as source (3.7)–(3.9). All claimed integral identities below concern these specified averages; the exact relation to evaluated moments is also proved below.

18.1 The actual harmonic curl has zero angular mean

For each retained source harmonic, its extended potential and pressure have cylindrical components of the form

$$C_j(r, z, t, Y) e^{i(n_j \theta + \psi_j(r, z, t, Y))}, \quad P_j(r, z, t, Y) e^{i(n_j \theta + \psi_j(r, z, t, Y))}, \quad n_j = k_{\beta_j} m_j p_j \in \mathbb{Z} \setminus \{0\}.$$

The coefficients include the actual cutoff χ , frame B , map L , coupled solution $y = (\Theta, \Omega)^T$, band powers and pressure formula. Their cylindrical components are independent of θ . In particular, no sign of Θ , Ω or the real part of a harmonic is prescribed. For a vector potential $\mathcal{A}$ the exact cylindrical curl is

$$(\text{curl } \mathcal{A})_r = r^{-1} \partial_\theta \mathcal{A}_z - \partial_z \mathcal{A}_\theta, \quad (\text{curl } \mathcal{A})_\theta = \partial_z \mathcal{A}_r - \mathcal{D}_r \mathcal{A}_z, \quad (\text{curl } \mathcal{A})_z = (\mathcal{D}_r + r^{-1}) \mathcal{A}_\theta - r^{-1} \partial_\theta \mathcal{A}_r.$$

These formulas include the derivatives of the cylindrical basis. Each displayed operation retains the same factor $e^{in_j \theta}$: the coefficients of $\mathcal{D}_r$ are independent of θ , and ∂_θ multiplies that factor by

in_j . Integration of the factor over the angular circle gives zero. Taking real parts and the actual finite harmonic sum proves

$$\langle v_r \rangle_\theta = \langle v_\theta \rangle_\theta = \langle v_z \rangle_\theta = \langle \pi \rangle_\theta = 0 \quad \text{at every } (r, z, t, Y). \quad (50)$$

This includes every curl and cutoff correction, not merely the principal amplitude. The same identities hold after physical evaluation because $Y(r, t)$ is independent of θ .

For every smooth periodic coefficient a , integration of its auxiliary derivatives gives

$$\langle \mathcal{D}_r a \rangle = \partial_r \langle a \rangle, \quad \langle \mathcal{D}_t a \rangle = \partial_t \langle a \rangle, \quad \langle \partial_z a \rangle = \partial_z \langle a \rangle, \quad \langle \partial_\theta a \rangle = 0. \quad (51)$$

There is no omitted variable coefficient in this assertion: $d_r r^{d_r-1} \mathbf{v}_r$ is independent of Y . Applying the first identity again proves $\langle \mathcal{D}_r^2 a \rangle = \partial_r^2 \langle a \rangle$. Explicitly, if $c(r) = d_r r^{d_r-1} \mathbf{v}_r$, then

$$\mathcal{D}_r^2 a = \partial_r^2 a + 2c \cdot \partial_Y \partial_r a + c' \cdot \partial_Y a + (c \cdot \partial_Y)^2 a;$$

each of the three additional terms has zero auxiliary average.

18.2 Exact removal of the linear and mean-flow terms

Put

$$\bar{u}_* := \langle u_* \rangle_\theta, \quad w_* := u_* - \bar{u}_*.$$

The field $\bar{u}_*$ may retain auxiliary dependence; none is discarded. Since it is independent of θ , (50) gives

$$\langle \bar{u}_* \otimes v + v \otimes \bar{u}_* \rangle = 0.$$

Define the actual symmetric cross-and-quadratic tensor

$$S := \langle w_* \otimes v + v \otimes w_* + v \otimes v \rangle. \quad (52)$$

Products of all original source waves with the added curl field are retained in this formula, including old and new curl remainders.

For two divergence-free fields a, b , Cartesian component expansion gives

$$\nabla \cdot (a \otimes b + b \otimes a) = (b \cdot \nabla) a + (a \cdot \nabla) b, \quad \nabla \cdot (v \otimes v) = (v \cdot \nabla) v.$$

For example $\sum_j \partial_j (a_i b_j) = \sum_j b_j \partial_j a_i + a_i \sum_j \partial_j b_j$; the last term is zero. The lifted derivatives obey the same product rule, so the identities hold on the extended domain before evaluation. The three advection terms in (48) are therefore exactly the divergence of $u_* \otimes v + v \otimes u_* + v \otimes v$.

For completeness, set $\mathcal{L}_0 = \mathcal{D}_r^2 + r^{-1} \mathcal{D}_r + r^{-2} \partial_\theta^2 + \partial_z^2$. The vector Laplacian and pressure gradient in the two components are

$$(\Delta v)_\theta = (\mathcal{L}_0 - r^{-2}) v_\theta + 2r^{-2} \partial_\theta v_r, \quad (\Delta v)_z = \mathcal{L}_0 v_z, \quad (\nabla \pi)_\theta = r^{-1} \partial_\theta \pi, \quad (\nabla \pi)_z = \partial_z \pi.$$

Equations (50)–(51) prove separately

$$\langle \mathcal{D}_t v \rangle_{\text{tan}} = 0, \quad -\nu_{\text{NS}} \langle \Delta v \rangle_{\text{tan}} = 0, \quad \langle \nabla \pi \rangle_{\text{tan}} = 0. \quad (53)$$

Thus viscosity and pressure were included before averaging; their vanishing here is a proved consequence of the actual carriers.

If T is symmetric, differentiation of $e_r = (\cos \theta, \sin \theta, 0)$, $e_\theta = (-\sin \theta, \cos \theta, 0)$ gives

$$\begin{aligned}(\nabla \cdot T)_\theta &= (\mathcal{D}_r + 2/r)T_{r\theta} + r^{-1}\partial_\theta T_{\theta\theta} + \partial_z T_{z\theta}, \\(\nabla \cdot T)_z &= (\mathcal{D}_r + 1/r)T_{rz} + r^{-1}\partial_\theta T_{\theta z} + \partial_z T_{zz}.\end{aligned}$$

Indeed $\partial_\theta e_r = e_\theta$ and $\partial_\theta e_\theta = -e_r$ supply the two radial connection terms in the first line and the single radial connection term in the second. Averaging these formulas and using (51), (52), and (53) proves the exact residual identities

$$F_2 := \langle \Delta R_\theta \rangle = (\partial_r + 2/r)S_{r\theta} + \partial_z S_{z\theta}, \quad F_1 := \langle \Delta R_z \rangle = (\partial_r + 1/r)S_{rz} + \partial_z S_{zz}. \quad (54)$$

In particular every mean-flow cross term vanishes as the divergence of its zero averaged tensor, even when the mean depends on Y .

18.3 The surviving axial fluxes and their signs

Define the two moments with the original physical radial measure:

$$M_2(z, t) = \int_0^\infty r^2 F_2(r, z, t) dr, \quad M_1(z, t) = \int_0^\infty r F_1(r, z, t) dr. \quad (55)$$

Their exact values are

$$M_2 = \partial_z J_2, \quad J_2 = \int_0^\infty r^2 \langle w_{*,z} v_\theta + v_z w_{*,\theta} + v_z v_\theta \rangle dr, \quad (56)$$

$$M_1 = \partial_z J_1, \quad J_1 = \int_0^\infty r \langle 2w_{*,z} v_z + v_z^2 \rangle dr. \quad (57)$$

To prove them, multiply (54) by the respective weight: $r^2(\partial_r + 2/r)S_{r\theta} = \partial_r(r^2 S_{r\theta})$ and $r(\partial_r + 1/r)S_{rz} = \partial_r(r S_{rz})$. The weighted boundary values at both ends of $(0, \infty)$ are zero, because each entry of S contains at least one compactly supported factor v . Differentiation under the radial integral is valid by the same compact support and smoothness. This proves (56)–(57) including their signs and factors.

There is no pointwise cancellation of the axial derivatives in these formulas. Since the added field has compact axial support, J_1 and J_2 have compact axial support and

$$\int_{\mathbb{R}} M_e(z, t) dz = 0 \quad (e = 1, 2). \quad (58)$$

This follows by evaluating the boundary values of J_e at the two axial ends. It does not assert $M_e(z, t) = 0$ at an individual axial point. The compact radial primitive in the covariance bridge must therefore retain its $b_e M_e$ terms with these actual M_e .

At fixed source state, replacing the added field and pressure by their negatives sends the old-wave cross terms in (56)–(57) to their negatives and leaves the quadratic terms unchanged. Thus negating an amplitude is not a sign reversal of either full moment. Under the full physical reflection $O = \text{diag}(1, -1, 1)$, applied to the source state and added field together as $a^\sharp(x, t) = Oa(Ox, t)$, the cylindrical radial and axial components retain their signs and the angular components reverse sign at $(r, -\theta, z, t)$. Changing the angular integration variable therefore proves

$$J_2^\sharp = -J_2, \quad M_2^\sharp = -M_2, \quad J_1^\sharp = J_1, \quad M_1^\sharp = M_1.$$

An explicit transverse-curl example shows why compactness and nonzero angular frequency alone do not force either moment to vanish. This example tests precisely those structural properties, and is not a replacement for the retained source state. Let H be any nonzero real $C_c^\infty((r_-, r_+))$ function with $0 < r_- < r_+$, let $g \in C_c^\infty(\mathbb{R})$ be nonconstant, let n be a nonzero integer, and put $f = rH'$. Use zero old state and the real potential

$$\mathcal{A}_r = fg \sin(n\theta), \quad \mathcal{A}_\theta = 0, \quad \mathcal{A}_z = Hg \cos(n\theta).$$

It has the required transverse representation: $\phi = n\theta$, $\xi = (n/r)e_\theta$, $C = (-ifg, 0, Hg)$ in cylindrical components, $a = i\xi \times C$, and $C = i\xi \times a/|\xi|^2$ because $\xi \cdot C = 0$. The curl is exactly

$$v_r = -\frac{nHg}{r} \sin(n\theta), \quad v_\theta = fg' \sin(n\theta) - H'g \cos(n\theta), \quad v_z = -\frac{nfg}{r} \cos(n\theta).$$

Consequently

$$M_2 = n g g' \int_0^\infty r^2 (H')^2 dr, \quad M_1 = n^2 g g' \int_0^\infty r (H')^2 dr.$$

For example $g(z) = \exp(-1/(1-z^2))$ on $|z| < 1$, extended by zero, has $g(z)g'(z) \neq 0$ at every $0 < |z| < 1$. Both integrals are strictly positive: a compactly supported smooth function with $H' = 0$ everywhere would be constant and hence zero. Thus both moments can be nonzero, with their displayed signed values, within the exact transverse-curl architecture.

18.4 Bounds in the actual fields and original chart units

At fixed (z, t) set

$$\|a\|_e^2 = \int_0^\infty r^e \langle |a|^2 \rangle dr, \quad e = 1, 2.$$

When an old-field norm is used below it can be restricted to the enclosing radial support of v and $\partial_z v$. This avoids imposing any decay condition on an unused exterior part of the old field. The following component bounds contain only actual fields:

$$|J_2| \leq \|w_{*,z}\|_2 \|v_\theta\|_2 + \|w_{*,\theta}\|_2 \|v_z\|_2 + \|v_z\|_2 \|v_\theta\|_2, \quad (59)$$

$$|J_1| \leq 2\|w_{*,z}\|_1 \|v_z\|_1 + \|v_z\|_1^2, \quad (60)$$

$$|M_2| \leq \|\partial_z w_{*,z}\|_2 \|v_\theta\|_2 + \|w_{*,z}\|_2 \|\partial_z v_\theta\|_2 + \|\partial_z v_z\|_2 \|w_{*,\theta}\|_2 \\ + \|v_z\|_2 \|\partial_z w_{*,\theta}\|_2 + \|\partial_z v_z\|_2 \|v_\theta\|_2 + \|v_z\|_2 \|\partial_z v_\theta\|_2, \quad (61)$$

$$|M_1| \leq 2(\|\partial_z w_{*,z}\|_1 \|v_z\|_1 + \|w_{*,z}\|_1 \|\partial_z v_z\|_1 + \|v_z\|_1 \|\partial_z v_z\|_1). \quad (62)$$

Each is the Cauchy–Schwarz inequality on the product measure $r^e dr d\theta / (2\pi) dY$, after applying the product rule in (56)–(57). For example, the derivative of $2w_{*,z}v_z + v_z^2$ is $2(\partial_z w_{*,z})v_z + 2w_{*,z}\partial_z v_z + 2v_z\partial_z v_z$, proving every coefficient in (62). For a (z, t) multi-index I , every further derivative is exactly obtained by replacing each product ab in J_e by

$$\partial^{I+(1,0)}(ab) = \sum_{K \leq I+(1,0)} \binom{I+(1,0)}{K} (\partial^K a) (\partial^{I+(1,0)-K} b)$$

inside its original radial integral. This identity follows inductively from the one-derivative product rule, and gives the corresponding finite-order moment bound by applying the same Cauchy–Schwarz inequality separately to every displayed summand.

The field v in these bounds is not an abstract replacement amplitude. For each actual band and harmonic write

$$t_j = \chi_j B_j L_j y_j, \quad c_j = \frac{i n_{\Phi,j} \times t_j}{k_{\beta_j} m_j |n_{\Phi,j}|^2}, \quad b_j = t_j + \text{curl}_{*,\text{coeff}} c_j, \\ v = \sum_j \text{Re}(Q_{\beta_j}^{-A} b_j e^{ik_{\beta_j} m_j \Phi_j}), \quad A = \frac{1}{2} + h, \quad D = \frac{1}{2} - h. \quad (63)$$

Here $\text{curl}_{*,\text{coeff}}$ differentiates the chart coefficient, including the cutoff, auxiliary chain rule and cylindrical basis, but not the displayed exponential. This is exactly the remainder of the physical potential $Q_{\beta_j}^{1/2-A} c_j e^{ik_{\beta_j} m_j \Phi_j}$: physical curl introduces the factor $Q_{\beta_j}^{-1/2}$, and differentiating the exponential produces t_j by the transverse triple-product identity. Explicitly, on the absolute auxiliary torus in a fixed band,

$$\mathcal{D}_R = \partial_R + \sqrt{Q} dr r^{d_r-1} \mathbf{v}_r \cdot \partial_Y, \quad r = \sqrt{Q} R, \quad \mathcal{D}_Z^* = \varepsilon \partial_Z.$$

For the actual coefficient $c = (c_r, c_\theta, c_z)$, which has $\partial_\theta c = 0$, the operator is exactly

$$\text{curl}_{*,\text{coeff}} c = (-\varepsilon \partial_Z c_\theta, \varepsilon \partial_Z c_r - \mathcal{D}_R c_z, (\mathcal{D}_R + R^{-1}) c_\theta).$$

Coefficients originally on a band auxiliary torus are pulled back to the absolute torus before applying this formula; this retains the integer-covering chain derivative in $\partial_Y c$. In the same fixed chart the exact axial derivative is

$$\partial_z v = \sum_j \text{Re} \left[Q_{\beta_j}^{-A-D} (\partial_{Z_j} b_j + ik_{\beta_j} m_j (\partial_{Z_j} \Phi_j) b_j) e^{ik_{\beta_j} m_j \Phi_j} \right]. \quad (64)$$

The band power is fixed during chart differentiation and $\partial_z = Q_{\beta_j}^{-D} \partial_{Z_j}$; auxiliary phase evaluation adds no axial derivative because of (49). Thus (63)–(64) substitute the actual retained pair and all its curl terms into the bounds. For example, writing $\|b\|_{e,\beta}^2 = \int R^e \langle |b|^2 \rangle dR$ in that band, the triangle inequality and $|\text{Re } z| \leq |z|$ give

$$\|v\|_e \leq \sum_j Q_{\beta_j}^{(e+1)/4-A} \|b_j\|_{e,\beta_j}, \quad (65)$$

$$\|\partial_z v\|_e \leq \sum_j Q_{\beta_j}^{(e+1)/4-A-D} \|\partial_{Z_j} b_j + ik_{\beta_j} m_j (\partial_{Z_j} \Phi_j) b_j\|_{e,\beta_j}. \quad (66)$$

The exponent $(e+1)/4$ is exactly the square root of the radial measure factor $r^e dr = Q^{(e+1)/2} R^e dR$. Distinct bands and harmonics remain distinct in these sums. No limiting estimate as $q \downarrow 0$ has been inferred from the finite compact-patch norms.

For a single fixed chart, put $F_e = Q^{-2A-1/2} E_e^*$ as in the covariance bridge and $S^* = Q^{2A} S$. Direct change of radial variable gives

$$M_e = Q^{e/2-2A} M_e^*, \quad M_e^* = \int R^e E_e^* dR, \quad M_e^* = \varepsilon \partial_Z \int R^e S_{z,\tau_e}^* dR, \quad \varepsilon = Q^h,$$

where $\tau_2 = \theta$ and $\tau_1 = z$. Indeed the axial flux has factor $Q^{(e+1)/2-2A}$, its axial derivative contributes Q^{-D} , and dividing by $Q^{e/2-2A}$ leaves $Q^{1/2-D} = Q^h$. This proves the precise relation to the source chart moment without altering its axial anisotropy.

18.5 Exact relation to moments after phase evaluation

Let

$$\mathcal{S} = \langle w_* \otimes v + v \otimes w_* + v \otimes v \rangle_\theta, \quad \mathcal{S}^\circ = \mathcal{S} - \langle \mathcal{S} \rangle_Y.$$

Thus $\langle \mathcal{S} \rangle_Y = \mathcal{S}$ and $\langle \mathcal{S}^\circ \rangle_Y = 0$. For the evaluated physical fields, denote the angular-only residual moments by M_e^{ev} . The proof of (50) holds after evaluation. The physical divergence, vector Laplacian and gradient identities therefore prove, without any auxiliary averaging,

$$M_e^{\text{ev}} = \partial_z \int_0^\infty r^e \mathcal{S}_{z,\tau_e}(r, z, t, Y(r, t)) dr.$$

The physical radial boundary terms still vanish by compact support; the chain rule is included in the physical radial derivative before that integration. Subtracting (56)–(57) proves the exact comparison

$$M_e^{\text{ev}} - M_e = \partial_z \int_0^\infty r^e \mathcal{S}_{z,\tau_e}^\circ(r, z, t, Y(r, t)) dr. \quad (67)$$

For example its absolute value is bounded by $\int r^e \sup_Y |\partial_z \mathcal{S}_{z,\tau_e}^\circ| dr$, because $Y(r, t)$ is independent of z . Zero auxiliary Haar average does not make the right side of (67) zero at the physical phase. The displayed map supplies precisely the remaining auxiliary-oscillation flux; it is not silently identified with the source fast mean.

19 Time derivatives of the actual signed third growth

This calculation is on the actual interval already constructed in *The actual third growth target on the moving second hold* and *Exact third growth on the moving second holding state*. It gives quantitative derivatives of that trajectory for the source-frame map. All derivatives in this section are with respect to coupled physical time t_c , and $\tau = t_c - t_b$ denotes elapsed coupled time. The notation does not use the cylindrical spatial radius for elapsed time.

19.1 Original data, evolving parent, and proved interval

Retain the source parameters

$$\begin{aligned} \nu &= \sqrt{A_0}, & \mu &= \lambda_0 = \lceil \sqrt{A_0}/2 \rceil, & K_j &= \mu \hat{\lambda}_j, \\ S_3 &= \frac{A_0}{\mu} \hat{\lambda}_3^{-k_3-6}, & P_3^* &= \frac{A_0}{\mu} \hat{\lambda}_3^{-7/8} = S_3 e^{L_3}. \end{aligned} \quad (68)$$

The physical common diffusivity is $d > 0$; it is distinct from ν . Use the actual second return t_b , its values $h_b, A_{1,b}, P_{2,b}$, and the original angles s_*, s_2, s_3 . Put

$$\begin{aligned} a &= \sin s_2, & \chi_b &= \sqrt{h_b^2 + a^2}, & \beta_b &= \arctan(a/h_b), & b &= \sin(s_3 + \beta_b), & g_b &= \cos(s_3 + \beta_b), \\ c &= bh_b - ag_b = \chi_b \sin s_3, & S_* &= A_0 \sin s_*, & C_* &= A_0 \cos s_*. \end{aligned} \quad (69)$$

Here $a, b, c, S_*, C_* > 0$, and the letter b is the original oriented geometric constant, whereas b_3 below is an amplitude coefficient. The moving parent is exactly

$$\begin{aligned} \chi^2 &= h^2 + a^2, & \dot{h} &= a\Omega_1, & \dot{\Omega}_1 &= aA_1/\chi - dK_1^2\Omega_1, \\ \dot{A}_1 &= -S_*\Omega_1 - dK_1^2A_1, & \dot{P}_2 &= -dK_2^2\chi^2P_2, & \dot{g} &= b\Omega_1, \\ g &= g_b + (b/a)(h - h_b), & N_3 &= b(A_1 + C_*) + gS_* + cK_2P_2, & R_3 &= g^2 + b^2. \end{aligned} \quad (70)$$

The retained feedback is $\dot{\alpha} = -a^2\Omega_1/\chi^2$ and $\Omega_2 = 0$. In particular the older temperature and first vorticity are not frozen. The cancellation in the complete numerator is

$$\dot{N}_3 = -dE_3, \quad E_3 = bK_1^2 A_1 + cK_2^3 \chi^2 P_2. \quad (71)$$

Write $N_i = N_3(t_b)$, $a_i = N_i/K_3$, $b_i = K_3 \sin s_3$, $\gamma_i = \sqrt{a_i b_i}$ and $u_i = \sqrt{b_i/a_i}$. The exact new equations, including their original signed seed, are

$$\begin{aligned} y &= \begin{pmatrix} \Theta_3 \\ \Omega_3 \end{pmatrix}, \quad \dot{y} = C_3 y, \quad C_3 = \begin{pmatrix} -\lambda_3 & a_3 \\ b_3 & -\lambda_3 \end{pmatrix}, \\ a_3 &= \frac{N_3}{K_3 R_3}, \quad b_3 = \frac{K_3 c}{\chi}, \quad \lambda_3 = dK_3^2 R_3, \quad y(t_b) = -S_3 \begin{pmatrix} 1 \\ u_i \end{pmatrix}. \end{aligned} \quad (72)$$

There are no new amplitude controls in this growth pair: $c_\vartheta = c_\omega = 0$. The activation of the compact wave is a separate original factor $\eta(\Gamma_3 \tau)$; it does not replace (72) by a forced coefficient trajectory.

For clarity, the following facts about this actual interval were proved in the third-threshold construction. They identify the trajectory to which the estimates below apply:

$$\begin{aligned} I_3 &= [t_b, t_3^a], \quad t_3^a - t_b = \tau_a < 14/\sigma_1, \quad P_3 = -\Theta_3 > 0, \quad P_3(t_b) = S_3, \quad P_3(t_3^a) = P_3^*, \\ u &= \Omega_3/\Theta_3, \quad \frac{u_i}{\sqrt{2}} \leq u \leq 2u_i, \quad \frac{a_i}{4} \leq a_3 \leq a_i, \quad \frac{b_i}{2} \leq b_3 \leq b_i, \\ 1 &\leq R_3 \leq \bar{R} := \left(\frac{265}{256}\right)^2 < 2, \quad 1 \leq \frac{\chi}{\chi_b} \leq \frac{259}{256}, \quad 0 < dK_3^2 \leq \frac{\gamma_i}{16\sqrt{2}}, \\ \frac{\gamma_i}{8\sqrt{2}} &\leq (\log P_3)' \leq 2\gamma_i, \quad 0 < A_1 \leq A_{1,b}, \quad 0 < P_2 \leq P_{2,b}, \quad 0 < \Omega_1 < 9\sigma_1, \\ h_b &\leq h \leq h_b + 144a, \quad g_b \leq g \leq g_b + 144b, \quad \frac{1}{2}N_i \leq N_3 \leq N_i, \quad 1 \leq \chi^2 < 10. \end{aligned} \quad (73)$$

The value of d is precisely the previously selected range $0 < d \leq d_{3,\max}$, retaining its K_3^{-2} dependence. Thus (73) does not assert a new unrestricted viscosity range or assume a further return.

19.2 Signed amplitudes and first derivatives with the diffusion retained

Theorem 1. *For every $t_c \in I_3$, both $\Theta_3(t_c)$ and $\Omega_3(t_c)$ are negative and strictly decreasing. Their exact component rates are*

$$\dot{\Theta}_3 = -P_3(a_3 u - \lambda_3), \quad \dot{\Omega}_3 = -P_3(b_3 - \lambda_3 u). \quad (74)$$

They obey the bounds

$$\begin{aligned} S_3 e^{\gamma_i \tau / (8\sqrt{2})} &\leq P_3(t_c) \leq \min\{P_3^*, S_3 e^{2\gamma_i \tau}\}, \\ \frac{u_i}{\sqrt{2}} P_3 &\leq |\Omega_3| \leq 2u_i P_3, \\ \left(\frac{\gamma_i}{4\sqrt{2}} - dK_3^2 \bar{R}\right) P_3 &\leq |\dot{\Theta}_3| \leq 2\gamma_i P_3, \\ u_i \left(\frac{\gamma_i}{2} - 2dK_3^2 \bar{R}\right) P_3 &\leq |\dot{\Omega}_3| \leq \gamma_i u_i P_3. \end{aligned} \quad (75)$$

In particular both lower derivative bounds are positive, and

$$\begin{aligned} P_3 \sqrt{1 + u_i^2/2} &\leq |y|_2 \leq P_3 \sqrt{1 + 4u_i^2} \leq P_3^* \sqrt{1 + 4u_i^2}, \\ |y|_2 &\leq \gamma_i P_3 \sqrt{4 + u_i^2} \leq \gamma_i P_3^* \sqrt{4 + u_i^2}. \end{aligned} \quad (76)$$

The original beginning and attained endpoint have the exact derivatives

$$\begin{aligned} \dot{y}(t_b) &= -S_3(\gamma_i - dK_3^2) \begin{pmatrix} 1 \\ u_i \end{pmatrix}, \\ y(t_3^a) &= -P_3^* \begin{pmatrix} 1 \\ u(t_3^a) \end{pmatrix}, \\ \dot{y}(t_3^a) &= -P_3^* \begin{pmatrix} a_3(t_3^a)u(t_3^a) - dK_3^2 R_3(t_3^a) \\ b_3(t_3^a) - dK_3^2 R_3(t_3^a)u(t_3^a) \end{pmatrix}. \end{aligned} \tag{77}$$

Proof. The signs and quotient in (73) give $y = -P_3(1, u)^T$. Substituting this expression in the exact pair proves (74). Integrating the already proved logarithmic rate from t_b and using the attained endpoint and strict increase of P_3 gives the first line of (75). The quotient box gives its second line. Since $a_i u_i = \gamma_i$ and $b_i = \gamma_i u_i$, direct multiplication of the positive boxes gives

$$\frac{\gamma_i}{4\sqrt{2}} \leq a_3 u \leq 2\gamma_i, \quad \frac{\gamma_i u_i}{2} \leq b_3 \leq \gamma_i u_i, \quad 0 < \lambda_3 \leq dK_3^2 \bar{R}.$$

Consequently the displayed lower bound for $a_3 u - \lambda_3$ follows. It is at least $\gamma_i/(8\sqrt{2}) > 0$, because $\bar{R} < 2$ and $dK_3^2 \leq \gamma_i/(16\sqrt{2})$. Similarly,

$$b_3 - \lambda_3 u \geq u_i(\gamma_i/2 - 2dK_3^2 \bar{R}) > \gamma_i u_i(1/2 - 1/(4\sqrt{2})) > \gamma_i u_i/4 > 0.$$

Thus both derivatives in (74) are strictly negative. Dropping their nonnegative diffusion losses only for the purpose of an upper bound gives the two upper derivative estimates; the exact rates continue to contain the losses. Squaring the two amplitude and two derivative bounds, adding them, and taking the positive square root proves (76). Finally $R_3(t_b) = 1$, $u(t_b) = u_i$, $a_3(t_b) = a_i$ and $b_3(t_b) = b_i$ give the first endpoint identity. Substitution of the proved target value gives the other two. In particular no later endpoint was substituted for the actual $u(t_3^a)$. $\square$

19.3 Explicit coefficient rates

Set $\bar{H} = h_b + 144a$, $\bar{G} = g_b + 144b$ and $\bar{W} = 9\sigma_1$. Differentiating the complete coefficients gives

$$\begin{aligned} \dot{a}_3 &= -a_3 \left(\frac{dE_3}{N_3} + \frac{2bg\Omega_1}{R_3} \right), \\ \dot{b}_3 &= -b_3 \frac{ah\Omega_1}{\chi^2}, \quad \dot{\lambda}_3 = 2dK_3^2 bg\Omega_1. \end{aligned} \tag{78}$$

Every quantity on the right comes from the moving parent. In particular the exact numerator derivative (71) is used before any upper estimate. Because $K_1 < K_2$, $\chi^2 < 10$ and all summands of N_3 are positive, $E_3 \leq 10K_2^2 N_3$. Thus

$$\begin{aligned} |\dot{a}_3| &\leq a_i(10dK_2^2 + 2b\bar{G}\bar{W}), \quad |\dot{b}_3| \leq b_i a \bar{H}\bar{W} \chi_b^{-2}, \\ |\dot{\lambda}_3| &\leq 2dK_3^2 b \bar{G}\bar{W}. \end{aligned} \tag{79}$$

These formulas also display the distinct inherited K_2^2 loss and new K_3^2 loss. They do not replace either by the source scaling parameter ν .

19.4 A closed all-orders derivative calculation

Here are finite recurrences whose values are explicit upper bounds from the actual birth data and the already proved interval. They use ordinary derivatives, with no change to physical time or amplitudes. For integers $n \geq 0$ and $m \geq 2$ define the product expression

$$\mathcal{P}_n(X^{(1)}, \dots, X^{(m)}) = \sum_{j_1 + \dots + j_m = n} \frac{n!}{j_1! \dots j_m!} X_{j_1}^{(1)} \dots X_{j_m}^{(m)}.$$

This notation is a finite sum, and repeated arguments mean repeated factors. Define the nonnegative arrays H, W, A, P, Q, G, N, J initially by

$$H_0 = \overline{H}, \quad W_0 = \overline{W}, \quad A_0^{\text{bd}} = A_{1,b}, \quad P_0 = P_{2,b}, \quad Q_0 = \chi_b^{-1}, \quad G_0 = \overline{G}, \quad N_0 = N_i, \quad J_0 = 1. \quad (80)$$

In the following recurrences the array A has entries A_n^{bd} ; the superscript distinguishes its zeroth entry from the unchanged physical constant A_0 in (68). Write

$$X_n = \mathcal{P}_n(H, H) + a^2 \mathbf{1}_{n=0}.$$

Having computed all eight arrays up to index n , compute index $n+1$ by the explicit formulas

$$\begin{aligned} H_{n+1} &= aW_n, \\ W_{n+1} &= a\mathcal{P}_n(A, Q) + dK_1^2 W_n, \\ A_{n+1}^{\text{bd}} &= S_* W_n + dK_1^2 A_n^{\text{bd}}, \\ P_{n+1} &= dK_2^2 \mathcal{P}_n(X, P), \\ Q_{n+1} &= a\mathcal{P}_n(H, W, Q, Q, Q), \\ G_{n+1} &= bW_n, \\ N_{n+1} &= dbK_1^2 A_n^{\text{bd}} + dcK_2^3 \mathcal{P}_n(X, P), \\ J_{n+1} &= 2b\mathcal{P}_n(G, W, J, J). \end{aligned} \quad (81)$$

For the phase-length array use

$$R_0^{\text{bd}} = \overline{R}, \quad R_n^{\text{bd}} = \mathcal{P}_n(G, G) \quad (n \geq 1).$$

The coefficient arrays are

$$\begin{aligned} \mathfrak{A}_0 &= a_i, \quad \mathfrak{B}_0 = b_i, \quad \mathfrak{D}_0 = dK_3^2 \overline{R}, \\ \mathfrak{A}_n &= K_3^{-1} \mathcal{P}_n(N, J), \quad \mathfrak{B}_n = K_3 c Q_n, \quad \mathfrak{D}_n = dK_3^2 R_n^{\text{bd}} \quad (n \geq 1). \end{aligned} \quad (82)$$

No optimization of these finite sums is needed for their validity. In particular $\mathfrak{A}_1$ may be replaced by the smaller of its value above and the first bound in (79), and similarly for the other two first coefficient derivatives.

The amplitude arrays start with the actual target and the sharper first-derivative bounds already proved:

$$\begin{aligned} T_0 &= P_3^*, \quad O_0 = 2u_i P_3^*, \quad T_1 = 2\gamma_i P_3^*, \quad O_1 = \gamma_i u_i P_3^*, \\ T_{n+1} &= \sum_{j=0}^n \binom{n}{j} (\mathfrak{D}_j T_{n-j} + \mathfrak{A}_j O_{n-j}), \quad n \geq 1, \\ O_{n+1} &= \sum_{j=0}^n \binom{n}{j} (\mathfrak{B}_j T_{n-j} + \mathfrak{D}_j O_{n-j}), \quad n \geq 1. \end{aligned} \quad (83)$$

The choice to begin these recurrences at $n = 1$ preserves the stronger first-derivative estimates instead of substituting a larger matrix-norm bound.

Theorem 2. *For every integer $n \geq 0$ the finite arrays above satisfy on I_3*

$$\begin{aligned} |\partial_{t_c}^n \Theta_3| &\leq T_n, & |\partial_{t_c}^n \Omega_3| &\leq O_n, & |\partial_{t_c}^n y|_2 &\leq \sqrt{T_n^2 + O_n^2}, \\ |\partial_{t_c}^n a_3| &\leq \mathfrak{A}_n, & |\partial_{t_c}^n b_3| &\leq \mathfrak{B}_n, & |\partial_{t_c}^n \lambda_3| &\leq \mathfrak{D}_n. \end{aligned} \quad (84)$$

The exact, signed amplitude derivative recurrence underlying these bounds is

$$\partial_{t_c}^{n+1} y = \sum_{j=0}^n \binom{n}{j} (\partial_{t_c}^j C_3) (\partial_{t_c}^{n-j} y), \quad n \geq 0. \quad (85)$$

In particular the exact second derivative contains the changing coefficient and the complete diffusion square:

$$\ddot{y} = \begin{pmatrix} \lambda_3^2 + a_3 b_3 - \dot{\lambda}_3 & \dot{a}_3 - 2\lambda_3 a_3 \\ \dot{b}_3 - 2\lambda_3 b_3 & \lambda_3^2 + a_3 b_3 - \dot{\lambda}_3 \end{pmatrix} y. \quad (86)$$

Proof. Introduce the exact auxiliary functions $q = \chi^{-1}$ and $j = R_3^{-1}$, both well-defined by (73). Direct differentiation of their specified inverses gives

$$\dot{q} = -ah\Omega_1 q^3, \quad \dot{j} = -2bg\Omega_1 j^2.$$

These identities retain the actual phase lengths; the symbols do not replace them by constant values. The complete equations for $(h, \Omega_1, A_1, P_2, q, g, N_3, j)$ are therefore the signed polynomial system consisting of (70), (71) and these two equations, with $\chi^2 = h^2 + a^2$ in the parent losses. Their zero-order absolute values are bounded respectively by $(H_0, W_0, A_0^{\text{bd}}, P_0, Q_0, G_0, N_0, J_0)$: the claims for q and j use $\chi \geq \chi_b$ and $R_3 \geq 1$.

For any m differentiable factors the ordinary n th derivative of their product is $\mathcal{P}_n$ applied to their derivative arrays. To prove this formula without assuming it, start with $n = 0$. Differentiating a term of order n once chooses one factor whose index increases. At indices summing to $n + 1$, the resulting coefficient is the sum of $n!/(j_1! \cdots (j_k - 1)! \cdots j_m!)$ over indices k with $j_k > 0$. It is $n!(j_1 + \cdots + j_m)/(j_1! \cdots j_m!) = (n + 1)!/(j_1! \cdots j_m!)$. This proves the formula by induction.

Apply this exact product derivative identity to the signed parent equations and then the triangle inequality. For instance

$$\partial_{t_c}^{n+1} N_3 = -dbK_1^2 \partial_{t_c}^n A_1 - dcK_2^3 \partial_{t_c}^n \{(h^2 + a^2)P_2\},$$

and its absolute value is bounded by the displayed recurrence for N_{n+1} . This example shows explicitly that neither inherited diffusion term is omitted. The other seven equations give exactly (81). Induction in n proves all eight parent derivative bounds. The identities $R_3 = g^2 + b^2$, $a_3 = N_3 j / K_3$, $b_3 = K_3 c q$ and $\lambda_3 = dK_3^2 R_3$ then prove (82) for $n \geq 1$ by the same product formula. Their zero-order bounds were already proved in (73).

Repeatedly differentiate the exact pair to obtain (85). Its componentwise absolute value gives (83), with the zero and first orders supplied by Theorem 1. Induction now proves the two amplitude bounds at every order, and the Euclidean norm bound follows by summing squares. Finally $\ddot{y} = (\dot{C}_3 + C_3^2)y$ and direct multiplication of the two by two matrix give (86). Every recurrence uses finitely many sums at each specified order. Thus each resulting bound is a determinate expression in the original physical constants and actual birth data, rather than an unevaluated supremum of a missing coefficient derivative. $\square$

19.5 Derivative transport through the specified source clock

Let $L_s > 0$ be the source slot length used for this actual interval, and define its specified clock by

$$\alpha_c = \tau_a/L_s, \quad t_c = t_b + \alpha_c v, \quad 0 \leq v \leq L_s.$$

The two actual endpoints are retained. Since α_c is constant for the fixed trajectory and slot, repeated chain differentiation gives, exactly,

$$\partial_v^n y(t_b + \alpha_c v) = \alpha_c^n \partial_{t_c}^n y(t_b + \alpha_c v), \quad |\partial_v^n y|_2 \leq \alpha_c^n \sqrt{T_n^2 + O_n^2}. \quad (87)$$

In a source chart with fixed label Q , exponent h_{src} , and $\partial_t v = Q^{-1-h_{\text{src}}}$, the derivative contribution acting on this amplitude alone is consequently

$$\partial_t^n y(t_b + \alpha_c v) = (\alpha_c Q^{-1-h_{\text{src}}})^n \partial_{t_c}^n y(t_b + \alpha_c v).$$

This identity holds with the stated fixed chart parameters and affine clock; derivatives of any additional frame, cutoff, or varying parameter must be included by the ordinary product rule. In particular neither the clock factor nor the physical frequency has been replaced by one. These finite derivative estimates establish no uniform terminal extension for an infinite sequence.

20 The actual third-growth pair in the source clock and frame

This calculation uses equations (6.2)–(6.12), (7.2)–(7.12), and Lemma 7.4 of the retained 166-page source. Its SHA-256 is displayed in two consecutive segments:

```
0e779481c4da40bd28d1e642e1d8ca57
447d129610df28dfa5a11e9af8ae228f.
```

The coupled input is the actual third-growth interval proved in `finite_stage/third_growth_entry.tex`. The source's pulse equation is written at physical viscosity one. We retain an arbitrary positive physical viscosity ν_{NS} when computing a candidate residual; when identifying the source's particular homogeneous pulse, we explicitly set $\nu_{\text{NS}} = 1$ and harmonic $m = 1$. The original coupled time scale $\nu = \sqrt{A_0}$, the physical coupled diffusivity d , and the source chart parameter $\varepsilon = Q^h$ are distinct. The source growth constant denoted λ_0 in $\lambda(v)$ below is the source's own growth constant. It is not identified with the coupled integer frequency $\lambda_0 = \mu = \lceil \sqrt{A_0}/2 \rceil$.

20.1 Exact physical differentiation of the source pulse clock

Fix a source label and one of its local lifts. Set

$$b_g = \sqrt{2} - 1, \quad v_r = (1, -b_g), \quad v_t = (b_g, 1), \quad J_g = \begin{pmatrix} 3 & 1 \\ 1 & 5 \end{pmatrix}, \quad T_g = 4 + \sqrt{2}, \quad \Lambda_g = 4 - \sqrt{2}.$$

Direct multiplication gives $J_g v_t = T_g v_t$ and $J_g v_r = \Lambda_g v_r$, while $v_t \cdot v_r = 0$ and $|v_t|^2 = |v_r|^2 = 1 + b_g^2$. The source retains

$$d_r = 2 \left((1 + h) \frac{\log \Lambda_g}{\log T_g} - h \kappa_s \right), \quad \kappa_s = 10^{-5},$$

$$Q = 2^{-\ell}, \quad i = \left\lfloor \log_{T_g} \frac{Q^{-1-h}}{S_*} \right\rfloor, \quad c_i = T_g^i Q^{1+h}.$$

On the local lift

$$Y_i - c_\gamma - k_{\text{copy}} = \xi_g v_r + \eta_g v_t, \quad v = \frac{\eta_g + r_0}{c_i}, \quad L_s = \frac{2r_0}{c_i}, \quad \eta_g = \frac{v_t \cdot (Y_i - c_\gamma - k_{\text{copy}})}{1 + b_g^2}.$$

The label and integer k_{copy} are held fixed on this lift. Consequently $L_i \eta_g = 0$, $N_i \eta_g = 1$, where $L_i = v_r \cdot \partial_{Y_i}$ and $N_i = v_t \cdot \partial_{Y_i}$. The clock has no explicit dependence on the slow variables R, Z, T . The evaluated chart derivatives are exactly

$$D_r = \partial_R + M_i d_r R^{d_r-1} L_i, \quad D_z = \varepsilon \partial_Z, \quad t_* = -\varepsilon \partial_T + c_i N_i, \quad M_i = \Lambda_g^i Q^{d_r/2}.$$

It follows by applying these displayed operators, without estimating any coefficient, that

$$D_r v = D_z v = 0, \quad t_* v = 1. \quad (88)$$

There is also an explicit physical verification. Evaluate at the source map $Y = v_r r^{d_r} + v_t t$ modulo the lattice. Locally the lift of $Y_i = J_g^i Y$ differs from $\Lambda_g^i v_r r^{d_r} + T_g^i v_t t$ by a constant lattice vector. The dual formula for η_g therefore gives $\eta_g = T_g^i t + C_\gamma$ with a locally constant C_γ . Thus, on the evaluated local rectangle,

$$\partial_r v = \partial_z v = \partial_\theta v = 0, \quad \partial_t v = \frac{T_g^i}{c_i} = Q^{-1-h}. \quad (89)$$

The ∂_t in this equation is the derivative of the evaluated physical function. On the extended domain the corresponding derivative is $\partial_t + N_{\text{abs}}$. These statements are identical after evaluation; neither drops the auxiliary derivative.

Let t_b and $r_a > 0$ be the actual second return time and the actual third target-crossing elapsed time. Define

$$t_c(v) = t_b + \alpha v, \quad \alpha = \frac{r_a}{L_s} > 0. \quad (90)$$

This parametrizes the entire proved coupled growth interval, with its original endpoints. In physical source variables its derivative is $\partial_t t_c = \alpha Q^{-1-h}$ and all three spatial derivatives vanish. The positive orientation of this clock asserts no sign for a velocity. The coupled amplitudes below are strictly negative.

20.2 The actual frame, its inverse, and the direct candidate

Write the original source phase normal as

$$n = \left(x_0 - v(pF_R + p_z G_R), \frac{p}{R}, p_z - \varepsilon v(pF_Z + p_z G_Z) \right).$$

All p, p_z, x_0, k, Q are the original fixed label constants, including $kp \in \mathbb{Z} \setminus \{0\}$. Set

$$K_a = \frac{n_{\text{tan}}}{|n_{\text{tan}}|}, \quad N_a = ((K_a)_z, -(K_a)_\theta), \quad s_a = \frac{n_r}{|n_{\text{tan}}|},$$

$$c(v) = c_0 \sqrt{1 + s(v)^2}, \quad s(v) = \sigma \left(\frac{u_*}{2} + \frac{u_* v}{L_s} \right).$$

Here $c_0 < 0$ is the unchanged source constant, and K_a, N_a are identified with their tangential three-vectors. In particular $c(v) \neq 0$. Because $p/R \neq 0$, every formula is defined throughout the source rectangle. The actual source frame is

$$U = [e_r - s_a K_a, N_a], \quad M = \begin{pmatrix} 1 & 1 \\ c & -c \end{pmatrix}, \quad B = UM.$$

The two columns of U are orthogonal, with squared lengths $g_a^2 = 1 + s_a^2$ and 1. Both are orthogonal to n . Since $\det M = -2c \neq 0$, B is a bijection from $\mathbb{R}^2$ onto $n^\perp$. The source left inverse, defined on all of $\mathbb{R}^3$, is

$$B_\ell = M^{-1} \begin{pmatrix} e_r^T \\ N_a^T \end{pmatrix}, \quad M^{-1} = \frac{1}{2} \begin{pmatrix} 1 & c^{-1} \\ 1 & -c^{-1} \end{pmatrix}. \quad (91)$$

Indeed the last two rows applied to U give I_2 .

The eigenvalues of

$$B^T B = \begin{pmatrix} g_a^2 + c^2 & g_a^2 - c^2 \\ g_a^2 - c^2 & g_a^2 + c^2 \end{pmatrix}$$

are $2g_a^2$ and $2c^2$, with orthogonal eigenvectors $(1, 1)$ and $(1, -1)$. The Euclidean operator norm and the inverse norm on the tangent plane are therefore exactly

$$\|B\| = \sqrt{2} \max(g_a, |c|), \quad \|(B|_{\mathbb{R}^2})^{-1}\|_{n^\perp \rightarrow \mathbb{R}^2} = \frac{1}{\sqrt{2} \min(g_a, |c|)}. \quad (92)$$

The ambient norm of the particular extension (91) is $(1/\sqrt{2}) \max(1, |c|^{-1})$; it need not equal the inverse norm restricted to $n^\perp$.

Retain the actual coupled data

$$K_3 = \mu \hat{\lambda}_3, \quad S_3 = \frac{A_0}{\mu} \hat{\lambda}_3^{-k_3-6}, \quad P_3^* = \frac{A_0}{\mu} \hat{\lambda}_3^{-7/8} = S_3 e^{L_3}, \quad u_i = \sqrt{\frac{b_i}{a_i}} = \frac{K_3}{\sigma_2 \sqrt{n_i}}.$$

On the interval (90), the already constructed solution is

$$y_3(t_c) = \begin{pmatrix} \Theta_3 \\ \Omega_3 \end{pmatrix} = -P(v) \begin{pmatrix} 1 \\ u(v) \end{pmatrix}, \quad S_3 \leq P(v) \leq P_3^*, \quad \frac{u_i}{\sqrt{2}} \leq u(v) \leq 2u_i,$$

with $P(0) = S_3$, $P(L_s) = P_3^*$ and $u(0) = u_i$. These inequalities follow from the actual evolving-parent system, rather than from a new frame approximation. The direct candidate $t_{\text{dir}} = By_3$ consequently has the exact components

$$\begin{aligned} (t_{\text{dir}})_r &= -P(1+u), \\ (t_{\text{dir}})_{\text{tan}} &= P s_a(1+u) K_a - P c(1-u) N_a, \\ |t_{\text{dir}}|^2 &= P^2 [(1+s_a^2)(1+u)^2 + c^2(1-u)^2], \\ (t_{\text{dir}})_r (t_{\text{dir}})_{\text{tan}} &= -P^2 s_a(1+u)^2 K_a + P^2 c(1-u)^2 N_a. \end{aligned} \quad (93)$$

These follow by multiplying $My_3 = -P(1+u, c(1-u))^T$ and using the orthogonality of the two columns of U . In particular

$$|(t_{\text{dir}})_r| \geq S_3(1+u_i/\sqrt{2}), \quad |t_{\text{dir}}| \leq \sqrt{2} \max(g_a, |c|) P_3^* \sqrt{1+4u_i^2}.$$

At the actual target,

$$\frac{P_3^* u_i}{\sqrt{2}} \leq |\Omega_3(t_b + r_a)| \leq 2P_3^* u_i, \quad P_3^* u_i = \frac{A_0}{\sigma_2 \sqrt{n_i}} \hat{\lambda}_3^{1/8}. \quad (94)$$

The last equality retains μ until it cancels against its identical factor in K_3 . Thus a decreasing temperature target does not give a decreasing two-component target.

20.3 The complete principal mismatch, including the cutoff

With a prime denoting v differentiation at fixed slow variables, retain

$$\mathbf{K} = \begin{pmatrix} 0 & -2F & 0 \\ 2F + RF_R & 0 & 0 \\ G_R & 0 & 0 \end{pmatrix}, \quad A_\Phi = -\mathbf{K} + \frac{n(n^T \mathbf{K} - (n')^T)}{|n|^2}, \quad H = B_\ell(A_\Phi B - B').$$

Differentiating $n^T B = 0$ shows $n^T B' = -(n')^T B$. Direct multiplication gives $n^T A_\Phi B = -(n')^T B$ as well. Therefore $A_\Phi B - B'$ takes values in $n^\perp$, and

$$A_\Phi B - B' = BH. \quad (95)$$

The source defines the exact error matrix $E = H - \text{diag}(\lambda, -\lambda)$ with $\lambda = \lambda_0/\sqrt{1+s^2}$. Its source estimate $\|E\| \leq C/S_*$ is not an assertion that E is zero.

The actual coupled matrix and arbitrary-viscosity source loss are

$$C_c = \begin{pmatrix} -d_c & a_3 \\ b_3 & -d_c \end{pmatrix}, \quad d_c = dK_3^2 R_3, \quad a_3 = \frac{N_3}{K_3 R_3}, \quad b_3 = \frac{K_3 c_{\text{det}}}{\chi_{\text{old}}},$$

$$\delta_\nu = \nu_{\text{NS}} m^2 \varepsilon k^2 |n|^2, \quad \Delta_\nu = \delta_\nu - \alpha d_c.$$

Here c_{det} and χ_{old} are the third-growth determinant constant and older phase length; these names distinguish them from the source $c(v)$ and the cutoff below. Every coupled quantity is evaluated at $t_c(v)$. The actual third growth has zero added amplitude controls, so $y'_3 = \alpha C_c y_3$ exactly.

Let $\chi = \eta(R, Z, T) \chi_g(\xi_g) \psi(v)$ be a smooth scalar cutoff with the source support, and set $t_\chi = \chi B y_3$. The homogeneous-pressure formula, with zero principal source, is

$$\pi_\chi = \frac{i}{km|n|^2} (n \cdot \mathbf{K} t_\chi - n' \cdot t_\chi).$$

The cutoff leaves $n^T t_\chi = 0$. Differentiating this identity and substituting the pressure gives

$$t'_\chi + \mathbf{K} t_\chi + \delta_\nu t_\chi + ikmn \pi_\chi = B \mathcal{E}_\chi, \quad \mathcal{E}_\chi = \chi(\alpha C_c - H + \delta_\nu I) y_3 + \chi' y_3. \quad (96)$$

For completeness, the normal component of the sum before pressure is $-n' \cdot t_\chi + n \cdot \mathbf{K} t_\chi$; multiplication by $ikmn$ times the displayed pressure cancels it. The remaining vector is $t'_\chi - A_\Phi t_\chi + \delta_\nu t_\chi$. Use (95) and $y'_3 = \alpha C_c y_3$ to obtain (96).

In original scalar entries this is

$$\mathcal{E}_\chi = -P \left\{ \chi \begin{pmatrix} \Delta_\nu - \lambda - E_{11} + (\alpha a_3 - E_{12})u \\ \alpha b_3 - E_{21} + (\Delta_\nu + \lambda - E_{22})u \end{pmatrix} + \chi' \begin{pmatrix} 1 \\ u \end{pmatrix} \right\}. \quad (97)$$

All entries are signed. In particular the damping mismatch is $\delta_\nu - \alpha dK_3^2 R_3$, not the sum of the losses. At fixed slow variables and fixed ξ_g , $\chi' = \eta \chi_g \psi'$. The physical derivative of χ also includes $-\varepsilon(\partial_T \eta) \chi_g \psi$ in $t_* \chi$; that slow term belongs to the complete physical residual and has not been included in, or removed by, the principal equation (96). A direct finite bound is

$$|\mathcal{E}_\chi| \leq P \sqrt{1+u^2} [|\chi|(|\Delta_\nu| + \lambda + \|E\| + \alpha \max(a_3, b_3)) + |\chi'|].$$

This is the triangle inequality and the exact norm $\left\| \begin{smallmatrix} 0 & a_3 \\ b_3 & 0 \end{smallmatrix} \right\| = \max(a_3, b_3)$, with $a_3, b_3 > 0$. It retains the carrier, frame, cutoff and all physical diffusion factors.

20.4 An explicit map of the actual seed to the actual source pulse

For this subsection only, take $m = 1$, $\nu_{\text{NS}} = 1$, so $\delta_1 = \varepsilon k^2 |n|^2$. This is precisely the source homogeneous pulse equation with its given background. Put

$$d_{\text{ref}} = \varepsilon k^2 B_s^2 (1 + s^2), \quad \mathcal{P}_\sigma(v) = \exp \left(\int_{L_s/2}^v (\lambda(w) - d_{\text{ref}}(w)) dw \right).$$

Every coefficient is finite on the closed interval. Thus $\mathcal{P}_\sigma(0) > 0$ exactly, without a Gaussian approximation. Source Lemma 7.4 specifies the initial coordinate vector

$$z_{\sigma,0} = \begin{pmatrix} \mathcal{P}_\sigma(0) \\ 0 \end{pmatrix}, \quad t_\sigma^h(0) = B(0)z_{\sigma,0}.$$

The fixed source label may affect this strictly positive scalar.

For any nonzero real target vector $z_0 = (z_1, z_2)^T$, define

$$L_0 = -\frac{1}{S_3(1 + u_i^2)} \begin{pmatrix} z_1 & -z_2 \\ z_2 & z_1 \end{pmatrix} \begin{pmatrix} 1 & u_i \\ -u_i & 1 \end{pmatrix}. \quad (98)$$

This is an explicit invertible linear map with

$$L_0[-S_3(1, u_i)^T] = z_0, \quad \det L_0 = \frac{|z_0|^2}{S_3^2(1 + u_i^2)} > 0,$$

and inverse

$$L_0^{-1} = -\frac{S_3}{|z_0|^2} \begin{pmatrix} 1 & -u_i \\ u_i & 1 \end{pmatrix} \begin{pmatrix} z_1 & z_2 \\ -z_2 & z_1 \end{pmatrix}. \quad (99)$$

Indeed the second matrix of (98) sends $(1, u_i)^T$ to $(1 + u_i^2, 0)^T$, and each matrix times its transpose is a positive scalar times I_2 . The same multiplication verifies the inverse and proves the exact norms

$$\|L_0\| = \frac{|z_0|}{S_3\sqrt{1 + u_i^2}}, \quad \|L_0^{-1}\| = \frac{S_3\sqrt{1 + u_i^2}}{|z_0|}. \quad (100)$$

In particular the source target gives

$$L_0 = -\frac{\mathcal{P}_\sigma(0)}{S_3(1 + u_i^2)} \begin{pmatrix} 1 & u_i \\ -u_i & 1 \end{pmatrix}, \quad \|L_0\| = \frac{\mathcal{P}_\sigma(0)\mu\hat{\lambda}_3^{k_3+6}}{A_0\sqrt{1 + K_3^2/(\sigma_2^2 n_i)}}.$$

No seed or source initial amplitude has been set to one.

Define the two identity-initial-value fundamental matrices by

$$X'_c = \alpha C_c X_c, \quad X_c(0) = I_2, \quad X'_s = (H - \delta_1 I_2) X_s, \quad X_s(0) = I_2.$$

Here is a direct existence, invertibility and uniqueness construction. For a continuous matrix D on $[0, L_s]$ let

$$X_D(v) = I_2 + \sum_{j \geq 1} \int_{0 \leq w_j \leq \dots \leq w_1 \leq v} D(w_1) \cdots D(w_j) dw_j \cdots dw_1.$$

If $\sup \|D\| = M_D$, the j th term is bounded by $(M_D L_s)^j / j!$. Uniform convergence and regrouping the integrals give $X_D = I_2 + \int_0^v D(w) X_D(w) dw$, hence the differential equation. The corresponding

right-ordered series for $Z' = -ZD$, $Z(0) = I_2$, converges by the same bound. Differentiation gives $(ZX_D)' = 0$, hence $ZX_D = I_2$. For square matrices this is invertibility. If two solutions agree initially, the difference satisfies the zero-initial-value integral equation; iteration bounds its norm by a fixed finite bound times $(M_D v)^j/j!$ for every j , which tends to zero. This proves uniqueness. Smooth coefficient dependence follows from the same series with the following explicit bound. On a compact coefficient-parameter neighborhood let $M_N \geq 1$ bound every coefficient derivative through order N . Differentiating a product of j factors N times allocates each derivative to one of j factors, so there are at most j^N terms, each of norm at most M_N^j . The differentiated j th integral is therefore bounded by $j^N(M_N L_s)^j/j!$, a summable sequence. Uniform convergence of these derivative series justifies termwise differentiation: integrate each first-derivative series along a parameter segment and interchange the uniform sum with that integral, then iterate through order N . The right-ordered inverse series has the same bound. Pulse derivatives then follow from the differential equations. This proves smoothness, with the explicit derivative equations below providing further bounds.

Now set

$$L(v) = X_s(v)L_0X_c(v)^{-1}, \quad L^{-1} = X_cL_0^{-1}X_s^{-1}. \quad (101)$$

Differentiation gives

$$L' = (H - \delta_1 I_2)L - \alpha LC_c.$$

The original solution is $y_3(v) = X_c(v)y_3(0)$. Consequently $L(v)y_3(v) = X_s(v)z_{\sigma,0}$, and

$$B(v)L(v)y_3(v) = t_\sigma^h(v) \quad (102)$$

by uniqueness of the source homogeneous equation and the exact initial seed calculation. Thus this finite map reaches the actual source pulse. It does not merely send the coupled pair into an unspecified transverse solution. Multiplying both sides by the identical source cutoffs preserves equality of the amplitudes, and applying identical linear curl and pressure constructions preserves equality of the resulting fields. Multiplication by the source prescribed scalar leading-wave amplitude also preserves that equality. The cutoff principal residual is then $\chi' B L y_3$; the remaining source construction uses its own estimates for that exact source object.

20.5 All scale costs of the finite intertwiner

Here and below $\|\cdot\|$ is the Euclidean matrix norm. Put $D_i = \text{diag}(1, u_i)$, $\kappa_i = \max(u_i, u_i^{-1})$, and

$$\beta_c(v) = \frac{1}{2} \left(a_3(t_c(v))u_i + \frac{b_3(t_c(v))}{u_i} \right).$$

The actual coefficient bounds $a_3 \leq a_i$, $b_3 \leq b_i$ and $a_i u_i = b_i/u_i = \gamma_i$ give $0 < \beta_c \leq \gamma_i$. Conjugating the coupled matrix by the constant D_i gives

$$D_i^{-1}C_c D_i = \begin{pmatrix} -d_c & a_3 u_i \\ b_3/u_i & -d_c \end{pmatrix}.$$

Its symmetric part has eigenvalues $-d_c \pm \beta_c$. For any solution w in these coordinates, differentiating $|w|^2$ therefore gives

$$2\alpha(-d_c - \beta_c)|w|^2 \leq (|w|^2)' \leq 2\alpha(-d_c + \beta_c)|w|^2.$$

Multiplication by the corresponding scalar integrating factors proves the following bounds, including the lower singular-value estimate needed for the inverse:

$$\|X_c(v)\| \leq \kappa_i e^{\int_0^v \alpha(\beta_c - d_c) dw}, \quad \|X_c(v)^{-1}\| \leq \kappa_i e^{\int_0^v \alpha(\beta_c + d_c) dw}.$$

For $D_s = H - \delta_1 I_2 = \text{diag}(\lambda, -\lambda) - \delta_1 I_2 + E$, its symmetric part is bounded between $(-\lambda - \delta_1 - \|E\|)I_2$ and $(\lambda - \delta_1 + \|E\|)I_2$. The identical energy argument gives

$$\|X_s(v)\| \leq e^{\int_0^v (\lambda - \delta_1 + \|E\|) dw}, \quad \|X_s(v)^{-1}\| \leq e^{\int_0^v (\lambda + \delta_1 + \|E\|) dw}.$$

Combining these four proved estimates with (101) yields

$$\begin{aligned} \|L(v)\| &\leq \kappa_i \|L_0\| e^{\int_0^v [\lambda - \delta_1 + \|E\| + \alpha(\beta_c + d_c)] dw}, \\ \|L(v)^{-1}\| &\leq \kappa_i \|L_0^{-1}\| e^{\int_0^v [\lambda + \delta_1 + \|E\| + \alpha(\beta_c - d_c)] dw}. \end{aligned} \quad (103)$$

All quantities on the right are the retained, finite coefficient paths. In particular the exact coupled diffusion integral at the endpoint is

$$\int_0^{L_s} \alpha d_c dv = dK_3^2 \int_0^{r_a} R_3(t_b + r) dr \in [dK_3^2 r_a, 2dK_3^2 r_a].$$

The equality is the substitution $r = \alpha v$; the interval is the proved actual phase-length bound. The determinant is also exact:

$$\det L(v) = \frac{|z_{\sigma,0}|^2}{S_3^2(1 + u_i^2)} \exp \left(\int_0^v [\text{tr } E - 2\delta_1 + 2\alpha d_c] dw \right). \quad (104)$$

For a differentiable 2 by 2 matrix, expanding its two-column determinant and inserting $X' = DX$ gives $(\det X)' = (\text{tr } D) \det X$. Its initial determinant is one. Apply this calculation to X_s and X_c , and use $\text{tr } H = \text{tr } E$ and $\text{tr } C_c = -2d_c$ to prove (104).

Higher pulse derivatives have the explicit recurrence

$$L^{(n+1)} = \sum_{j=0}^n \binom{n}{j} \left[D_s^{(j)} L^{(n-j)} - L^{(n-j)} D_c^{(j)} \right], \quad D_c = \alpha C_c.$$

For a coefficient parameter derivative ∂_ξ the exact inhomogeneous equation is

$$(\partial_\xi L)' = D_s \partial_\xi L - (\partial_\xi L) D_c + (\partial_\xi D_s) L - L (\partial_\xi D_c).$$

Its initial value is the derivative of the explicit (98). To see directly that this produces a finite bound, multiply on the left by X_s^{-1} and on the right by X_c , differentiate, and integrate the last two terms with these same multipliers. The preceding norm estimates bound every resulting integral on $[0, L_s]$. Iteration proves the analogous assertion for every finite mixed derivative; all product-rule terms and initial derivatives are retained. This also proves smoothness on each compact source slow neighborhood where the frame is defined.

The factor $\mathcal{P}_\sigma(0)/S_3$ and the two explicit propagation costs in (103) can be very large or very small. There is no supplied relation identifying L_s with L_3 . Thus these equations prove a finite invertible map and exact matching to the source object, while exposing the cost of that map. They do not prove a uniform weighted-class estimate for L or for the direct candidate $L = I$. The source's particular pulse retains its own source estimates because of the equality (102). Transport of the full viscosity-one witness to another physical viscosity uses the already proved affine orthogonal and viscosity map. Alternatively one may replace δ_1 by δ_ν in this finite ODE construction, but the resulting changed-background homogeneous pulse is not asserted to be the source's published witness.

21 Actual third-growth budgets in the physical curl and force

We now use the actual third-growth pair and the original source frame of the preceding calculations, with the direct coefficient choice $L = I_2$. This fixes a particular candidate within the earlier general bridge. All source band, box, sign and nonzero harmonic indices below are retained, and only a finite collection is used on the present compact interior patch. The cutoff of each term has compact support inside its source rectangle and active annulus, with smooth extension by zero. The original coupled datum, its selected return time t_b and its growth duration r_a are fixed before assigning these source labels; they have no dependence on the source spatial or auxiliary variables. The source time and the coupled time have the exact dictionary proved above:

$$t_{c,j} = t_b + \alpha_j v_j, \quad \alpha_j = \frac{r_a}{L_{s,j}}, \quad \nabla_x t_{c,j} = 0, \quad \partial_t t_{c,j} = \alpha_j Q_j^{-1-h}.$$

These derivative identities concern the complete physical phase evaluation on each lifted rectangle. They remain valid when differentiating a field supported strictly inside that rectangle: the field vanishes on a collar where its extension would otherwise be needed.

Write $P(t_c) = -\Theta_3(t_c)$, $u(t_c) = \Omega_3(t_c)/\Theta_3(t_c)$. The actual finite third growth gives

$$S_3 \leq P \leq P_3^*, \quad \frac{u_i}{\sqrt{2}} \leq u \leq 2u_i, \quad u_i = \sqrt{b_i/a_i} = \frac{K_3}{\sigma_2 \sqrt{n_i}}, \quad \gamma_i = \sqrt{a_i b_i}.$$

The original quantities remain

$$K_3 = \mu \hat{\lambda}_3, \quad S_3 = \frac{A_0}{\mu} \hat{\lambda}_3^{-k_3-6}, \quad P_3^* = \frac{A_0}{\mu} \hat{\lambda}_3^{-7/8}, \quad \nu = \sqrt{A_0}.$$

In particular the following two component budgets have different original frequency powers:

$$\begin{aligned} Y_1 &= P_3^* = \frac{A_0}{\mu} \hat{\lambda}_3^{-7/8}, \\ Y_2 &= 2u_i P_3^* = \frac{2A_0}{\sigma_2 \sqrt{n_i}} \hat{\lambda}_3^{1/8}, \\ V_1 &= 2\gamma_i P_3^*, \quad V_2 = \gamma_i u_i P_3^* = \frac{\gamma_i A_0}{\sigma_2 \sqrt{n_i}} \hat{\lambda}_3^{1/8}. \end{aligned} \tag{105}$$

Here the indices 1,2 mean the two amplitude components, not physical vector directions or radial moment weights. The derivative calculation above proves $|y_a(t_c)| \leq Y_a$ and $|\dot{y}_a(t_c)| \leq V_a$ on the entire original interval, including its original negative seed. For the second component at the actual threshold, the sharper two-sided conclusion is

$$\frac{A_0}{\sigma_2 \sqrt{2n_i}} \hat{\lambda}_3^{1/8} \leq |\Omega_3(t_3^a)| \leq \frac{2A_0}{\sigma_2 \sqrt{n_i}} \hat{\lambda}_3^{1/8}.$$

All these identities follow by multiplying the original quotient bounds by P and using $K_3 = \mu \hat{\lambda}_3$; the factor μ is canceled only in the explicitly displayed product $u_i P_3^*$, not removed from the underlying objects or their definitions.

21.1 Two complete spatial shape fields for each retained harmonic

For harmonic j , retain $k_j m_j \neq 0$, where k_j is the source carrier and m_j its integer harmonic. Neither equals the coupled integer k_3 by definition. Let B_j be the source frame, n_j its normal, and

χ_j the full real slow, transverse and pulse cutoff. All are the original specified functions on that label. Put, for $a = 1, 2$ and the standard vectors $e_a \in \mathbb{R}^2$,

$$\begin{aligned} T_{ja} &= \chi_j B_j e_a, & C_{ja} &= \frac{i n_j \times T_{ja}}{k_j m_j |n_j|^2}, \\ W_{ja} &= \text{Re}[(T_{ja} + \text{curl}_{*,j,\text{coeff}} C_{ja}) e^{ik_j m_j \Phi_j}], \\ \varpi_{ja} &= \text{Re} \left[\frac{i \{n_j^T K_j T_{ja} - (n'_j)^T T_{ja}\}}{k_j m_j |n_j|^2} e^{ik_j m_j \Phi_j} \right], \\ Z_{ja} &= Q_j^{-A} W_{ja}, & \Pi_{ja} &= Q_j^{-2A} \varpi_{ja}, & A &= \frac{1}{2} + h. \end{aligned} \tag{106}$$

Vectors in a cross product have their cylindrical components in the oriented orthonormal physical basis. The coefficient curl is the full operator from the preceding moment calculation. Equivalently,

$$Z_{ja} = \text{Re curl}_x [Q_j^{1/2-A} C_{ja} e^{ik_j m_j \Phi_j}].$$

The transverse triple product proves this equivalence, including every cutoff derivative. Thus each Z_{ja} is a smooth compactly supported divergence-free physical vector field; the same divergence identity holds before auxiliary evaluation, with the commuting lifted derivatives. The pressure coefficient is linear in T_{ja} and includes the full moving-normal term.

The actual added physical fields, with the two original amplitudes, are exactly

$$w_y = \sum_j \sum_{a=1}^2 y_a(t_{c,j}) Z_{ja}, \quad \pi_y = \sum_j \sum_{a=1}^2 y_a(t_{c,j}) \Pi_{ja}. \tag{107}$$

Indeed $\nabla_x t_{c,j} = 0$ gives $\text{curl}(y_a(t_{c,j})\mathcal{A}) = y_a(t_{c,j}) \text{curl} \mathcal{A}$. It also gives $\nabla(y_a(t_{c,j})\Pi) = y_a(t_{c,j})\nabla\Pi$. The pressure formula in (106) is linear, so it agrees with the selected pressure for the amplitude $\chi_j B_j y(t_{c,j})$. This proves (107); in particular it does not discard a spatial derivative of a rapidly growing amplitude. That derivative has been calculated and is zero in these precise source coordinates.

The exact spatial derivative is

$$\partial_z w_y = \sum_{j,a} Q_j^{-A-D} y_a(t_{c,j}) \partial_{Z_j} W_{ja}, \quad D = \frac{1}{2} - h,$$

where $\partial_{Z_j} W_{ja}$ differentiates its coefficient, its exponential phase and all source frame and cutoff factors. The absolute auxiliary evaluation map is independent of z . There is no derivative of the actual coupled time in this formula.

21.2 Quantitative radial moments with the original component powers

For a physical field f , use the precise norm

$$\|f\|_{[e]}^2 = \int_0^\infty r^e \langle |f|^2 \rangle dr, \quad e = 1, 2,$$

with the normalized angular and auxiliary averages of the moment calculation. In a fixed band let $\|F\|_{[e],j}^2 = \int R_j^e \langle |F|^2 \rangle dR_j$. All norms are at the corresponding fixed slow axial coordinate and physical time. Define the actual finite shape budgets

$$\begin{aligned} \mathcal{V}_e &= \sum_{j,a} Y_a Q_j^{(e+1)/4-A} \|W_{ja}\|_{[e],j}, \\ \mathcal{Z}_e &= \sum_{j,a} Y_a Q_j^{(e+1)/4-A-D} \|\partial_{Z_j} W_{ja}\|_{[e],j}. \end{aligned} \tag{108}$$

The factors Y_1, Y_2 are the two explicit original-frequency expressions in (105), not a common unspecified constant. Triangle inequality, the pointwise amplitude bounds, and $r^e dr = Q_j^{(e+1)/2} R_j^e dR_j$ prove

$$\|w_y\|_{[e]} \leq \mathcal{V}_e, \quad \|\partial_z w_y\|_{[e]} \leq \mathcal{Z}_e.$$

No orthogonality of distinct harmonics has been assumed in these bounds. One may evaluate every norm by the explicit source fields (106); compactness makes each finite.

Let $w_* = u_* - \langle u_* \rangle_\theta$ be the actual old wave, including its curl corrections, and set

$$\mathcal{W}_e = \|w_*\|_{[e]}, \quad \mathcal{W}_{e,z} = \|\partial_z w_*\|_{[e]},$$

restricting the old-field radial integrals to a fixed enclosing support of the new fields and their derivatives. The exact moments already proved are

$$\begin{aligned} M_2 &= \partial_z \int r^2 \langle w_{*,z} w_{y,\theta} + w_{y,z} w_{*,\theta} + w_{y,z} w_{y,\theta} \rangle dr, \\ M_1 &= \partial_z \int r \langle 2w_{*,z} w_{y,z} + w_{y,z}^2 \rangle dr. \end{aligned}$$

Apply the product rule to every term and then Cauchy–Schwarz on its original weighted product measure. This gives the concrete simultaneous bounds

$$|M_e| \leq 2(\mathcal{W}_{e,z} \mathcal{V}_e + \mathcal{W}_e \mathcal{Z}_e + \mathcal{V}_e \mathcal{Z}_e), \quad e = 1, 2. \quad (109)$$

For M_2 , the first two cross products contribute one term each to each of the first two summands, and the self product contributes two terms to the last. For M_1 the explicit coefficient 2 and the derivative of the square give exactly those same three coefficients. Thus (109) retains the complete cross-and-quadratic dependence on the original component budgets.

21.3 The full physical residual with all clock and diffusion costs

Fix the actual finite source state (u_*, p_*) used in the bridge and an arbitrary physical viscosity $\nu_{\text{NS}} > 0$ in the residual. It remains distinct from the coupled d and $\nu = \sqrt{A_0}$. Define the explicitly differentiated shape residual

$$\mathcal{K}_{ja} = \partial_t Z_{ja} + (u_* \cdot \nabla) Z_{ja} + (Z_{ja} \cdot \nabla) u_* - \nu_{\text{NS}} \Delta Z_{ja} + \nabla \Pi_{ja}.$$

With $C_3(t_c) = \begin{pmatrix} -dK_3^2 R_3 & a_3 \\ b_3 & -dK_3^2 R_3 \end{pmatrix}$, the full residual increment of (107) is

$$\begin{aligned} \Delta R_y &= \sum_{j,a} y_a(t_{c,j}) \mathcal{K}_{ja} \\ &\quad + \sum_{j,a} \alpha_j Q_j^{-1-h} (C_3 y)_a(t_{c,j}) Z_{ja} \\ &\quad + \sum_{j,a,k,b} y_a(t_{c,j}) y_b(t_{c,k}) (Z_{ja} \cdot \nabla) Z_{kb}. \end{aligned} \quad (110)$$

To prove it, substitute (107) into the complete Navier–Stokes residual and subtract $R_{\nu_{\text{NS}}}(u_*, p_*)$. The time product rule gives $\partial_t y_a = \alpha_j Q_j^{-1-h} (C_3 y)_a$. Every spatial derivative of $y_a(t_{c,j})$ vanishes, so neither advection nor diffusion nor pressure adds an unrecorded derivative of that factor. The three linear spatial terms, the time derivative of Z_{ja} and the pressure gradient are precisely $\mathcal{K}_{ja}$.

The remaining quadratic product is the last full sum in (110). This proves the identity with every old-wave interaction, cutoff, curl term and pressure included. Distinct labels with disjoint source supports have zero corresponding products; no such cancellation is used for harmonics of the same label. The baseline $R_{\nu_{\text{NS}}}(u_*, p_*)$ must be added back to recover the candidate's total residual.

The source units make every factor in this identity explicit. Let $\nabla_{*,j}$ and $\Delta_{*,j}$ be the full cylindrical spatial operators, including basis derivatives and the lifted auxiliary derivatives, with $\nabla_x = Q_j^{-1/2} \nabla_{*,j}$ and $\Delta_x = Q_j^{-1} \Delta_{*,j}$ on that chart. Put $U_{*,j} = Q_j^A u_*$ and $\mathbf{t}_{*,j} = Q_j^{1+h} \partial_t$ with the full physical chain rule. Its extended expression is $-\varepsilon_j \partial_{T_j} + c_{i,j} N_{i,j}$ as in source (6.6). Define

$$\mathfrak{K}_{ja} = \mathbf{t}_{*,j} W_{ja} + (U_{*,j} \cdot \nabla_{*,j}) W_{ja} + (W_{ja} \cdot \nabla_{*,j}) U_{*,j} - \nu_{\text{NS}} \varepsilon_j \Delta_{*,j} W_{ja} + \nabla_{*,j} \varpi_{ja}.$$

Each term is obtained by differentiating the explicit shapes in (106), and

$$\mathcal{K}_{ja} = Q_j^{-2A-1/2} \mathfrak{K}_{ja}.$$

For the time term the exponent identity is $A + 1 + h = 2A + 1/2$; for viscosity it is $Q_j^{-A-1} = Q_j^{-2A-1/2} \varepsilon_j$. The nonlinear and pressure terms have the same stated residual unit. This proves every factor rather than replacing the source or coupled viscosity by a common normalized number.

Using Euclidean vector norms and the induced norm of the differentiated vector map, (110) and (105) give the following pointwise bound in actual shape fields:

$$\begin{aligned} |\Delta R_y| &\leq \sum_{j,a} Q_j^{-2A-1/2} (Y_a |\mathfrak{K}_{ja}| + \alpha_j V_a |W_{ja}|) \\ &\quad + \sum_{j,a,k,b} Y_a Y_b Q_j^{-A} Q_k^{-A-1/2} |W_{ja}| \|\nabla_{*,k} W_{kb}\|. \end{aligned} \tag{111}$$

Both Q_j and Q_k remain in every mixed-band term. The expression $\nabla_{*,k} W_{kb}$ includes the cylindrical frame connection, so the bound controls the actual Cartesian directional derivative. The clock satisfies the explicit original-stage bounds

$$\frac{L_3}{2\gamma_i L_{s,j}} \leq \alpha_j \leq \frac{8\sqrt{2}L_3}{\gamma_i L_{s,j}}.$$

Consequently its derivative costs in the first line can, if desired, be bounded using the original constants by

$$\alpha_j V_1 \leq \frac{16\sqrt{2}L_3}{L_{s,j}} \frac{A_0}{\mu} \hat{\lambda}_3^{-7/8}, \quad \alpha_j V_2 \leq \frac{8\sqrt{2}L_3}{L_{s,j}} \frac{A_0}{\sigma_2 \sqrt{n_i}} \hat{\lambda}_3^{1/8}.$$

These are finite quantitative costs for the actual inserted third stage. They do not assert decay of the resulting residual at a singular limit.

The source pulse equation (7.5) itself belongs to its viscosity-one source object. At a general ν_{NS} in the displayed candidate residual, the principal wave damping is $\nu_{\text{NS}} m_j^2 \varepsilon_j k_j^2 |n_j|^2$. Alternatively, the exact viscosity transport already proved can be applied to the complete viscosity-one source field and this complete candidate. That transport rescales support, force, residual and vorticity together. Merely changing this one damping coefficient does not assert that the unchanged source background is a solution with the same force at the changed viscosity.

Equations (106), (107), (109) and (110) are the explicit substitution of the actual third-growth amplitudes into the source-coordinate construction. Every surviving moment and residual has an actual differentiated field and a retained-scale bound. The exact intertwiner above supplies the relation to the source's homogeneous pulse; neither this direct candidate nor that finite coordinate identity is silently promoted to a completed infinite modified coupled construction.

Part II

The actual source pressure and five-row correction

22 The actual wave increment and the source's three compatibility targets

The source is the retained 166-page manuscript. Its SHA-256 is given in these two consecutive segments:

```
0e779481c4da40bd28d1e642e1d8ca57
447d129610df28dfa5a11e9af8ae228f.
```

The source operations used here are (8.3)–(8.27) and the complete four-step cycle of Proposition 9.6, on printed pages 89–111. Its coefficient classes are Definitions 6.4–6.5 and Proposition 6.6, printed pages 69–72. The source estimates quoted below are attributed to those statements. The following calculation of the actual candidate targets and all pressure-induced changes is a direct finite calculation. It does not infer source iteration exponents from finite smoothness. The source correction cycle is its viscosity-one cycle. The physical increment identities below separately retain any fixed $\nu_{\text{NS}} > 0$; using another physical viscosity in those identities does not identify the resulting candidate with the source's witness.

22.1 The actual covariance increment before pressure reconstruction

Use right-handed cylindrical coordinates (r, θ, z) . An overbar means the normalized angular average followed by normalized auxiliary Haar average, taken at fixed slow variables before phase evaluation. The old source state has actual zero-angular-mean wave w_* , including all its curl corrections. The new candidate is the proved finite field

$$w_y = \sum_{j,a} y_a(t_{c,j}) Z_{ja}, \quad \pi_y = \sum_{j,a} y_a(t_{c,j}) \Pi_{ja}, \quad \langle w_y \rangle_\theta = \langle \pi_y \rangle_\theta = 0.$$

The Z_{ja}, Π_{ja} are the complete physical shapes from the third-growth quantitative calculation; y_a is the actual coupled pair, with $t_{c,j} = t_b + (r_a/L_{s,j})v_j$. No amplitude factor is removed from an auxiliary average: it can depend on $v_j(Y)$. Write the exact double-averaged covariance increment as

$$C_{ab} = \overline{w_{*,a} w_{y,b} + w_{y,a} w_{*,b} + w_{y,a} w_{y,b}}, \quad a, b \in \{r, \theta, z\}. \quad (112)$$

Equivalently, substitute the displayed finite sum for both occurrences of w_y inside this average. In particular C retains every old-wave cross term and every same-label quadratic term. It is symmetric. All its radial integrals below are finite because the new fields are smooth and supported in a compact interior annulus. The old field need only be restricted to this support.

The old source finite correction state has the two exact mean-velocity constraints

$$\int_0^\infty r^2 \bar{v} dr = 0, \quad \int_0^\infty r \bar{\gamma} dr = 0.$$

These are source (9.10) in physical variables. The present initial addition w_y leaves (β, v, γ) unchanged because its angular mean is zero. The temporal mean operation has zero auxiliary mean,

and the first two rows of the five-row operation below give zero increments of these integrals. Thus all specified operations preserve their initially zero values. Preservation, by itself, does not change a nonzero entering value to zero.

The already calculated raw tangential residual moments are

$$M_2 = \partial_z J_2, \quad M_1 = \partial_z J_1, \quad J_2 = \int_0^\infty r^2 C_{z\theta} dr, \quad J_1 = \int_0^\infty r C_{zz} dr.$$

Here the candidate has been added with its zero-angular-mean pressure π_y , so the mean pressure is still the old mean pressure. The source algorithm next reconstructs the mean pressure from the updated velocity. This operation changes the axial residual moment.

To compute it, first let $\mathcal{C}_{ab}$ denote the angular average of the tensor inside (112), retaining its auxiliary dependence. Let $\mathcal{D}_r$ be the physical radial derivative on the extended domain, including the auxiliary phase-map derivative. Its auxiliary average is ordinary ∂_r . Since the mean velocity is unchanged by w_y , subtracting the radial conservative equations gives

$$\Delta g_r = -(\mathcal{D}_r + r^{-1})\mathcal{C}_{rr} - \partial_z \mathcal{C}_{zr} + r^{-1}\mathcal{C}_{\theta\theta}. \quad (113)$$

Indeed the radial transport of a divergence-free velocity is $(\mathcal{D}_r + r^{-1})u_r^2 + r^{-1}\partial_\theta(u_\theta u_r) + \partial_z(u_z u_r) - u_\theta^2/r$. Angular averaging removes the angular derivative and all products containing exactly one old mean factor and one zero-angular-mean wave. The source defines g_r as the negative of the mean radial residual apart from the pressure gradient, which gives exactly the signs in (113). The time and viscous averages of w_y are zero, so no physical viscosity has been dropped in this subtraction. The formula remains true for any fixed $\nu_{\text{NS}} > 0$ in the physical residual.

Set $g = \overline{\Delta g_r}$. Auxiliary derivative integrals vanish, and hence

$$g = -(\partial_r + r^{-1})C_{rr} - \partial_z C_{zr} + r^{-1}C_{\theta\theta}. \quad (114)$$

The increment of the source's pressure defect is therefore the following explicit new scalar, rather than either raw tangential moment:

$$\Delta P = \int_0^\infty g dr = \int_0^\infty \frac{C_{\theta\theta} - C_{rr}}{r} dr - \partial_z \int_0^\infty C_{zr} dr. \quad (115)$$

The equality integrates the derivative $\partial_r C_{rr}$; both endpoints vanish by compact support. The weighted second moment of this same source is

$$\int_0^\infty r^2 g dr = \int_0^\infty r(C_{rr} + C_{\theta\theta}) dr - \partial_z \int_0^\infty r^2 C_{zr} dr. \quad (116)$$

For its first term, integration by parts gives $-\int r^2 \partial_r C_{rr} = 2 \int r C_{rr}$, after which the explicit $-C_{rr}/r$ contribution subtracts one copy. These two identities retain the radial and axial coupling exactly.

22.2 Compact mean pressure and the corrected axial flux moment

Retain the source concentration scale and its full axial derivative:

$$z = q^D \eta, \quad 1 - t = q(1 - \eta^2), \quad A = \frac{1}{2} + h, \quad D = \frac{1}{2} - h, \quad q_z = \frac{2\eta q^{1-D}}{1 - 2h\eta^2}.$$

Choose the source fixed bump $\hat{\rho}$ in its reserved mean interval I_m , with $\int \hat{\rho}(x) dx = 1$, and put

$$\rho_{\text{phys}} = q^{-1/2} \hat{\rho}(r/\sqrt{q}), \quad c_{\rho, \text{phys}} = \frac{1}{2} \int r^2 \rho_{\text{phys}} dr = \frac{q}{2} \int x^2 \hat{\rho}(x) dx.$$

Thus $(c_{\rho,\text{phys}})_z = (q_z/q)c_{\rho,\text{phys}}$; it is generally not constant in z .

For precision the source's compact physical primitive is as follows. The fixed slow cutoff χ_m is zero on the inner collar and one on the outer collar of the active annulus. For an auxiliary-dependent scalar f define

$$\begin{aligned} If(r, Y) &= \int_0^r f(s, Y + (s^{d_r} - r^{d_r})v_r) ds, \\ Jf(r, Y) &= \int_0^\infty f(s, Y + (s^{d_r} - r^{d_r})v_r) ds, \quad I_c f = If - \chi_m Jf. \end{aligned}$$

Here $d_r = 2((1+h)\rho_g - h\kappa_s)$ is the exact source exponent, $\rho_g = \log(4 - \sqrt{2})/\log(4 + \sqrt{2})$, $v_r = (1, -(\sqrt{2}-1))$, and $\kappa_s = 10^{-5}$. The derivative $\mathcal{D}_r$ of the shifted integral differentiates the integration endpoint while cancelling the derivative of its $-r^{d_r}v_r$ shift. Consequently $\mathcal{D}_r I_c f = f - (\partial_r \chi_m) Jf$. Also $\overline{Jf} = \int \overline{f} dr$, by translation invariance of Haar measure. This proves the mean identities used next directly.

The source reconstruction after the actual wave addition changes the physical mean pressure by

$$\Delta p_m = I_c(\Delta g_r - \rho_{\text{phys}} \Delta P).$$

Its source has zero radial auxiliary-mean integral by (115). Therefore

$$\partial_r \overline{\Delta p_m} = g - \rho_{\text{phys}} \Delta P, \quad \int r \overline{\Delta p_m} dr = -\frac{1}{2} \int r^2 g dr + c_{\rho,\text{phys}} \Delta P. \quad (117)$$

The first identity follows from the displayed primitive formula; its cutoff remainder has zero auxiliary average. The second integrates $\partial_r(r^2 \overline{\Delta p_m})$ between the two zero endpoints. The full radial residual increment after this pressure update is $-\rho_{\text{phys}} \Delta P - (\partial_r \chi_m) J(\Delta g_r - \rho_{\text{phys}} \Delta P)$. The second term is retained. The source proves its terminal flatness under its all-order uniform class bounds; finite smoothness alone is not that proof.

Use different symbols for the source's two flux defects and the raw candidate fluxes J_2, J_1 . Their increments are exactly

$$\begin{aligned} \Delta J_\theta &= J_2, \\ \Delta J_z &= J_1 - \frac{1}{2} \int r^2 g dr \\ &= \int r \left(C_{zz} - \frac{1}{2} C_{rr} - \frac{1}{2} C_{\theta\theta} \right) dr + \frac{1}{2} \partial_z \int r^2 C_{zr} dr. \end{aligned} \quad (118)$$

These are the differences of source (8.15): the mean-velocity terms are unchanged and the full wave covariance changes by C . The axial pressure identity (117) now proves the exact moments after pressure reconstruction:

$$M_{2,\text{new}} = \partial_z \Delta J_\theta = M_2, \quad M_{1,\text{new}} = \partial_z (\Delta J_z + c_{\rho,\text{phys}} \Delta P) = M_1 + \partial_z \left(-\frac{1}{2} \int r^2 g dr + c_{\rho,\text{phys}} \Delta P \right). \quad (119)$$

These are increment identities. The old source moments and defects are added to obtain the full updated state. No hypothesis that the old defects vanish was used: identical old mean-velocity terms cancel in the subtraction. The product involving $c_{\rho,\text{phys}}$ stays inside ∂_z , including its nonzero derivative.

22.3 New quantitative budgets for the three actual inputs

Retain the previously proved quantities $\mathcal{W}_e, \mathcal{W}_{e,z}, \mathcal{V}_e, \mathcal{Z}_e$ for $e = 1, 2$. They satisfy

$$\|w_*\|_{[e]} = \mathcal{W}_e, \quad \|\partial_z w_*\|_{[e]} = \mathcal{W}_{e,z}, \quad \|w_y\|_{[e]} \leq \mathcal{V}_e, \quad \|\partial_z w_y\|_{[e]} \leq \mathcal{Z}_e.$$

In particular $\mathcal{V}_e, \mathcal{Z}_e$ retain separately the actual amplitude bounds $Y_1 = (A_0/\mu)\hat{\lambda}_3^{-7/8}$ and $Y_2 = 2A_0\hat{\lambda}_3^{1/8}/(\sigma_2\sqrt{n_i})$ and their physical source-band factors. Define merely for the following displayed estimates

$$B_e = 2\mathcal{W}_e\mathcal{V}_e + \mathcal{V}_e^2, \quad D_e = \mathcal{W}_{e,z}\mathcal{V}_e + \mathcal{W}_e\mathcal{Z}_e + \mathcal{V}_e\mathcal{Z}_e.$$

Choose $r_- > 0$ enclosing the radial support of the new fields and their axial derivatives from below. Cauchy–Schwarz on the original weighted product measures gives

$$|\Delta P| \leq r_-^{-2}B_1 + 2r_-^{-1}D_1, \quad |\Delta J_\theta| \leq B_2, \quad |\Delta J_z| \leq B_1 + D_2. \quad (120)$$

Here is the full estimate. The matrix $\text{diag}(-1, 1, 0)$ has operator norm one, so $|C_{\theta\theta} - C_{rr}|$ is bounded after averaging by $2|w_*||w_y| + |w_y|^2$. Replacing r^{-1} by $r_-^{-2}r$ and applying Cauchy–Schwarz gives the first part $r_-^{-2}B_1$ of the pressure bound. Differentiating C_{zr} produces the six actual cross and self product terms. Their integral with weight r^e is at most $2D_e$. Replace the unweighted integral by r_-^{-1} times the r -weighted integral to obtain $2r_-^{-1}D_1$ in (115). The same product estimate without differentiation bounds $\int r^2 C_{z\theta}$ by B_2 . Finally $\text{diag}(-1/2, -1/2, 1)$ has norm one. Apply it to the first integral in (118) to get B_1 ; the last differentiated integral is bounded by $\frac{1}{2}(2D_2) = D_2$. This proves all three claims without replacing the two amplitude components by a new unspecified constant.

22.4 The exact five-row map for these targets

The source map acts on the *current* physical triple (P, J_θ, J_z) , after adding (115) and (118) to the old source triple and reconstructing pressure. It does not take (M_2, M_1) as its input. The five equations, in physical variables, are

$$\begin{aligned} \int r^2 \Delta v \, dr &= 0, & \int r \gamma_d \, dr &= 0, \\ \int \frac{2V}{r} \Delta v \, dr &= -P, \\ \int r^2 (G \Delta v + V \gamma_d) \, dr &= -J_\theta, \\ \int (2r G \gamma_d - r V \Delta v) \, dr &= -J_z. \end{aligned} \quad (121)$$

In the last row there is exactly one factor r multiplying $V \Delta v$. The pressure-source term contributes $-\frac{1}{2} \int r^2 (2V/r) \Delta v \, dr = - \int r V \Delta v \, dr$, which proves that factor rather than treating the fifth row as an independent constraint with an arbitrary weight.

Retain the source's reserved patch $x = r/\sqrt{q} \in I_m$. Its actual summed base satisfies source (5.44) and (8.24):

$$\begin{aligned} V_{\text{phys}} &= q^{-A} a(\eta) x^{-1-2\lambda}, & G_{\text{phys}} &= 0, \\ a(\eta) &= 2^{1/2+\lambda} c_{\text{patch}} (1 + \eta^2)^{-1}, & \lambda &> 0, \quad |a(\eta)| \geq a_0 > 0. \end{aligned}$$

The source fixed λ here is its reserved-profile exponent, not a frequency of the coupled problem. In a fixed Q chart, the same physical base is

$$V_* = (Q/q)^A a(\eta) (\sqrt{Q/q} R)^{-1-2\lambda}, \quad G_* = 0.$$

The factor $(Q/q)^A$ has been retained.

Choose a nonnegative nonzero bump $\varphi_0 \in C_c^\infty(I_m)$ supported near an interior point and a fixed spacing $d_b > 0$ so small that the three profiles

$$\varphi_j(x) = e^{-jd_b} \varphi_0(e^{-jd_b} x), \quad j = 0, 1, 2,$$

have disjoint supports in the patch. This is possible because the patch contains an open interval about that point: choose three distinct scaled centers there, then decrease the support radius below half the minimum separation. The symbol d_b distinguishes this source bump spacing from the coupled physical diffusivity d . For every real p , set

$$\mu_p = \int x^p \varphi_0(x) dx > 0, \quad \int x^p \varphi_j(x) dx = \mu_p e^{jd_b p}.$$

The equality is the substitution $x = e^{jd_b} \xi$ and retains the prefactor in φ_j .

Define the two explicit moment matrices

$$A_\theta = \begin{pmatrix} \mu_2 & \mu_2 e^{2d_b} & \mu_2 e^{4d_b} \\ 2a\mu_{-2-2\lambda} & 2a\mu_{-2-2\lambda} e^{d_b(-2-2\lambda)} & 2a\mu_{-2-2\lambda} e^{2d_b(-2-2\lambda)} \\ -a\mu_{-2\lambda} & -a\mu_{-2\lambda} e^{-2d_b\lambda} & -a\mu_{-2\lambda} e^{-4d_b\lambda} \end{pmatrix},$$

$$A_z = \begin{pmatrix} \mu_1 & \mu_1 e^{d_b} \\ a\mu_{1-2\lambda} & a\mu_{1-2\lambda} e^{d_b(1-2\lambda)} \end{pmatrix}, \quad a = a(\eta).$$

Both inverses exist at every source parameter value. Indeed the angular matrix is a diagonal row factor times the matrix with rows $(1, t_p, t_p^2)$ for $p = 2, -2 - 2\lambda, -2\lambda$ and $t_p = e^{d_b p}$. Subtracting the first row from the other two and factoring their differences gives

$$\det[1, t_p, t_p^2]_{p=p_1, p_2, p_3} = (t_{p_2} - t_{p_1})(t_{p_3} - t_{p_1})(t_{p_3} - t_{p_2}).$$

All three powers are distinct for $\lambda > 0$. The diagonal row factor has determinant

$$-2a^2 \mu_2 \mu_{-2-2\lambda} \mu_{-2\lambda} \neq 0.$$

Likewise

$$\det A_z = a\mu_1 \mu_{1-2\lambda} (e^{d_b(1-2\lambda)} - e^{d_b}) \neq 0.$$

This also gives an explicit finite inverse norm bound $\|A^{-1}\| \leq \|\text{adj } A\|/|\det A|$ for either matrix. No uniform limit as $\lambda \downarrow 0$ is taken.

Use the exact row normalizations

$$P_q = q^{2A} P, \quad J_{\theta, q} = q^{2A-3/2} J_\theta, \quad J_{z, q} = q^{2A-1} J_z,$$

and define the five coefficients by

$$\mathbf{u} = A_\theta^{-1}(0, -P_q, -J_{z, q})^T, \quad \mathbf{s} = A_z^{-1}(0, -J_{\theta, q})^T. \quad (122)$$

Then the actual physical velocity increments are

$$\Delta v = q^{-A} \sum_{j=0}^2 u_j \varphi_j(r/\sqrt{q}), \quad \gamma_d = q^{-A} \sum_{j=0}^1 s_j \varphi_j(r/\sqrt{q}). \quad (123)$$

The first and second velocity moments have factors $q^{3/2-A}$ and q^{1-A} , respectively. The three inhomogeneous rows have factors q^{-2A} , $q^{3/2-2A}$, and q^{1-2A} , respectively. Substituting the original base into (121) and changing variables $r = \sqrt{q}x$ therefore gives exactly the two linear systems (122). This proves all five equations, uniqueness in the selected five-dimensional span, and smooth dependence on the actual target values. For instance $\partial_\xi A^{-1} = -A^{-1}(\partial_\xi A)A^{-1}$ follows by differentiating $AA^{-1} = I$; repeated product rules give every coefficient derivative with the actual q, η factors retained.

The desired axial increment has zero weighted integral and is auxiliary independent. Its exact compact azimuthal potential and radial increment are

$$\Psi(r, z, t) = \frac{1}{r} \int_0^r s \gamma_d(s, z, t) ds, \quad \Delta\beta = -\partial_z \Psi, \quad \Delta\gamma = (\partial_r + r^{-1})\Psi = \gamma_d.$$

The zero moment in (121) makes Ψ vanish beyond its outer support; its lower-end definition makes it vanish below the inner support. Differentiation proves $(\partial_r + r^{-1})\Delta\beta + \partial_z \Delta\gamma = 0$. The support includes the intervals between bump supports, which still lie within the reserved patch. To display the axial scaling cost explicitly, put $\Psi_j(x) = x^{-1} \int_0^x \xi \varphi_j(\xi) d\xi$. Then

$$\begin{aligned} \Psi &= q^{1/2-A} \sum_{j=0}^1 s_j \Psi_j(x), \\ \Delta\beta &= -q^{1/2-A} \sum_{j=0}^1 \left[(\partial_z s_j) \Psi_j + \frac{q_z}{q} s_j \left((1/2 - A) \Psi_j - \frac{x}{2} \Psi_j' \right) \right]. \end{aligned} \quad (124)$$

The two individual Ψ_j can have x^{-1} tails. The exact linear relation $\sum_j s_j \mu_1 e^{j d_b} = 0$ cancels those tails in their sum and, after differentiation, in (124). No individual compactness was assumed.

22.5 What the five-row operation actually leaves

Let (β, v, γ) be the current mean correction, after whatever preceding wave and mean updates have actually been performed. Keep the base (b, V, G) and the full current wave w fixed during this five-row step. Use the three full current defects as its targets. Write $\mathcal{D}_r$ for the physical extended radial derivative and $\mathcal{D}_t$ for the physical extended time derivative. With $\Delta_0 = \mathcal{D}_r^2 + r^{-1}\mathcal{D}_r + \partial_z^2$, direct subtraction of the full radial source gives

$$\begin{aligned} R_g &:= g_{r,\text{new}} - g_r - \frac{2V}{r} \Delta v \\ &= -\mathcal{D}_t \Delta\beta - (\mathcal{D}_r + r^{-1})(2b\Delta\beta + 2\beta\Delta\beta + (\Delta\beta)^2) \\ &\quad - \partial_z(b\Delta\gamma + G\Delta\beta + \beta\Delta\gamma + \gamma\Delta\beta + \Delta\beta\Delta\gamma) \\ &\quad + \frac{2v\Delta v + (\Delta v)^2}{r} + \nu_{\text{NS}}(\Delta_0 - r^{-2})\Delta\beta. \end{aligned} \quad (125)$$

Here all increments are auxiliary independent, so their $\mathcal{D}_t$ is ordinary ∂_t . The products with the old means still retain their auxiliary dependence. Equation (125) keeps the actual physical viscosity; the source unit-viscosity formula (8.26) is its case $\nu_{\text{NS}} = 1$.

The full updated defects, with pressure recomputed from the new radial source, are exactly

$$\begin{aligned} P_{\text{new}} &= \int \overline{R_g} dr, \\ (J_\theta)_{\text{new}} &= \int r^2 \overline{\gamma \Delta v + v \Delta \gamma + \Delta \gamma \Delta v} dr, \\ (J_z)_{\text{new}} &= \int r^2 \overline{2\gamma \Delta \gamma + (\Delta \gamma)^2} dr - \frac{1}{2} \int r^2 \overline{R_g} dr. \end{aligned} \quad (126)$$

To verify these formulas, expand every quadratic product in the definitions of the current and new defects. The radial source increment is $(2V/r)\Delta v + R_g$. The third row of (121) cancels the old P with the integral of the displayed linear term. The azimuthal flux increment contains the base-linear terms $G\Delta v + V\Delta \gamma$, cancelled by the fourth row because $\Delta \gamma = \gamma_d$. The axial defect contains $2G\Delta \gamma$ and the radial-source contribution $-\frac{1}{2}r(2V\Delta v)$ after integration. These are precisely the fifth row. All remaining products are the terms displayed in (126); the full wave covariance is unchanged during this mean step. This proves the formulas without assuming their right-hand sides vanish.

The two velocity moments remain exactly zero. The new residual moments therefore have the source form

$$\int r^2 \overline{E_{\theta,\text{new}}} dr = \partial_z (J_\theta)_{\text{new}}, \quad \int r \overline{E_{z,\text{new}}} dr = \partial_z ((J_z)_{\text{new}} + c_{\rho,\text{phys}} P_{\text{new}}).$$

The pressure and $c_{\rho,\text{phys}}$ are the actual reconstructed objects. Thus the five equations cancel specified linear defect contributions and expose the remaining nonlinear and differential terms. Equations (125)–(126) supply the next actual inputs rather than a claim of complete cancellation.

22.6 Exact source operation dictionary and the bounds it requires

For the source's estimates, $Q = 2^{-\ell}$, $\varepsilon = Q^h$, $S_* = \ell^2$, $R = r/\sqrt{Q}$, $Z = z/Q^D$, $T = (1-t)/Q$. Use a coarsest applicable common auxiliary torus H as in source Lemma 6.2; different representatives agree by its integer covering maps. The original radial edge weights are

$$\begin{aligned} \zeta(X) &= \exp \left(-\frac{a_a}{\log^2(X/X_a)} - \frac{a_b}{\log^2(X_b/X)} \right), \\ \delta(X) &= \min\{1, \log(X/X_a), \log(X_b/X)\}, \quad X = \frac{QR^2}{2q}, \end{aligned}$$

with the source fixed $a_a, a_b > 0$. In the following bounds the band, label, representative and harmonic are held fixed when differentiating. The constants may depend on the finite correction stage and derivative multi-index, but are uniform over bands, labels, points and rectangle copies.

A mean coefficient $f \in M^\alpha$ has no angular dependence, descends to the common torus, extends smoothly by zero outside the active shell, and satisfies for each multi-index I in (R, Z, T, H)

$$|\partial^I f| \leq C_I \varepsilon^\alpha S_*^{b_I} \zeta \delta^{-d_I}.$$

It can have nonzero auxiliary modes and has no rectangle or pulse envelope restriction. A moment coefficient $F(Z, T) \in S^\alpha$ satisfies $|\partial_{Z,T}^I F| \leq C_I \varepsilon^\alpha S_*^{b_I}$. A labelled nonzero harmonic amplitude $a_{\gamma,m} \in W^\alpha$ has no angular dependence, the source slow, transverse and shell support, smooth zero extension across those boundaries, compatible lift representatives, and

$$|\partial^I a_{\gamma,m}| \leq C_I \varepsilon^\alpha S_*^{b_I} \sqrt{\zeta} \delta^{-d_I} \mathcal{P}_\gamma(v), \quad 0 \leq v \leq L_s.$$

The prescribed source phase has $k_\gamma p_\gamma \in \mathbb{Z} \setminus \{0\}$ and $k_\gamma = \lceil \varepsilon^{-1/2} \rceil$. The harmonic set is finite at a fixed stage and independent of band. Equal label-harmonic terms are grouped before testing the bounds. The pulse envelope is the original positive exponential in (7.12), with its proved Gaussian estimates. Actual cut-off waves additionally have support where the source pulse cutoff ψ is supported and smooth zero extension in v . No divisibility by ψ is required.

The source's anisotropic derivative mappings, with the phase factored out, are

$$\begin{aligned} \partial_{R,Z,T} : C^\alpha &\rightarrow C^\alpha, & D_r : C^\alpha &\rightarrow C^{\alpha-\kappa_s}, & D_z = \varepsilon \partial_Z : C^\alpha &\rightarrow C^{\alpha+1}, \\ c_{i_0} N_{i_0} : C^\alpha &\rightarrow C^\alpha, & -\varepsilon \partial_T : C^\alpha &\rightarrow C^{\alpha+1}, & C &= M, W. \end{aligned}$$

Products obey $M^\alpha M^\beta \subset M^{\alpha+\beta}$ and $M^\alpha W^\beta \subset W^{\alpha+\beta}$. Same-label wave products have this summed order, in M for zero output harmonic and W otherwise; distinct supported labels have zero products. The zero-harmonic global M statement here is for the actual temporally cut-off waves. Before that cutoff, the source gives the mean bounds on the labelled rectangle, not a global M assertion. These are source Proposition 6.6, not deductions from the candidate's finite norm bounds.

Every radial target must first carry the correct measure and weight. For $f_* = Q^{a_f} f_{\text{phys}}$,

$$\int R^e \overline{f_*} dR = Q^{a_f - (e+1)/2} \int r^e \overline{f_{\text{phys}}} dr.$$

For the present inputs this reads

$$\begin{aligned} P_* &= Q^{2A} P, & (J_\theta)_* &= Q^{2A-3/2} J_\theta, & (J_z)_* &= Q^{2A-1} J_z, & (c_\rho)_* &= Q^{-1} c_{\rho,\text{phys}}, \\ (M_2)_* &= Q^{2A-1} M_2, & (M_1)_* &= Q^{2A-1/2} M_1. \end{aligned}$$

In particular (119) becomes exactly $\varepsilon \partial_Z \Delta J_{\theta,*}$ and $\varepsilon \partial_Z (\Delta J_{z,*} + c_{\rho,*} \Delta P_*)$. The factors match because $h + D = 1/2$. This is the extra axial factor used in source (9.12); using the raw J_1 after pressure reconstruction would omit part of its actual input.

For clarity, the following are the four ordered operations of source Proposition 9.6, with the full current velocity and pressure reconstructed after each operation.

1. For each supported nonzero harmonic coefficient $f_{\gamma,m}$ of the full current residual, solve the original transverse pulse equation with zero entrance data and source $-f_{\gamma,m}$. In frame coordinates its coefficient matrix is $H_\gamma - m^2 \varepsilon k_\gamma^2 |n_\gamma|^2 I$. Multiply the solved potential and pressure by ψ and take the full curl. Source Proposition 7.2 maps W^α sources to W^α velocity amplitudes and $W^{\alpha+1/2}$ pressure. Proposition 9.1 assigns the remaining supported linear residual order $\alpha + 1/2 - 3\kappa_s$ and retains its Gaussian cutoff tails additively. The actual candidate's full residual formula determines the input coefficients. Following this wave change, replace w_y by the actual total new wave increment in (112), recompute g_r and reconstruct pressure. Its old-wave cross terms must change with the updated wave.
2. Let F_2, F_1 be the auxiliary-averaged physical tangential residuals of that updated state. Their two moments are the updated versions of (119), including the old defects. For each $e = 1, 2$ retain the fixed normalized physical bump $b_e = q^{-(e+1)/2} \widehat{b}_e(r/\sqrt{q})$ with $\int x^e \widehat{b}_e(x) dx = 1$. Form

$$\mathfrak{M}_e = \int r^e F_e dr, \quad \sigma_e = -r^{-e} \int_0^r s^e (F_e(s) - b_e(s) \mathfrak{M}_e) ds.$$

The zero total weighted integral proves compact support and $(\partial_r + e/r) \sigma_e = -F_e + b_e \mathfrak{M}_e$. Use $\Sigma = Q^{2A}(\sigma_2, \sigma_1)$ as the input to the source's signed amplitude derivative (7.31), with

its fixed positive primary amplitudes. Source Proposition 7.6 maps an auxiliary-independent M^α stress to $W^{\alpha-1/2}$. Take its actual curl, retain the old-wave cross and quadratic correction tensor (9.13), and reconstruct pressure. The source does not set $b_e \mathfrak{M}_e$ to zero.

If $\mathfrak{M}_e = \partial_z \mathfrak{J}_e$, its bump term satisfies the exact derivative identity

$$b_e \partial_z \mathfrak{J}_e = \partial_z (b_e \mathfrak{J}_e) - (\partial_z b_e) \mathfrak{J}_e, \quad \partial_z b_e = -\frac{q_z}{2q} ((e+1)b_e + r \partial_r b_e).$$

It follows by differentiating the original q -scaled bump. Thus trading the moment bump for an axial tensor flux leaves the displayed radial-profile derivative term. For the axial equation, $\mathfrak{J}_1 = J_z + c_{\rho, \text{phys}} P$, including both product derivatives in (119).

3. For the zero-auxiliary-mean part of the current angularly invariant residual, first convert the physical residual to the fixed source chart. In this step a star denotes that chart representative:

$$E_* = Q^{2A+1/2} E_{\text{phys}}, \quad E_{a,*}^\circ = E_{a,*} - \langle E_{a,*} \rangle_Y, \\ (\Delta \beta_*, \Delta v_*, \Delta \gamma_*) = Q^A (\Delta \beta_{\text{phys}}, \Delta v_{\text{phys}}, \Delta \gamma_{\text{phys}}).$$

The source temporal update in these exact units is

$$\Delta v_* = -c_{i_0}^{-1} N_{i_0}^{-1} E_{\theta,*}^\circ, \quad \gamma_{d,*} = -c_{i_0}^{-1} N_{i_0}^{-1} E_{z,*}^\circ, \quad N_{i_0}^{-1} F = \sum_{k \neq 0} \frac{\widehat{F}(k)}{2\pi i v_t \cdot k} e^{2\pi i k \cdot H}.$$

Here $v_t = (\sqrt{2} - 1, 1)$ and $c_{i_0} = T_g^{i_0} Q^{1+h}$, with $c_{i_0}^{-1} \leq C S_*$. The source Diophantine bound gives $\|N^{-1} F\|_{C^m} \leq C_m \|F\|_{C^{m+4}}$. Let $T_{1,*}$ and $A_{1,*}$ be the source's radial potential and its cutoff remainder in this chart. Realize $\gamma_{d,*}$ by $T_{1,*} \gamma_{d,*}$: $\Delta \beta_* = -\varepsilon \partial_z T_{1,*} \gamma_{d,*}$ and $\Delta \gamma_* = \gamma_{d,*} - A_{1,*} \gamma_{d,*}$. The actual axial remainder $-A_{1,*} \gamma_{d,*}$ stays in the new velocity. The exact normalized cancellations are

$$c_{i_0} N_{i_0} \Delta v_* = -E_{\theta,*}^\circ, \\ c_{i_0} N_{i_0} \Delta \gamma_* = -E_{z,*}^\circ - c_{i_0} N_{i_0} A_{1,*} \gamma_{d,*}.$$

Convert every resulting velocity component back by Q^{-A} before reconstructing the physical pressure or taking physical products and defects. In particular its physical angular update is exactly

$$\Delta v_{\text{phys}} = -Q^{-A} c_{i_0}^{-1} Q^{2A+1/2} N_{i_0}^{-1} E_{\theta, \text{phys}}^\circ = -T_g^{-i_0} N_{i_0}^{-1} E_{\theta, \text{phys}}^\circ.$$

The last equality uses $A+1/2 = 1+h$ and the original $c_{i_0} = T_g^{i_0} Q^{1+h}$. It is also $-N_{\text{abs}}^{-1} E_{\theta, \text{phys}}^\circ$, since $N_{\text{abs}} = T_g^{i_0} N_{i_0}$ on the representative. This proves the conversion without applying a normalized inverse directly to a physical residual.

4. Apply (122) to the three defects of this updated state, use (123) and its actual radial component (124), and reconstruct pressure again. The exact remaining defects are (125)–(126). Source Lemma 8.7 maps S^α targets to M^α tangential increments and $M^{\alpha+1}$ radial increments. Source Lemma 8.8 gives the remaining defect class $S^{\alpha+0.9-2\kappa_s}$ using the actual bounds $v, \gamma \in M^{0.9}$, $\beta \in M^{1.9}$, $\alpha \geq 0.9$, and its reserved-patch base derivative bounds.

The source cycle assumes its full finite correction-state bounds $w \in W^{1/2}$, $w - w_0^{\text{tan}} \in W^{0.68}$, $v, \gamma, p_m \in M^{0.9}$, $\beta \in M^{1.9}$, with supported nonzero residual order $B_j = 1/2 + \sigma_j$, mean and defect order $C_j^* = 1 + \sigma_j$, and $\sigma_j = 1/5 + j/10$. Its ordered estimate proves gains $B_{j+1} - B_j = C_{j+1}^* - C_j^* =$

1/10. These are requirements and conclusions for the source's construction. The explicit candidate budgets and equations above do not by themselves identify those exponents.

There is a precise finite-domain statement available now. Write the full current residual as $R_{\text{old}} + \Delta R$, retaining the old source state and every term of the computed increment. The added candidate uses finitely many labels, with its cutoffs supported strictly inside the source rectangle and radial annulus. On those compact coefficient supports, $\varepsilon > 0$, $\zeta > 0$, $\delta > 0$ and $\mathcal{P}_\gamma(v) > 0$ have positive minima for each fixed label. By local finiteness of the retained source-label family, only finitely many old labels intersect this fixed compact perturbation support. Every differentiated coefficient of ΔR is finite by its explicit smooth shapes, including the retained old-state cross terms there. For any fixed α and derivative I , the maximum of its derivative divided by $\varepsilon^\alpha \sqrt{\zeta} \mathcal{P}_\gamma$ on these finitely many supports is therefore a concrete finite constant. Extend the finite *increment* family by zero on unused labels. This proves its finite coefficient inequalities without assigning an asymptotic gain to the amplitude. It does not set any old source coefficient to zero: R_{old} retains its actual locally finite family and its attributed source-class bounds. The finite-sum closure in source Proposition 6.6 combines those bounds with the proved increment bounds for each full current input. The same compact-support argument applies to the added mean sources and the reserved bump operators. Thus the operations above are defined on the actual old-plus-increment inputs. Their constants can contain the inverse of a very small Gaussian envelope or source scale; no uniform control for an increasing family of coupled stages or bands follows from this compactness argument. The original component budgets and the full nonlinear remainders remain the quantities to propagate when investigating such a family.

Part III

Compact completion of the full averaged force

23 Compact completion of the full averaged residual

This section computes a compact symmetric stress and a compact pressure for the complete averaged residual of the actual finite curl candidate. It retains the original radial bumps depending on $q(z, t)$, both axial fluxes, the radial force, and the full cylindrical connection terms. The result is a stress–pressure representation of a computed force; it does not by itself modify a velocity or identify the constructed stress with a covariance of additional waves.

23.1 The actual extended fields and all three averaged components

Use right-handed cylindrical coordinates $x = (r \cos \theta, r \sin \theta, z)$, with oriented orthonormal basis (e_r, e_θ, e_z) and $r > 0$. The retained finite source state is (u_*, p_*) ; the actual added field and its selected pressure are $(v, \pi) = (w_y, \pi_y)$. On a finite collection of source labels the added potential has the specified form

$$\begin{aligned} \mathcal{A}_y &= \sum_j \operatorname{Re} \left[Q_j^{1/2-A} \frac{i n_j \times t_j}{k_j m_j |n_j|^2} e^{ik_j m_j \Phi_j} \right], & v &= \operatorname{curl}_{\mathcal{D}} \mathcal{A}_y, \\ t_j &= \chi_j B_j y_3(t_{c,j}), & y_3 &= \begin{pmatrix} \Theta_3 \\ \Omega_3 \end{pmatrix}, & t_{c,j} &= t_b + \alpha_j v_j, & n_j^T B_j &= 0, & k_j m_j p_j &\in \mathbb{Z} \setminus \{0\}. \end{aligned} \tag{127}$$

Here y_3 is the actual signed third-growth trajectory, the cutoffs and source frame are the retained functions, and Q_j is constant on each source chart. The chosen pressure is the real harmonic sum with chart coefficient

$$Q_j^{-2A} \frac{i \{n_j^T K_j t_j - (n'_j)^T t_j\}}{k_j m_j |n_j|^2} e^{ik_j m_j \Phi_j}.$$

All coefficient components are independent of θ ; their angular dependence is the displayed nonzero integer harmonic. Their radial and axial supports lie in a fixed compact set with r bounded away from zero on the finite time interval under consideration. Smooth zero extension at each source cutoff is retained.

The source absolute auxiliary coordinate is $Y \in \mathbb{T}^2$ and the physical evaluation is

$$\begin{aligned} Y(r, t) &= v_r r^{d_r} + v_t t \pmod{\mathbb{Z}^2}, & v_r &= (1, -b_g), & v_t &= (b_g, 1), & b_g &= \sqrt{2} - 1, \\ \rho_g &= \frac{\log(4 - \sqrt{2})}{\log(4 + \sqrt{2})}, & \kappa_s &= 10^{-5}, & d_r &= 2((1 + h)\rho_g - h\kappa_s), \\ \mathcal{D}_r &= \partial_r + d_r r^{d_r-1} v_r \cdot \partial_Y, & \mathcal{D}_t &= \partial_t + v_t \cdot \partial_Y, & \mathcal{D}_z &= \partial_z. \end{aligned} \tag{128}$$

The angular derivative is ∂_θ . The exponent h here is the unchanged source chart exponent. These lifted derivatives commute, obey the product rule, and satisfy $\mathcal{D}_r r = 1$. Equivalently in Cartesian coordinates they are $\partial_{x_i} + \sum_\alpha (\partial_{x_i} Y_\alpha) \partial_{Y_\alpha}$; their commutators vanish because mixed derivatives of Y commute. Thus evaluation at the physical map preserves every differentiated identity below.

The actual finite source has its retained representation $u_* = \operatorname{curl}_{\mathcal{D}} \mathcal{A}_* + B_* e_\theta$, with $\partial_\theta B_* = 0$. Commutation of the lifted derivatives proves $\operatorname{div}_{\mathcal{D}} \operatorname{curl}_{\mathcal{D}} = 0$. Also $\operatorname{div}_{\mathcal{D}} (B_* e_\theta) = 0$. Consequently both u_* and v are divergence free at every independent Y , not only after physical evaluation.

Let the normalized averages act on cylindrical components before evaluation:

$$\langle f \rangle_\theta = \frac{1}{2\pi} \int_0^{2\pi} f d\theta, \quad \langle f \rangle_Y = \int_{\mathbb{T}^2} f dY, \quad \langle f \rangle = \langle \langle f \rangle_\theta \rangle_Y, \quad \int_{\mathbb{T}^2} 1 dY = 1.$$

Every cylindrical curl operation retains the nonzero angular harmonic in (127). For example its radial and axial components are $r^{-1}\partial_\theta \mathcal{A}_z - \partial_z \mathcal{A}_\theta$ and $(\mathcal{D}_r + r^{-1})\mathcal{A}_\theta - r^{-1}\partial_\theta \mathcal{A}_r$. Integrating the integer harmonic proves

$$\langle v \rangle_\theta = 0, \quad \langle \pi \rangle_\theta = 0 \quad \text{at every } (r, z, t, Y). \quad (129)$$

This statement includes the complete coefficient curl and every cutoff derivative, rather than only the principal amplitude.

For a smooth periodic coefficient f , integration of its auxiliary derivatives gives

$$\langle \mathcal{D}_r f \rangle = \partial_r \langle f \rangle, \quad \langle \mathcal{D}_t f \rangle = \partial_t \langle f \rangle, \quad \langle \partial_z f \rangle = \partial_z \langle f \rangle, \quad \langle \partial_\theta f \rangle = 0. \quad (130)$$

Applying the first identity twice also treats $\mathcal{D}_r^2$; the derivative of its variable coefficient has zero auxiliary average because that coefficient is independent of Y .

At arbitrary fixed physical viscosity $\nu_{\text{NS}} > 0$, the exact residual increment is

$$\Delta R = \mathcal{D}_t v + (u_* \cdot \nabla_{\mathcal{D}})v + (v \cdot \nabla_{\mathcal{D}})u_* + (v \cdot \nabla_{\mathcal{D}})v - \nu_{\text{NS}} \Delta_{\mathcal{D}} v + \nabla_{\mathcal{D}} \pi. \quad (131)$$

This is the residual of the actual modified state minus the source state's residual at the same viscosity. Put

$$\bar{u}_* = \langle u_* \rangle_\theta, \quad w_* = u_* - \bar{u}_*, \quad S = \langle w_* \otimes v + v \otimes w_* + v \otimes v \rangle. \quad (132)$$

It is a smooth real symmetric axisymmetric tensor compactly supported in the annular and axial set at issue. Each summand contains a factor of the actual compact field v .

Theorem 3. *The complete averaged residual is the full Cartesian divergence of the actual tensor S :*

$$\langle \Delta R \rangle = F_r e_r + F_2 e_\theta + F_1 e_z = \text{div } S, \quad (133)$$

where

$$\begin{aligned} F_r &= (\partial_r + r^{-1})S_{rr} - r^{-1}S_{\theta\theta} + \partial_z S_{rz}, \\ F_2 &= (\partial_r + 2r^{-1})S_{r\theta} + \partial_z S_{z\theta}, \\ F_1 &= (\partial_r + r^{-1})S_{rz} + \partial_z S_{zz}. \end{aligned} \quad (134)$$

Proof. For divergence-free fields, direct Cartesian component expansion rewrites the three advection terms of (131) as the lifted divergence of $u_* \otimes v + v \otimes u_* + v \otimes v$. The mean-flow cross tensor has zero angular average by (129), even when $\bar{u}_*$ depends on Y . Thus its averaged divergence is zero by (130) and the cylindrical divergence formula.

To check the linear radial component explicitly, let

$$\mathcal{L}_0 = \mathcal{D}_r^2 + r^{-1}\mathcal{D}_r + r^{-2}\partial_\theta^2 + \partial_z^2.$$

The complete cylindrical vector Laplacian is

$$\begin{aligned} (\Delta_{\mathcal{D}} v)_r &= (\mathcal{L}_0 - r^{-2})v_r - 2r^{-2}\partial_\theta v_\theta, \\ (\Delta_{\mathcal{D}} v)_\theta &= (\mathcal{L}_0 - r^{-2})v_\theta + 2r^{-2}\partial_\theta v_r, \quad (\Delta_{\mathcal{D}} v)_z = \mathcal{L}_0 v_z. \end{aligned}$$

Each average is zero by (129) and (130). The radial pressure average is $\langle \mathcal{D}_r \pi \rangle = \partial_r \langle \pi \rangle = 0$; the angular and axial components vanish as well. The averaged time derivative is zero. These verifications retain the radial connection term and physical viscosity before taking the average.

Finally, for a symmetric tensor independent of θ in cylindrical components, differentiation of $e_r = (\cos \theta, \sin \theta, 0)$ and $e_\theta = (-\sin \theta, \cos \theta, 0)$ gives exactly the three expressions in (134). One direct verification is to write its Cartesian matrix as $O_\theta S O_\theta^T$, where $O_\theta = [e_r, e_\theta, e_z]$, and use $\partial_x = \cos \theta \partial_r - r^{-1} \sin \theta \partial_\theta$ and $\partial_y = \sin \theta \partial_r + r^{-1} \cos \theta \partial_\theta$. The derivatives of O_θ give $-S_{\theta\theta}/r$ in the radial row, $2S_{r\theta}/r$ in the angular row, and S_{rz}/r in the axial row. Averaging the lifted version replaces $\mathcal{D}_r$ by ∂_r . This proves the full Cartesian identity. $\square$

23.2 The compact fluxes and the original varying radial bumps

For $e = 1, 2$, write $\tau_1 = z$, $\tau_2 = \theta$, and set

$$M_e(z, t) = \int_0^\infty r^e F_e(r, z, t) dr, \quad J_e(z, t) = \int_0^\infty r^e S_{z\tau_e}(r, z, t) dr. \quad (135)$$

Multiplication of the corresponding row of (134) by r^e makes its radial term $\partial_r(r^e S_{r\tau_e})$. The radial boundary values vanish by compact annular support. Therefore

$$M_e = \partial_z J_e, \quad \int_{\mathbb{R}} M_e(z, t) dz = 0 \quad (e = 1, 2). \quad (136)$$

Both J_e are smooth and compactly supported in z . No claim that either M_e vanishes pointwise is used.

Retain the original q -dependent bumps, for $q(z, t) > 0$ on the compact source patch:

$$b_e(r, z, t) = q(z, t)^{-(e+1)/2} \widehat{b}_e(r/\sqrt{q(z, t)}), \quad \widehat{b}_e \in C_c^\infty((0, \infty)), \quad \int_0^\infty x^e \widehat{b}_e(x) dx = 1. \quad (137)$$

The fixed profiles may be taken to be the already selected ones. The change of variable $r = \sqrt{q} x$ proves $\int r^e b_e dr = 1$ for every (z, t) . Choose a fixed annulus containing the supports of S and of the bumps on the relevant compact (z, t) set. This is possible because q has positive lower and finite upper bounds there. Outside the compact axial set where a flux or force is nonzero all the formulas below extend by zero.

The exact derivatives and cumulative bump are

$$\begin{aligned} \partial_z b_e &= -\frac{q_z}{2q} ((e+1)b_e + r \partial_r b_e), \\ \beta_e(r, z, t) &= \int_0^r s^e b_e(s, z, t) ds = \int_0^{r/\sqrt{q}} x^e \widehat{b}_e(x) dx, \\ \partial_z \beta_e &= -\frac{q_z}{2q} r^{e+1} b_e(r, z, t), \quad \int_0^\infty r^e \partial_z b_e dr = 0. \end{aligned} \quad (138)$$

The first identity is the product and chain rule. The second is the same radial substitution as the normalization. Differentiating its upper limit proves the third. Differentiating the normalization under the uniformly compact radial integral proves the fourth. In particular none of these identities assigns a sign to q_z .

23.3 Corrected radial primitives and the complete symmetric tensor

Define the actual corrected primitives

$$\sigma_e(r, z, t) = -r^{-e} \int_0^r s^e (F_e(s, z, t) - \partial_z(b_e(s, z, t)J_e(z, t))) ds, \quad e = 1, 2. \quad (139)$$

Their exact radial equations are

$$(\partial_r + e/r)\sigma_e = -F_e + \partial_z(b_e J_e). \quad (140)$$

This follows by multiplying (139) by r^e and using the fundamental theorem of calculus.

For comparison with the initially proposed primitive, put

$$\sigma_e^{\text{old}} = -r^{-e} \int_0^r s^e (F_e - b_e \partial_z J_e) ds.$$

Every missing term can be displayed exactly:

$$\sigma_e = \sigma_e^{\text{old}} + r^{-e} J_e \partial_z \beta_e = \sigma_e^{\text{old}} - \frac{r q_z}{2q} b_e J_e. \quad (141)$$

Indeed expanding $\partial_z(b_e J_e)$ in (139) proves the first equality, and (138) proves the second. With the old primitive the angular or axial stress divergence below would instead be $-F_e - (\partial_z b_e)J_e$. Thus the correction is necessary for the retained varying bump. It vanishes in the special case $q_z = 0$.

There is an exact expression entirely in the original tensor entries:

$$\begin{aligned} C_e(r, z, t) &= \int_0^r s^e S_{z\tau_e}(s, z, t) ds - \beta_e(r, z, t) J_e(z, t), \\ \sigma_e &= -S_{r\tau_e} - r^{-e} \partial_z C_e. \end{aligned} \quad (142)$$

To prove it, integrate the corresponding component of (134) only up to r , obtaining $\int_0^r s^e F_e ds = r^e S_{r\tau_e} + \partial_z \int_0^r s^e S_{z\tau_e} ds$. Substitute this identity and $\int_0^r s^e b_e ds = \beta_e$ into (139). Below both radial supports $C_e = 0$; above both supports its two summands are J_e and J_e , so it is again zero. This proves its compact support as well as the exact relation of the constructed radial stress to the actual covariance tensor.

Theorem 4. *The real symmetric axisymmetric tensor with cylindrical entries*

$$\begin{aligned} T_{rr} &= 0, & T_{r\theta} &= T_{\theta r} = \sigma_2, & T_{rz} &= T_{zr} = \sigma_1, \\ T_{z\theta} &= T_{\theta z} = -b_2 J_2, & T_{zz} &= -b_1 J_1, & T_{\theta\theta} &= rF_r + r\partial_z \sigma_1 \end{aligned} \quad (143)$$

is smooth and compactly supported in an annular cylinder and obeys the full Cartesian identity

$$\text{div } T = -(F_r e_r + F_2 e_\theta + F_1 e_z). \quad (144)$$

If the term rF_r is omitted only in $T_{\theta\theta}$, the exact divergence is instead $-(0, F_2, F_1)$ in cylindrical components.

Proof. The integral of the numerator in (139) over the entire radial half-line is

$$M_e - \partial_z \left(J_e \int_0^\infty s^e b_e ds \right) = \partial_z J_e - \partial_z J_e = 0.$$

Thus the numerator primitive is zero below its support and zero above it. Division by r^e introduces no singularity because it is zero on a fixed neighborhood of the axis. All derivatives under its fixed compact radial integral exist, and the normalization identity holds smoothly in (z, t) , so its extension by zero is smooth. Compact axial support follows because both F_e and J_e vanish outside the fixed compact axial set. The same statements hold for all entries of (143). Passing to the Cartesian matrix $O_\theta T O_\theta^T$ preserves smoothness away from the axis; near the axis it is identically zero. Hence this is a smooth compactly supported Cartesian tensor.

Apply the full symmetric axisymmetric divergence formula already proved in (134). Its three rows are

$$\begin{aligned} (\operatorname{div} T)_r &= -\frac{rF_r + r\partial_z\sigma_1}{r} + \partial_z\sigma_1 = -F_r, \\ (\operatorname{div} T)_\theta &= (\partial_r + 2/r)\sigma_2 - \partial_z(b_2J_2) = -F_2, \\ (\operatorname{div} T)_z &= (\partial_r + 1/r)\sigma_1 - \partial_z(b_1J_1) = -F_1. \end{aligned}$$

The last two identities are (140), including every derivative of b_e . This proves the Cartesian vector identity and, on omitting rF_r , its stated radial-zero variant. $\square$

23.4 Compact pressure, tracefree stress, and the exact gauge relation

The trace and its prescribed pressure are explicitly

$$\operatorname{tr} T = rF_r + r\partial_z\sigma_1 - b_1J_1, \quad p_T = \frac{1}{3}(rF_r + r\partial_z\sigma_1 - b_1J_1), \quad T^0 = T - p_TI_3. \quad (145)$$

Then $\operatorname{tr} T^0 = 0$, and every field is smooth and compactly supported in the same annular cylinder. The exact full cancellation is

$$F + \operatorname{div} T^0 + \nabla p_T = 0. \quad (146)$$

To check its sign, for any scalar p Cartesian differentiation gives $\operatorname{div}(pI_3) = \nabla p$. Therefore $\operatorname{div} T^0 = \operatorname{div} T - \nabla p_T = -F - \nabla p_T$, which proves the displayed equation. In components the new diagonal entries are

$$T_{rr}^0 = -p_T, \quad T_{\theta\theta}^0 = rF_r + r\partial_z\sigma_1 - p_T, \quad T_{zz}^0 = -b_1J_1 - p_T.$$

The axial stress divergence alone changes by $-\partial_z p_T$; the pressure gradient contributes $+\partial_z p_T$. Thus the combined stress–pressure correction preserves the exact tangential equations. There is no claim that their stress term alone stays unchanged.

More generally $T - pI_3$ with any smooth compact scalar p satisfies $F + \operatorname{div}(T - pI_3) + \nabla p = 0$. The choice in (145) is the unique one making this stress tracefree. The map from symmetric tensors to tracefree-tensor-scalar pairs,

$$T \longmapsto \left(T - \frac{1}{3} \operatorname{tr} T I_3, \frac{1}{3} \operatorname{tr} T\right),$$

has exact inverse $(U, p) \mapsto U + pI_3$ when $\operatorname{tr} U = 0$. It intertwines div with $(U, p) \mapsto \operatorname{div} U + \nabla p$. This supplies the full map between the two presentations, including compact support and signs.

23.5 All total force and torque components and the divergence image

The Cartesian components of the averaged force are

$$F = (F_r \cos \theta - F_2 \sin \theta, F_r \sin \theta + F_2 \cos \theta, F_1).$$

Integration in θ with the physical volume element $dx = r dr d\theta dz$ gives

$$\begin{aligned} \int_{\mathbb{R}^3} F dx &= 2\pi e_z \int_{\mathbb{R}} M_1(z, t) dz = 0, \\ \int_{\mathbb{R}^3} x \times F dx &= 2\pi e_z \int_{\mathbb{R}} M_2(z, t) dz = 0. \end{aligned} \tag{147}$$

For the torque calculation, direct multiplication in the oriented basis gives

$$x \times F = -zF_2e_r + (zF_r - rF_1)e_\theta + rF_2e_z.$$

The first two terms integrate to zero in θ , and the third gives the second equation of (147). Thus all three force and all three torque components vanish. The last equalities use the already proved compact flux identities (136). The same torque vanishes about every fixed origin x_0 , because subtracting x_0 changes it by $-x_0 \times \int F dx = 0$. On a finite axial slab $a < z < b$, the axial force and torque are respectively $2\pi(J_1(b, t) - J_1(a, t))$ and $2\pi(J_2(b, t) - J_2(a, t))$. The quantities need not vanish on an individual slice.

There is also a direct Cartesian stress verification. Since $F_j = -\partial_k T_{jk}$ and T is compact, integration by parts gives

$$\int F_j dx = 0, \quad \int x_i F_j dx = \int T_{ji} dx = - \int S_{ji} dx. \tag{148}$$

Symmetry makes the antisymmetric part of the first-moment matrix zero, which is precisely all three torque components. Its symmetric part is not asserted to vanish. The second equality with S uses $F = \text{div } S$. Equivalently $\text{div}(T + S) = 0$; multiplying this equation by x_i and integrating proves $\int (T + S)_{ji} dx = 0$. In the tracefree-pressure presentation the same first moment is $\int T_{ji}^0 dx + \delta_{ij} \int p_T dx$. All these identities persist under any finite time differentiation on a common compact support, by differentiation under the integral.

For completeness, this construction proves the exact image of compact symmetric-stress divergence in the axisymmetric class. Let an open axial interval I_z contain the actual axial supports, with $q > 0$ smooth on a neighborhood of its relevant compact subsets at the fixed time. In the following two spaces all supports are compact subsets of $\{r > 0, z \in I_z\}$. Thus the original bump is defined everywhere it is used, and integrating between axial support endpoints stays inside I_z . Let $\mathcal{F}_0$ be the space of smooth real axisymmetric compact forces supported away from $r = 0$ and satisfying exactly

$$\int_{\mathbb{R}} \int_0^\infty r F_1 dr dz = 0, \quad \int_{\mathbb{R}} \int_0^\infty r^2 F_2 dr dz = 0.$$

Let $\mathcal{S}_c$ be the space of smooth real compact symmetric axisymmetric Cartesian tensors supported away from $r = 0$. Define $\mathcal{D} : \mathcal{S}_c \rightarrow \mathcal{F}_0$ by $\mathcal{D}(U) = -\text{div } U$. Integration by parts proves that its values satisfy both defining identities of $\mathcal{F}_0$, and axisymmetry makes the other force and torque components vanish as above.

For any $F \in \mathcal{F}_0$, compute its actual radial moments and set

$$J_e(z) = \int_{-\infty}^z \int_0^\infty r^e F_e(r, s) dr ds.$$

This is a smooth compact axial function: its derivative is M_e , it vanishes below the axial support, and the defining zero total integral makes it vanish above that support. Use the fixed bumps (137),

the corrected primitive (139), and all entries of (143). The proof of Theorem 4 uses only these identities and proves a compact tensor $\mathcal{C}_b F$ with

$$\mathcal{D}\mathcal{C}_b F = F \quad (F \in \mathcal{F}_0). \quad (149)$$

Thus the two total moments are necessary and sufficient, with an explicit right inverse. For the actual force in (133), this primitive in z equals the already computed J_e from S , because both have derivative M_e and vanish at the negative axial end. No additional compatibility condition has been imposed on the actual candidate.

For fixed q and bump profiles, all operations defining $\mathcal{C}_b$ are linear. Hence $\mathcal{P}_b = \mathcal{C}_b \mathcal{D}$ is a projection on $\mathcal{S}_c$: (149) gives $\mathcal{P}_b^2 = \mathcal{P}_b$ and $\ker \mathcal{P}_b = \ker \mathcal{D}$. In particular the completed actual tensor is $T = \mathcal{P}_b(-S)$, and $T + S$ is a compact divergence-free symmetric tensor. This gives an exact relation to the actual averaged covariance, not merely equality of its two radial moments.

23.6 Exact source chart, bump, stress, and pressure powers

Fix one source chart and retain all of its constants:

$$r = \sqrt{Q} R, \quad z = Q^D Z, \quad 1 - t = QT, \quad A = \frac{1}{2} + h, \quad D = \frac{1}{2} - h, \quad \varepsilon = Q^h. \quad (150)$$

The dyadic label Q is fixed under differentiation. The averaged quantities are independent of Y and θ , so in this subsection the complete normalized spatial derivatives are ∂_R , $R^{-1}\partial_\theta$ and $\varepsilon\partial_Z$, with the same cylindrical basis derivatives as before. Define the exact chart representatives

$$\begin{aligned} S^* &= Q^{2A} S, & F^* &= Q^{2A+1/2} F, & T^* &= Q^{2A} T, & \Sigma_e &= Q^{2A} \sigma_e, \\ J_e^* &= Q^{2A-(e+1)/2} J_e = \int_0^\infty R^e S_{\tau_e}^* dR, & M_e^* &= Q^{2A-e/2} M_e = \varepsilon \partial_Z J_e^*, \\ q_* &= q/Q, & b_e^* &= Q^{(e+1)/2} b_e(\sqrt{Q}R, z, t) = q_*^{-(e+1)/2} \widehat{b}_e(R/\sqrt{q_*}). \end{aligned} \quad (151)$$

The two moment powers follow directly from $r^e dr = Q^{(e+1)/2} R^e dR$ and $\partial_z = Q^{-D} \partial_Z$; the remaining factor in M_e^* is $Q^{1/2-D} = \varepsilon$. In particular the scaled bump satisfies $\int R^e b_e^* dR = 1$.

Every term of the corrected primitive has the same exact stress unit:

$$\begin{aligned} \Sigma_e &= -R^{-e} \int_0^R X^e (F_e^* - \varepsilon \partial_Z (b_e^* J_e^*)) dX, \\ (\partial_R + e/R) \Sigma_e &= -F_e^* + \varepsilon \partial_Z (b_e^* J_e^*), \\ \Sigma_e &= \Sigma_e^{\text{old}} - \frac{\varepsilon R \partial_Z q_*}{2q_*} b_e^* J_e^*. \end{aligned} \quad (152)$$

To verify the first line, insert $F_e = Q^{-2A-1/2} F_e^*$ and $b_e J_e = Q^{-2A} b_e^* J_e^*$ in (139), and change the integration variable to $s = \sqrt{Q} X$. The second follows by differentiation. For the third, the physical correction in (141) has factor $\sqrt{Q} q_z/q$; using $q_z = Q^{1-D} \partial_Z q_*$ gives $\sqrt{Q} q_z/q = \varepsilon \partial_Z q_*/q_*$. This verifies the original q derivative, including its sign. Also, with $C_e^* = Q^{2A-(e+1)/2} C_e$,

$$\Sigma_e = -S_{r\tau_e}^* - \varepsilon R^{-e} \partial_Z C_e^*.$$

The completed stress and its pressure become exactly

$$\begin{aligned} T_{rr}^* &= 0, & T_{r\theta}^* &= \Sigma_2, & T_{rz}^* &= \Sigma_1, & T_{z\theta}^* &= -b_2^* J_2^*, & T_{zz}^* &= -b_1^* J_1^*, \\ T_{\theta\theta}^* &= R F_r^* + \varepsilon R \partial_Z \Sigma_1, & p_T^* &= Q^{2A} p_T = \frac{1}{3} (R F_r^* + \varepsilon R \partial_Z \Sigma_1 - b_1^* J_1^*), \\ (T^0)^* &= T^* - p_T^* I_3, & F^* + \text{div}_*(T^0)^* + \nabla_* p_T^* &= 0. \end{aligned} \quad (153)$$

For example the radial row is $-T_{\theta\theta}^*/R + \varepsilon\partial_Z\Sigma_1 = -F_r^*$, and the tangential rows use (152). The pressure identity follows from the same Cartesian isotropic-tensor calculation as before. Physical divergence and pressure gradient both have factor $Q^{-2A-1/2}$ because $\nabla_x = Q^{-1/2}\nabla_*$ on the chart. Thus the radial residual, hoop stress, both surviving axial fluxes, varying-bump correction and compact pressure all have their exact source units. No decay or uniform estimate at a singular endpoint is inferred from these finite compact-support identities.

Part IV

All-order actual moment and moving-scale costs

24 Original-scale derivative costs of the actual compatibility defects

The state and increment in this section are exactly the finite source state and actual third-growth curl constructed in the preceding reader. Write $u_* = \langle u_* \rangle_\theta + w_*$, $v = w_y$, and

$$C = \langle w_* \otimes v + v \otimes w_* + v \otimes v \rangle,$$

where the outer brackets are the normalized angular and auxiliary averages before physical evaluation. Every component below is in the original oriented cylindrical orthonormal basis. The full moment completion and source pressure calculation prove the formulas used here; we also derive the scalar compatibility formulas directly to fix all signs. Throughout this section the unadorned symbols P, J_θ, J_z denote the defect increments $\Delta P, \Delta J_\theta, \Delta J_z$ caused by this actual wave addition. They are not the full old-plus-new source defects. The unchanged old defects must be added when applying the source correction to the whole updated state.

24.1 Actual mixed derivatives of the third-growth curl

Retain the fixed finite set of source labels j , the actual clocks $t_{c,j} = t_b + \alpha_j v_j$, and the two full shape fields W_{ja} . The original data t_b, r_a, S_3, u_i are independent of these spatial labels. In particular

$$v = \sum_{j,a} Q_j^{-A} y_a(t_{c,j}) W_{ja}, \quad A = \frac{1}{2} + h, \quad D = \frac{1}{2} - h, \quad \partial_t t_{c,j} = \alpha_j Q_j^{-1-h}, \quad \partial_z t_{c,j} = 0.$$

Let $Y_{1,n} = T_n$ and $Y_{2,n} = O_n$ be the complete finite derivative arrays already proved for the actual coupled solution. Thus $Y_{1,0} = (A_0/\mu)\hat{\lambda}_3^{-7/8}$, $Y_{2,0} = 2A_0\hat{\lambda}_3^{1/8}/(\sigma_2\sqrt{n_i})$, $Y_{1,1} = 2\gamma_i Y_{1,0}$, and $Y_{2,1} = \gamma_i A_0\hat{\lambda}_3^{1/8}/(\sigma_2\sqrt{n_i})$. Higher entries are those closed recurrences in the retained moving-parent data; no independent amplitude bound is introduced here.

For a pair $I = (a, b) \in \mathbb{N}^2$, put $\partial^I = \partial_z^a \partial_t^b$ and $I_z = I + (1, 0)$. On a fixed chart the full time operator is $\mathbf{t}_{*,j} = -\varepsilon_j \partial_{T_j} + c_{i,j} N_{i,j}$. The exact mixed derivative formula is

$$\begin{aligned} \partial_z^a \partial_t^b v = & \sum_{j,c} \sum_{s=0}^b \binom{b}{s} Q_j^{-A-aD-b(1+h)} \alpha_j^s y_c^{(s)}(t_{c,j}) \\ & \cdot \partial_{Z_j}^a \mathbf{t}_{*,j}^{b-s} W_{jc}. \end{aligned} \tag{154}$$

All operators on the shapes include their phase, frame and cutoff derivatives. The shape time operator is the full derivative after auxiliary evaluation, not a derivative that holds the fast phase fixed. To prove the identity, differentiate the physical product b times in time. If s of the derivatives act on y_c , its factor is $(\alpha_j Q_j^{-1-h})^s y_c^{(s)}$; the other $b-s$ derivatives give $Q_j^{-(b-s)(1+h)} \mathbf{t}_{*,j}^{b-s} W_{jc}$. There are $\binom{b}{s}$ such choices. The a axial derivatives act only on the shape and give $Q_j^{-aD} \partial_{Z_j}^a$. The axial derivative commutes with the stated time operator, since its coefficients are fixed label constants. Combining the powers proves (154) without dropping any derivative.

For each $e \in \{-1, 0, 1, 2\}$, define the precise weighted norm

$$\|f\|_{[e]}^2 = \int_0^\infty r^e \langle |f|^2 \rangle dr, \quad \|F\|_{[e],j}^2 = \int_0^\infty R_j^e \langle |F|^2 \rangle dR_j.$$

Every field being integrated is supported in a compact annulus away from zero, so the negative weight introduces no singular integral. Define the explicit actual-shape budget

$$\begin{aligned} \mathcal{V}_{e,(a,b)} = \sum_{j,c} \sum_{s=0}^b \binom{b}{s} \alpha_j^s Y_{c,s} Q_j^{(e+1)/4-A-aD-b(1+h)} \\ \cdot \|\partial_{Z_j}^a \mathbf{t}_{*,j}^{b-s} W_{jc}\|_{[e],j}. \end{aligned} \quad (155)$$

Then $\|\partial^I v\|_{[e]} \leq \mathcal{V}_{e,I}$. Indeed the measure identity $r^e dr = Q_j^{(e+1)/2} R_j^e dR_j$, valid also for $e = -1$, and the triangle inequality applied to (154) give this bound. There is no inter-label orthogonality assumption. Put $\mathcal{W}_{e,I} = \|\partial^I w_*\|_{[e]}$, restricted to a fixed radial set enclosing the supports of the actual new fields and all derivatives at issue. The old wave includes its existing curl corrections. These are norms of the specified source state, not missing amplitude variables.

24.2 A contraction estimate that retains every product derivative

For $J = (j_1, j_2) \leq I = (a, b)$, set $\binom{I}{J} = \binom{a}{j_1} \binom{b}{j_2}$, and define

$$\begin{aligned} \mathcal{B}_{e,I} = \sum_{J \leq I} \binom{I}{J} (\mathcal{W}_{e,J} \mathcal{V}_{e,I-J} + \mathcal{V}_{e,J} \mathcal{W}_{e,I-J} \\ + \mathcal{V}_{e,J} \mathcal{V}_{e,I-J}). \end{aligned} \quad (156)$$

If E is any fixed real symmetric three-by-three matrix in cylindrical components, then

$$\left| \partial^I \int r^e (E : C) dr \right| \leq \|E\|_{\text{op}} \mathcal{B}_{e,I}. \quad (157)$$

For a single outer product, $E : (f \otimes g) = f^T E g$, whose absolute value is at most $\|E\|_{\text{op}} |f| |g|$. The repeated product rule gives exactly the sum over $J \leq I$ in (156). Cauchy-Schwarz on the original weighted angular-auxiliary-radial measure gives each displayed product of norms. Compact smooth support justifies differentiation under the integral. Axial and time derivatives do not differentiate the cylindrical basis. This proves (157) for every specified order. For the time derivative, the full lifted physical operator is $\mathcal{D}_t = \partial_t + N_{\text{abs}}$, with N_{abs} a constant torus direction. The Haar integral of $N_{\text{abs}} f$ is zero by periodicity. Therefore $\partial_t \langle f \rangle = \langle \mathcal{D}_t f \rangle$. The same identity applies repeatedly, and axial differentiation commutes with this average and this constant torus direction. Consequently the ordinary mixed derivatives of the averaged tensor in (157) are exactly the averages of the full physical derivatives estimated in (154).

24.3 The actual three defects and the pressure-corrected axial flux

Since the wave increment has zero angular mean and is divergence-free, the change of the radial mean pressure source is

$$g = -(\partial_r + r^{-1}) C_{rr} - \partial_z C_{zr} + r^{-1} C_{\theta\theta}.$$

The three actual source compatibility-defect increments are therefore

$$\begin{aligned} P &= \int \frac{C_{\theta\theta} - C_{rr}}{r} dr - \partial_z \int C_{zr} dr, \\ J_\theta &= \int r^2 C_{z\theta} dr, \\ J_z &= \int r (C_{zz} - \tfrac{1}{2} C_{rr} - \tfrac{1}{2} C_{\theta\theta}) dr + \tfrac{1}{2} \partial_z \int r^2 C_{zr} dr. \end{aligned} \tag{158}$$

The first formula is $\int g dr$, using the zero boundary integral of $\partial_r C_{rr}$. For the last, integration by parts gives

$$\int r^2 g dr = \int r(C_{rr} + C_{\theta\theta}) dr - \partial_z \int r^2 C_{zr} dr.$$

Substitute it into the source definition $J_z = \int r C_{zz} dr - \tfrac{1}{2} \int r^2 g dr$. This proves all three identities in (158). In particular the axial source defect is not replaced by the raw flux $\int r C_{zz} dr$.

Let the source pressure bump be the original

$$\rho(r, z, t) = q^{-1/2} \hat{\rho}(r/\sqrt{q}), \quad \int \hat{\rho}(x) dx = 1, \quad c_0 = \tfrac{1}{2} \int x^2 \hat{\rho}(x) dx.$$

It has $\int \rho dr = 1$ and $c_\rho = \tfrac{1}{2} \int r^2 \rho dr = c_0 q$. The auxiliary average of the compact source pressure increment is

$$\overline{\Delta p_m}(r) = \int_0^r (g(s) - \rho(s)P) ds.$$

Its derivative is $g - \rho P$, and its integrand has zero total radial integral, giving compact support. Its weighted pressure moment is $\int r \overline{\Delta p_m} dr = -\tfrac{1}{2} \int r^2 g dr + c_\rho P$. Here is the exact map to the full auxiliary-dependent source operation. Let Δg_r be the angularly averaged source increment before auxiliary averaging, so that $\langle \Delta g_r \rangle_Y = g$, and put $f = \Delta g_r - \rho P$. The source's shifted primitive uses

$$\begin{aligned} \mathcal{I}f(r, Y) &= \int_0^r f(s, Y + (s^{d_r} - r^{d_r})v_r) ds, \\ \mathcal{J}f(r, Y) &= \int_0^\infty f(s, Y + (s^{d_r} - r^{d_r})v_r) ds, \quad \mathcal{I}_c f = \mathcal{I}f - \chi_m \mathcal{J}f. \end{aligned}$$

The fixed source exponent and direction are the original d_r, v_r , and χ_m is its inner-zero, outer-one radial cutoff. The full pressure increment is $\Delta p_m = \mathcal{I}_c f$. Haar translation invariance gives $\langle \mathcal{J}f \rangle_Y = \int (g - \rho P) dr = 0$ and $\langle \mathcal{I}f \rangle_Y = \int_0^r (g - \rho P) ds$. This proves the asserted averaged pressure formula from the full map. The lifted radial derivative cancels the shift derivative and yields

$$\mathcal{D}_r \Delta p_m - \Delta g_r = -\rho P - (\partial_r \chi_m) \mathcal{J}f.$$

The second term has zero auxiliary average but need not vanish before physical evaluation. No estimate for its singular-time flatness is inferred from its zero average. Consequently the double-averaged axial residual-moment increment after this reconstruction is

$$\partial_z K_z, \quad K_z = J_z + c_0 q P.$$

The full product, including the derivatives of q , is retained.

The following simultaneous quantitative estimates hold at every finite mixed order:

$$\begin{aligned}
|\partial^I P| &\leq \mathcal{B}_{-1,I} + \frac{1}{2}\mathcal{B}_{0,I_z}, \\
|\partial^I J_\theta| &\leq \frac{1}{2}\mathcal{B}_{2,I}, \\
|\partial^I J_z| &\leq \mathcal{B}_{1,I} + \frac{1}{4}\mathcal{B}_{2,I_z}, \\
|\partial^I K_z| &\leq \mathcal{B}_{1,I} + \frac{1}{4}\mathcal{B}_{2,I_z} \\
&\quad + |c_0| \sum_{J \leq I} \binom{I}{J} |\partial^J q| (\mathcal{B}_{-1,I-J} + \frac{1}{2}\mathcal{B}_{0,(I-J)_z}).
\end{aligned} \tag{159}$$

For the first line the diagonal contraction matrix is $\text{diag}(-1, 1, 0)$, of norm one, whereas the rz entry is the contraction against the symmetric matrix with rz, zr entries $1/2$, of norm $1/2$. For the second line use the analogous $z\theta$ matrix. For the third the diagonal matrix is $\text{diag}(-1/2, -1/2, 1)$, of norm one, and the coefficient $1/2$ multiplying C_{zr} contributes an additional factor $1/2$. These facts and (157) prove the first three bounds. The exact product rule for qP proves the fourth. Replacing I by I_z gives bounds for the two residual moments that actually enter the source radial inverse. Thus every pressure, radial and axial coupling has a bound in the original shape fields and third-growth constants.

24.4 Exact derivatives of the original moving scale and bumps

The source scale is determined by

$$q - z^2 q^{2h} = 1 - t, \quad z = q^D \eta, \quad D = \frac{1}{2} - h, \quad 0 < h < 1/100, \quad -1 < \eta < 1.$$

Set $B(\eta) = 1 - 2h\eta^2 > 1 - 2h > 0$. Implicit differentiation of the first equation, followed by the displayed formula for z , gives

$$\begin{aligned}
q_z &= \frac{2\eta q^A}{B}, & q_t &= -\frac{1}{B}, \\
\eta_z &= \frac{q^{-D}(1 - \eta^2)}{B}, & \eta_t &= \frac{D\eta q^{-1}}{B}.
\end{aligned} \tag{160}$$

For example $q_z = 2zq^{2h}/B$, and $D + 2h = A$. Differentiating $\eta = zq^{-D}$ gives the last two formulas; the identity $2D + 2h = 1$ gives the factor $1 - \eta^2$. This proves the derivative dictionary without assigning a sign to q_z .

Put $x = r/\sqrt{q}$. For a smooth function $f(x, \eta)$ define the explicit differential expressions

$$\mathcal{Z}_\sigma f = \frac{2\sigma\eta f - \eta x f_x + (1 - \eta^2)f_\eta}{B}, \quad \mathcal{T}_\sigma f = \frac{-\sigma f + \frac{1}{2}x f_x + D\eta f_\eta}{B}.$$

Equations (160) and the ordinary chain rule prove

$$\partial_z(q^\sigma f) = q^{\sigma-D}\mathcal{Z}_\sigma f, \quad \partial_t(q^\sigma f) = q^{\sigma-1}\mathcal{T}_\sigma f, \quad \partial_r(q^\sigma f) = q^{\sigma-1/2}f_x. \tag{161}$$

For the last formula q, η are independent of r . For the first two, differentiation of x gives respectively $-\eta x q^{-D}/B$ and $x q^{-1}/(2B)$, proving every term.

In particular let the original radial moment bump be

$$b_e = q^{-(e+1)/2} \widehat{b}_e(x), \quad \int x^e \widehat{b}_e(x) dx = 1.$$

For any $k, a, b \geq 0$, its derivative is exactly

$$\partial_r^k \partial_z^a \partial_t^b b_e = q^{-(e+1+k)/2-aD-b} f_{e;k,a,b}(x, \eta). \quad (162)$$

Here the coefficient is given by a finite recursion: start at $\sigma = -(e+1+k)/2$ and $f = \widehat{b}_e^{(k)}$; repeat $f \leftarrow \mathcal{Z}_\sigma f$, $\sigma \leftarrow \sigma - D$, a times, and then $f \leftarrow \mathcal{T}_\sigma f$, $\sigma \leftarrow \sigma - 1$, b times. The result is $f_{e;k,a,b}$. This proves the identity by induction using (161). Smooth mixed derivatives commute, so this fixed order computes the stated derivative. Every coefficient is explicitly a finite rational expression in x, η , the retained constants, and derivatives of the specified bump, with nonzero denominator a power of B . On its compact x support and $|\eta| \leq 1$ its maximum is finite and computable directly from that expression. Thus (162) supplies the original power of q at every order. The pressure bump ρ uses exactly this recursion with $e = 0$. Likewise every derivative in (159) is explicit: $\partial_z^a \partial_t^b q = q^{1-aD-b} Q_{a,b}(\eta)$, where the same recursion starts with $\sigma = 1, f = 1$.

For completeness, the weighted cumulative bump has the exact form

$$\beta_e(r, z, t) = \int_0^r s^e b_e(s, z, t) ds = \int_0^{r/\sqrt{q}} x^e \widehat{b}_e(x) dx.$$

Its axial derivative is

$$\partial_z \beta_e = -\frac{q_z}{2q} r^{e+1} b_e. \quad (163)$$

This follows by differentiating its upper endpoint, which is $-rq_z/(2q^{3/2})$. Inserting the definition of b_e gives the displayed original physical factors. Therefore replacing the primitive of $F_e - b_e \partial_z J_e$ by the exact primitive of $F_e - \partial_z(b_e J_e)$ changes the stress coefficient by

$$-\frac{rq_z}{2q} b_e J_e.$$

This is the moving-bump term required in the complete stress construction. Its sign and all further derivatives are fixed by (160)–(163); it is not set to zero by holding the physical source scale fixed during axial differentiation.

These calculations supply finite quantitative inputs for the actual source pressure and moment operations. They do not identify the distinct direct candidate with a completed infinite source correction sequence: the complete source-operation formulas retain the new velocity, covariance, pressure and nonlinear defects that such a step actually creates.

Part V

Physical mean velocity and derivative costs

25 Physical derivative and full vector-force costs of the five-row mean operation

This section completes the finite mean operation on the actual current source state. All coordinates are the original right-handed cylindrical coordinates. The physical viscosity is the fixed number $\nu_{\text{NS}} > 0$. Write the full current velocity as $U = (b + \beta, V + v, G + \gamma) + w$, where w includes every current wave and curl correction. During the operation below this wave is held fixed. The three targets are the full current defects

$$\mathcal{P} = P_{\text{old}} + \Delta P, \quad \mathcal{J}_\theta = (J_\theta)_{\text{old}} + \Delta J_\theta, \quad \mathcal{J}_z = (J_z)_{\text{old}} + \Delta J_z. \quad (164)$$

Here the increments are the pressure-corrected actual wave increments of (158). If other finite source operations have already been performed, “old” means their actual entering state and the increments are recomputed for the actual addition. Thus none of the following formulas substitutes an increment for a full target.

Retain, without a change of scale or profile constant,

$$\begin{aligned} q - z^2 q^{2h} &= 1 - t, \quad z = q^D \eta, \quad x = r/\sqrt{q}, \quad A = \tfrac{1}{2} + h, \quad D = \tfrac{1}{2} - h, \\ 0 < h < 1/100, \quad q > 0, \quad -1 < \eta < 1, \quad B(\eta) &= 1 - 2h\eta^2, \\ V &= q^{-A} a(\eta) x^{-1-2\lambda}, \quad G = 0 \quad \text{on the reserved mean patch,} \\ a(\eta) &= \frac{2^{1/2+\lambda} c_{\text{patch}}}{1 + \eta^2}, \quad \lambda > 0, \quad |a(\eta)| \geq a_0 > 0. \end{aligned} \quad (165)$$

The symbol $a(\eta)$ in this display is a scalar base coefficient; below the bold symbol $\mathbf{a}$ denotes the vector increment. The original constant c_{patch} can have either nonzero sign.

25.1 Exact inverse profiles and the five physical rows

Use precisely the bumps and spacing of the five-row operation:

$$\varphi_j(x) = e^{-jd_b} \varphi_0(e^{-jd_b} x), \quad \mu_p = \int_0^\infty x^p \varphi_0(x) dx > 0, \quad t_p = e^{dbp}, \quad db > 0.$$

The nonnegative nonzero smooth bump φ_0 and its three dilates have support in the reserved open interval $I_m \Subset (0, \infty)$. The substitution $x = e^{jd_b} \xi$ proves exactly $\int x^p \varphi_j(x) dx = \mu_p t_p^j$. Number matrix rows and columns starting at zero, and define

$$\begin{aligned} V_\theta &= [t_p^j]_{p=2, -2-2\lambda, -2\lambda; j=0, 1, 2}, \quad B_\theta = V_\theta^{-1}, \\ V_z &= [t_p^j]_{p=1, 1-2\lambda; j=0, 1}, \quad B_z = V_z^{-1}, \\ A_\theta &= \text{diag}(\mu_2, 2a\mu_{-2-2\lambda}, -a\mu_{-2\lambda}) V_\theta, \quad A_z = \text{diag}(\mu_1, a\mu_{1-2\lambda}) V_z. \end{aligned} \quad (166)$$

For three row values t_0, t_1, t_2 , subtraction of the first row from each later row, followed by factoring, gives $\det V_\theta = (t_1 - t_0)(t_2 - t_0)(t_2 - t_1)$. For two row values the determinant is $t_1 - t_0$. The original exponents in each displayed matrix are distinct because $\lambda > 0$, and the exponential with $db > 0$ is injective. The row factors are nonzero. This proves both inverses exist at the actual fixed source

parameters. Explicitly each entry of either B is its corresponding transposed cofactor divided by the stated determinant; this is a finite formula in the original exponentials.

Define the following actual profiles, including their η dependence:

$$\begin{aligned} V_P(x, \eta) &= -\frac{1}{2a(\eta)\mu_{-2-2\lambda}} \sum_{j=0}^2 B_\theta[j, 1] \varphi_j(x), \\ V_{J_z}(x, \eta) &= \frac{1}{a(\eta)\mu_{-2\lambda}} \sum_{j=0}^2 B_\theta[j, 2] \varphi_j(x), \\ G_{J_\theta}(x, \eta) &= -\frac{1}{a(\eta)\mu_{1-2\lambda}} \sum_{j=0}^1 B_z[j, 1] \varphi_j(x). \end{aligned} \tag{167}$$

The signs, the factor two, and all three negative powers in the moment denominators are part of the formulas. Multiplying the columns by the exact rows in (166) proves the eight scalar profile identities

$$\begin{array}{c|cc} & V_P & V_{J_z} \\ \hline \int x^2(\cdot) dx & 0 & 0 \\ 2a \int x^{-2-2\lambda}(\cdot) dx & -1 & 0 \\ -a \int x^{-2\lambda}(\cdot) dx & 0 & -1 \end{array} \quad \begin{array}{c|c} & G_{J_\theta} \\ \hline \int x(\cdot) dx & 0 \\ a \int x^{1-2\lambda}(\cdot) dx & -1 \end{array}. \tag{168}$$

There are eight entries in these two tables; each follows from $V_\theta B_\theta = I_3$ or $V_z B_z = I_2$, with the row factor left in place. The complete matrix identities include all nine and all four scalar inverse identities, respectively.

The source coefficients are

$$\mathbf{u} = A_z^{-1}(0, -q^{2A}\mathcal{P}, -q^{2A-1}\mathcal{J}_z)^T, \quad \mathbf{s} = A_z^{-1}(0, -q^{2A-3/2}\mathcal{J}_\theta)^T.$$

Since $(DV)^{-1} = V^{-1}D^{-1}$, substitution of these coefficients in the original physical increments gives

$$\delta v = q^A \mathcal{P} V_P + q^{A-1} \mathcal{J}_z V_{J_z}, \quad \delta \gamma = q^{A-3/2} \mathcal{J}_\theta G_{J_\theta}. \tag{169}$$

For example the fifth-row coefficient in $\mathbf{u}$ is $(-q^{2A-1}\mathcal{J}_z)/(-a\mu_{-2\lambda})$, which proves the positive sign in V_{J_z} . Multiplication by the original outer factor q^{-A} proves its exponent $A-1$. The other two terms follow by the same displayed diagonal products.

Changing variables $r = \sqrt{q}x$ in each integral, and using (168), now proves the full five equations

$$\begin{aligned} \int r^2 \delta v dr &= 0, & \int r \delta \gamma dr &= 0, & \int \frac{2V}{r} \delta v dr &= -\mathcal{P}, \\ \int r^2 (G \delta v + V \delta \gamma) dr &= -\mathcal{J}_\theta, \\ \int (2r G \delta \gamma - r V \delta v) dr &= -\mathcal{J}_z. \end{aligned} \tag{170}$$

Indeed the pressure term with V_P has scale $q^{-A}q^A = 1$; the angular flux with G_{J_θ} has scale $q^{3/2}q^{-A}q^{A-3/2} = 1$; and the fifth row with V_{J_z} has scale $qq^{-A}q^{A-1} = 1$. Every other term is one of the zero entries in the tables. The two homogeneous velocity moments use the first rows of those tables. This verifies the physical equations, including the single factor r multiplying $V\delta v$ in the fifth row.

25.2 Compact potential, induced radial velocity, and support

Define the combined potential profile

$$H_\theta(x, \eta) = \frac{1}{x} \int_0^x \xi G_{J_\theta}(\xi, \eta) d\xi. \quad (171)$$

Choose $0 < x_- < x_+$ such that the convex hull of all the bump supports lies in $[x_-, x_+] \subseteq I_m$. For $x < x_-$ the integral in (171) is zero. For $x > x_+$ it is the zero axial moment in (168). It follows that H_θ , together with every derivative, is supported in this fixed compact interval. Its smoothness at the two collars follows from the fact that the integrand is identically zero there and that its total integral is zero. In particular it extends smoothly by zero to $x \leq 0$.

The primitive of an individual φ_j is $\psi_j = x^{-1} \int_0^x \xi \varphi_j(\xi) d\xi$, which equals $\mu_1 e^{j d_b} / x$ past its support. Thus an individual primitive can have a tail. The exact identity $\sum_{j=0}^1 B_z[j, 1] e^{j d_b} = 0$ cancels these tails in $H_\theta = -(a\mu_{1-2\lambda})^{-1} \sum_j B_z[j, 1] \psi_j$. The identity holds for every η , so its η derivatives also vanish. This proves compactness for the combined profile and all derivatives without assuming compactness for the summands.

The physical potential and radial increment are exactly

$$\begin{aligned} \Psi(r, z, t) &= \frac{1}{r} \int_0^r s \delta\gamma(s, z, t) ds = q^{A-1} \mathcal{J}_\theta H_\theta(x, \eta), \\ \delta\beta &= -\partial_z \Psi \\ &= -q^{A-1} (\partial_z \mathcal{J}_\theta) H_\theta - q^{A-1-D} \mathcal{J}_\theta \mathcal{Z}_{A-1} H_\theta. \end{aligned} \quad (172)$$

The substitution in the integral supplies one factor q and its outer $1/r$ supplies $q^{-1/2}$, giving the exponent $A - 1$. The operator in the second line is explicitly

$$\mathcal{Z}_\sigma f = \frac{2\sigma\eta f - \eta x f_x + (1 - \eta^2) f_\eta}{B(\eta)}.$$

The chain rule (161) proves the last line, including the η derivative of the original factor $1/a$. Direct differentiation of $x H_\theta$ gives $(\partial_x + x^{-1}) H_\theta = G_{J_\theta}$, so $(\partial_r + r^{-1}) \Psi = \delta\gamma$. Commuting the smooth physical r, z derivatives proves

$$(\partial_r + r^{-1}) \delta\beta + \partial_z \delta\gamma = 0. \quad (173)$$

The three velocity increments and Ψ are auxiliary independent and angularly independent; their support lies in $\sqrt{q}[x_-, x_+]$, including any intervals between the bumps.

25.3 Every mixed physical derivative, with original scale powers

For clarity the original differential dictionary is

$$q_z = \frac{2\eta q^A}{B}, \quad q_t = -\frac{1}{B}, \quad \eta_z = \frac{q^{-D}(1 - \eta^2)}{B}, \quad \eta_t = \frac{D\eta q^{-1}}{B}, \quad \mathcal{T}_\sigma f = \frac{-\sigma f + \frac{1}{2} x f_x + D\eta f_\eta}{B}.$$

These equations follow by implicit differentiation of (165); in particular $1 - 2hz^2 q^{2h-1} = B$ and $D + 2h = A$. Differentiating $x = r q^{-1/2}$ supplies $x_z = -\eta x q^{-D}/B$ and $x_t = x q^{-1}/(2B)$. Consequently

$$\partial_r(q^\sigma F) = q^{\sigma-1/2} F_x, \quad \partial_z(q^\sigma F) = q^{\sigma-D} \mathcal{Z}_\sigma F, \quad \partial_t(q^\sigma F) = q^{\sigma-1} \mathcal{T}_\sigma F.$$

Here $F = F(x, \eta)$. These are physical derivatives of the moving scale; the time loss is the original one in this scale.

For $k, m, n \geq 0$ define $F_{\sigma;k,m,n}$ by the following finite recursion. Start with $f = \partial_x^k F$ and $s = \sigma - k/2$. Repeat $f \leftarrow \mathcal{Z}_s f$, $s \leftarrow s - D$ exactly m times; then repeat $f \leftarrow \mathcal{T}_s f$, $s \leftarrow s - 1$ exactly n times. The final f is $F_{\sigma;k,m,n}$. For any actual scalar target $X = X(z, t)$, the exact product formula is

$$\partial_r^k \partial_z^m \partial_t^n (q^\sigma X F) = \sum_{i=0}^m \sum_{j=0}^n \binom{m}{i} \binom{n}{j} (\partial_z^i \partial_t^j X) \cdot q^{\sigma-k/2-(m-i)D-(n-j)} F_{\sigma;k,m-i,n-j}(x, \eta). \quad (174)$$

To prove it, first apply k radial derivatives, which do not act on X, q, η . The repeated product rule for the m axial and n time derivatives assigns i, j derivatives to X , with the two binomial multiplicities displayed. Apply the three chain rules above successively to the remaining factor. Each axial derivative lowers its current exponent by D , and each time derivative lowers it by one, while its current exponent is the index in the next operator. This is precisely the stated recursion. Smooth mixed derivatives commute, so the chosen radial-then-axial- then-time order computes the derivative written in the formula.

This proves every mixed derivative of $\delta v, \delta \gamma, \Psi$ by using, respectively, the two terms and the one term in (169) and (172). It proves every derivative of $\delta \beta$ either by applying (174) to its two displayed terms or by

$$\partial_r^k \partial_z^m \partial_t^n \delta \beta = -\partial_r^k \partial_z^{m+1} \partial_t^n (q^{A-1} \mathcal{J}_\theta H_\theta). \quad (175)$$

This equality retains, in particular, the derivative of the full target and the additional original factor q^{-D} .

The recursion is an explicit finite rational expression in x, η, h , the retained profile derivatives, and powers of B^{-1} . The other profile constants are exactly the inverse Vandermonde entries and moments in (167). In fact the reciprocal base coefficient has the exact derivatives

$$\partial_\eta^s (a^{-1}) = \frac{1}{2^{1/2+\lambda} c_{\text{patch}}} \begin{cases} 1 + \eta^2 & s = 0, \\ 2\eta & s = 1, \\ 2 & s = 2, \\ 0 & s \geq 3. \end{cases} \quad (176)$$

Also $\varphi_j^{(k)}(x) = e^{-(k+1)jdb} \varphi_0^{(k)}(e^{-jdb} x)$. These two formulas give every mixed derivative of the three profiles without an unspecified inverse-matrix derivative constant.

For a similarly explicit derivative of H_θ , write $F_0(x, \eta) = \int_0^x \xi G_{J_\theta}(\xi, \eta) d\xi$. For $\ell \geq 1$ the fundamental theorem of calculus and the product rule give

$$\partial_x^\ell F_0 = x \partial_x^{\ell-1} G_{J_\theta} + (\ell - 1) \partial_x^{\ell-2} G_{J_\theta},$$

where the second term is defined to be zero for $\ell = 1$. Therefore, for $x > 0$,

$$\partial_x^k \partial_\eta^s H_\theta = \sum_{\ell=0}^k \binom{k}{\ell} (-1)^{k-\ell} (k-\ell)! x^{-1-k+\ell} \partial_x^\ell \partial_\eta^s F_0. \quad (177)$$

The $\ell = 0$ factor is the explicitly specified integral; every other factor is the displayed profile derivative. For example its absolute integral is at most $\int_{x_-}^{x_+} \xi |\partial_\eta^s G_{J_\theta}(\xi, \eta)| d\xi$. This supplies finite computable profile costs on the exact support. The bound $B \geq 1 - 2h > 0$, together with $x \geq x_- > 0$, proves that each fixed recursively specified coefficient is bounded on $[x_-, x_+] \times [-1, 1]$. The original moments, spacing, h , c_{patch} , and λ remain in those coefficients.

25.4 Explicit costs in full target derivative arrays

For $I = (m, n)$, put $I_z = (m + 1, n)$. Use the original actual-shape arrays $\mathcal{B}_{e,I}$ from (156), with their two retained coupled amplitude components and old-wave cross terms. Define the full target majorants

$$\begin{aligned} P_I &= |\partial_z^m \partial_t^n P_{\text{old}}| + \mathcal{B}_{-1,I} + \frac{1}{2} \mathcal{B}_{0,I_z}, \\ J_{\theta,I} &= |\partial_z^m \partial_t^n (J_{\theta})_{\text{old}}| + \frac{1}{2} \mathcal{B}_{2,I}, \\ J_{z,I} &= |\partial_z^m \partial_t^n (J_z)_{\text{old}}| + \mathcal{B}_{1,I} + \frac{1}{4} \mathcal{B}_{2,I_z}. \end{aligned} \quad (178)$$

The triangle inequality in (164) and the proved increment bounds (159) give $|\partial^I \mathcal{P}| \leq P_I$, $|\partial^I \mathcal{J}_{\theta}| \leq J_{\theta,I}$, and $|\partial^I \mathcal{J}_z| \leq J_{z,I}$. Thus the old absolute derivatives appear explicitly in every cost.

For a nonnegative target array X , define the following pointwise, fully specified finite sum:

$$\begin{aligned} \mathfrak{C}_{\sigma,F;k,m,n}[X](r, z, t) &= \sum_{i=0}^m \sum_{j=0}^n \binom{m}{i} \binom{n}{j} X_{(i,j)}(z, t) \\ &\quad \cdot q^{\sigma-k/2-(m-i)D-(n-j)} |F_{\sigma;k,m-i,n-j}(x, \eta)|. \end{aligned} \quad (179)$$

All powers are powers of the actual positive q , even when their exponents are negative. Define

$$\begin{aligned} A_{\theta;k,m,n} &= \mathfrak{C}_{A,V_P;k,m,n}[P] + \mathfrak{C}_{A-1,V_{J_z};k,m,n}[J_z], \\ A_{z;k,m,n} &= \mathfrak{C}_{A-3/2,G_{J_{\theta}};k,m,n}[J_{\theta}], \\ S_{k,m,n} &= \mathfrak{C}_{A-1,H_{\theta};k,m,n}[J_{\theta}], \\ A_{r;k,m,n} &= \mathfrak{C}_{A-1,H_{\theta};k,m,n}[J_{\theta}^{+z}] + \mathfrak{C}_{A-1-D,\mathcal{Z}_{A-1}H_{\theta};k,m,n}[J_{\theta}], \quad (J_{\theta}^{+z})_{(i,j)} = J_{\theta,(i+1,j)}. \end{aligned} \quad (180)$$

Equations (174)–(172) and the triangle inequality prove, for every k, m, n ,

$$\begin{aligned} |\partial_r^k \partial_z^m \partial_t^n \delta v| &\leq A_{\theta;k,m,n}, & |\partial_r^k \partial_z^m \partial_t^n \delta \gamma| &\leq A_{z;k,m,n}, \\ |\partial_r^k \partial_z^m \partial_t^n \Psi| &\leq S_{k,m,n}, & |\partial_r^k \partial_z^m \partial_t^n \delta \beta| &\leq A_{r;k,m,n}. \end{aligned} \quad (181)$$

The alternative upper bound $S_{k,m+1,n}$ for the last quantity follows directly from (175). For example the zero-order costs read

$$\begin{aligned} |\delta v| &\leq q^A P_{0,0} |V_P| + q^{A-1} J_{z,0,0} |V_{J_z}|, \\ |\delta \gamma| &\leq q^{A-3/2} J_{\theta,0,0} |G_{J_{\theta}}|, & |\Psi| &\leq q^{A-1} J_{\theta,0,0} |H_{\theta}|, \\ |\delta \beta| &\leq q^{A-1} J_{\theta,1,0} |H_{\theta}| + q^{A-1-D} J_{\theta,0,0} |\mathcal{Z}_{A-1} H_{\theta}|. \end{aligned}$$

These display separately the target derivative and moving-scale radial cost. Replacing each profile value in (179) by its maximum over its fixed x, η support supplies a pointwise bound depending only on z, t and the stated fixed profile data.

The same formula gives weighted radial costs without a loss hidden in a constant. For any real e , replace the final profile absolute value in (179) by

$$q^{(e+1)/4} \left(\int_{x_-}^{x_+} x^e |F_{\sigma;k,m-i,n-j}(x, \eta)|^2 dx \right)^{1/2}.$$

Call the resulting sum $\mathfrak{C}^{[e]}$. The exact measure identity $r^e dr = q^{(e+1)/2} x^e dx$, followed by the triangle inequality in this weighted L^2 space, proves all four bounds (181) with the left sides replaced by their $[e]$ norms and the right sides by the corresponding $\mathfrak{C}^{[e]}$ sums. The support is bounded away from zero, so these integrals are finite for every fixed real e .

25.5 Actual pressure reconstruction and the surviving defects

Set $\mathbf{a} = (a_r, a_\theta, a_z) = (\delta\beta, \delta v, \delta\gamma)$. Before pressure is changed, let δg_r denote the exact change in the source's angularly averaged negative radial residual. The angular average here retains auxiliary dependence. The actual pressure-defect change is

$$\delta P_{\text{step}} = \int_0^\infty \overline{\delta g_r} dr = P_{\text{new}} - \mathcal{P}. \quad (182)$$

This is the defect change caused by the present mean operation; it is different from ΔP in (164). The source pressure bump is still $\rho = q^{-1/2}\hat{\rho}(x)$, $\int \hat{\rho} dx = 1$. The actual pressure increment for this operation is

$$\delta p = \mathcal{I}_c f, \quad f = \delta g_r - \rho \delta P_{\text{step}}. \quad (183)$$

The original physical shifted primitive is

$$\begin{aligned} \mathcal{I}f(r, Y) &= \int_0^r f(s, Y + (s^{d_r} - r^{d_r})v_r, z, t) ds, \\ \mathcal{J}f(r, Y) &= \int_0^\infty f(s, Y + (s^{d_r} - r^{d_r})v_r, z, t) ds, \\ \mathcal{I}_c f &= \mathcal{I}f - \chi_m \mathcal{J}f, \\ d_r &= 2((1+h)\rho_g - h\kappa_s), \quad \rho_g = \frac{\log(4 - \sqrt{2})}{\log(4 + \sqrt{2})}, \quad \kappa_s = 10^{-5}, \quad v_r = (1, -(\sqrt{2} - 1)). \end{aligned}$$

Here χ_m is the actual inner-zero, outer-one source cutoff; the two suppressed arguments of each integrand are exactly z, t . The pressure entering the physical force is the physical evaluation of this source function, with its original phase evaluation and all its derivatives. Thus it includes any nonzero auxiliary modes of the source reconstruction.

To verify the reconstruction identity, the source physical radial operator $\mathcal{D}_r$ cancels the derivative of the term $-r^{d_r}v_r$ in the shift. Endpoint differentiation gives $\mathcal{D}_r \mathcal{I}f = f$ and $\mathcal{D}_r \mathcal{J}f = 0$, and hence

$$\mathcal{D}_r \delta p - \delta g_r = -\rho \delta P_{\text{step}} - (\partial_r \chi_m) \mathcal{J}f. \quad (184)$$

Haar translation invariance and (182) give $\overline{\mathcal{J}f} = 0$, but this does not set $\mathcal{J}f$ itself to zero. In the inner collar $\mathcal{I}f = 0$ and $\chi_m = 0$; in the outer collar $\mathcal{I}f = \mathcal{J}f$ and $\chi_m = 1$. This proves compact support of the actual pressure and all derivatives. In particular its gradient is retained in the force calculation below.

For completeness the exact remaining source terms can be written entirely in the original current mean fields. With $\Delta_0 = \mathcal{D}_r^2 + r^{-1}\mathcal{D}_r + \partial_z^2$, direct expansion of the radial conservative source gives

$$\begin{aligned} R_g &:= \delta g_r - \frac{2V}{r} \delta v \\ &= -\partial_t \delta \beta - (\mathcal{D}_r + r^{-1})(2b\delta\beta + 2\beta\delta\beta + (\delta\beta)^2) \\ &\quad - \partial_z(b\delta\gamma + G\delta\beta + \beta\delta\gamma + \gamma\delta\beta + \delta\beta\delta\gamma) \\ &\quad + r^{-1}(2v\delta v + (\delta v)^2) + \nu_{\text{NS}}(\Delta_0 - r^{-2})\delta\beta. \end{aligned} \quad (185)$$

Indeed the radial conservative quadratic expression before the change is $-(\mathcal{D}_r + r^{-1})(b + \beta)^2 - \partial_z((b + \beta)(G + \gamma)) + (V + v)^2/r$, together with the wave covariance, time term, and viscosity. Subtract it from its value after the mean change. The unchanged wave covariance cancels; each remaining quadratic product expands to the terms displayed. The term $2V\delta v/r$ has been written

separately in the definition of R_g . The ordinary time derivative on the increment is exact because it is auxiliary independent.

The five physical rows then give the exact new defects

$$\begin{aligned} P_{\text{new}} &= \int \overline{R_g} dr, \\ (J_\theta)_{\text{new}} &= \int r^2 \overline{\gamma \delta v + v \delta \gamma + \delta \gamma \delta v} dr, \\ (J_z)_{\text{new}} &= \int r^2 \overline{2\gamma \delta \gamma + (\delta \gamma)^2} dr - \frac{1}{2} \int r^2 \overline{R_g} dr. \end{aligned} \quad (186)$$

For the first line add $\mathcal{P}$ to $\int (2V\delta v/r + \overline{R_g}) dr$ and use the third row. For the second, expanding $(G+\gamma+\delta\gamma)(V+v+\delta v) - (G+\gamma)(V+v)$ gives base-linear terms $G\delta v + V\delta\gamma$, cancelled by the fourth row, and exactly the three remaining terms displayed. For the third, expand $(G+\gamma+\delta\gamma)^2 - (G+\gamma)^2$ and the radial-source moment $-\frac{1}{2} \int r^2 \delta g_r dr$; the base-linear terms are the fifth row. This proves all three equalities. Neither their right sides nor the cutoff term in (184) has been removed.

25.6 Exact full cylindrical physical force increment

For the actual physical fields, put $\mathcal{R}(U, p) = \partial_t U + (U \cdot \nabla)U - \nu_{\text{NS}}\Delta U + \nabla p$. Ordinary derivatives in this formula are derivatives after the physical source evaluation; they therefore include all phase, frame, cutoff and scale derivatives of the actual full U . Expanding the two quadratic products proves the vector identity

$$\delta \mathcal{R} = \partial_t \mathbf{a} + (U \cdot \nabla) \mathbf{a} + (\mathbf{a} \cdot \nabla)U + (\mathbf{a} \cdot \nabla) \mathbf{a} - \nu_{\text{NS}}\Delta \mathbf{a} + \nabla \delta p. \quad (187)$$

The pressure here is precisely (183).

To derive every cylindrical component, the basis derivatives are $\partial_\theta e_r = e_\theta$, $\partial_\theta e_\theta = -e_r$, and $\partial_\theta e_z = 0$. Thus for two arbitrary vector fields X, Y the component vector of $(X \cdot \nabla)Y$ is

$$X_r \partial_r Y + \frac{X_\theta}{r} \partial_\theta Y + X_z \partial_z Y + \frac{X_\theta}{r} (-Y_\theta, Y_r, 0).$$

Apply this equality to $(U, \mathbf{a})$, $(\mathbf{a}, U)$, and $(\mathbf{a}, \mathbf{a})$. The scalar components of $\mathbf{a}$ have zero angular derivative, whereas those of the actual U generally do not. Writing $L_0 = \partial_r^2 + r^{-1} \partial_r + \partial_z^2$, the complete result is

$$\begin{aligned} \delta \mathcal{R}_r &= \partial_t a_r + U_r \partial_r a_r + U_z \partial_z a_r + a_r \partial_r U_r + a_z \partial_z U_r + \frac{a_\theta}{r} \partial_\theta U_r \\ &\quad + a_r \partial_r a_r + a_z \partial_z a_r - \frac{2U_\theta a_\theta + a_\theta^2}{r} - \nu_{\text{NS}}(L_0 - r^{-2})a_r + \partial_r \delta p, \end{aligned} \quad (188)$$

$$\begin{aligned} \delta \mathcal{R}_\theta &= \partial_t a_\theta + U_r \partial_r a_\theta + U_z \partial_z a_\theta + a_r \partial_r U_\theta + a_z \partial_z U_\theta + \frac{a_\theta}{r} \partial_\theta U_\theta \\ &\quad + a_r \partial_r a_\theta + a_z \partial_z a_\theta + \frac{U_r a_\theta + a_r U_\theta + a_r a_\theta}{r} - \nu_{\text{NS}}(L_0 - r^{-2})a_\theta + \frac{1}{r} \partial_\theta \delta p, \end{aligned} \quad (189)$$

$$\begin{aligned} \delta \mathcal{R}_z &= \partial_t a_z + U_r \partial_r a_z + U_z \partial_z a_z + a_r \partial_r U_z + a_z \partial_z U_z + \frac{a_\theta}{r} \partial_\theta U_z \\ &\quad + a_r \partial_r a_z + a_z \partial_z a_z - \nu_{\text{NS}}L_0 a_z + \partial_z \delta p. \end{aligned} \quad (190)$$

The self terms in the radial equation include $-a_\theta^2/r$; those in the angular equation include $a_r a_\theta/r$. The old-wave cross terms occur in the actual U and in its angular derivatives in all three equations.

The mean pressure is angularly invariant, so its angular derivative is zero for this source operation; its radial and axial derivatives are the actual reconstructed ones and are generally nonzero.

For the viscosity formula, applying the scalar cylindrical Laplacian to the moving basis yields, for a general vector field X ,

$$\Delta X = (\Delta_s X_r - r^{-2} X_r - 2r^{-2} \partial_\theta X_\theta, \Delta_s X_\theta - r^{-2} X_\theta + 2r^{-2} \partial_\theta X_r, \Delta_s X_z), \quad \Delta_s = L_0 + r^{-2} \partial_\theta^2.$$

This follows by differentiating the two basis identities twice and applying the product rule. Substituting the angularly independent components of $\mathbf{a}$ gives exactly $((L_0 - r^{-2})a_r, (L_0 - r^{-2})a_\theta, L_0 a_z)$, proving all three displayed viscous terms. The radial expression before the pressure term, when angularly averaged in the extended source domain and negated, is precisely δg_r : this follows from the definition of the negative radial source residual and linearity of the average. It is also an explicit formula for the input to (183) in terms of the same full velocity and mean increments.

25.7 Finite all-order force costs with every inverse-radius term

Here are explicit derivative bounds for all three components, including angular derivatives of the actual current velocity. Let $\alpha = (k, \ell, m, n) \in \mathbb{N}^4$ index the scalar component derivative $\partial^\alpha = \partial_r^k \partial_\theta^\ell \partial_z^m \partial_t^n$, and let e_r, e_θ, e_z, e_t be the four coordinate multi-indices. Extend the arrays in (180) by

$$A_{i,\alpha} = \begin{cases} A_{i;k,m,n} & \ell = 0, \\ 0 & \ell > 0, \end{cases} \quad U_{i,\alpha} = |\partial^\alpha U_i|.$$

The pressure array Π will be the following explicit majorant for the derivatives of the actual specified function (183), after physical evaluation.

The original physical radial phase satisfies $Y_{\text{phys}}(s, z, t) - Y_{\text{phys}}(r, z, t) = (s^{d_r} - r^{d_r})v_r$. Substituting this exact equality in the two shifted integrals proves the evaluated formula

$$\begin{aligned} \delta p(r, z, t) &= \int_0^r f_{\text{phys}}(s, z, t) ds - \chi_m(r, z, t) \int_0^\infty f_{\text{phys}}(s, z, t) ds, \\ f_{\text{phys}} &= (\delta g_r)_{\text{phys}} - \rho \delta P_{\text{step}}. \end{aligned} \tag{191}$$

Every original axial and temporal phase dependence is retained in f_{phys} . The second integral is independent of r after this evaluation. This equality does not assert that it is zero.

Fix any compact interior physical domain for the finite operation, and choose one physical interval $[R_-, R_+] \subseteq (0, \infty)$ containing the supports of δg_r and ρ for that domain and every auxiliary phase. Put $L_R = R_+ - R_-$, with these actual endpoints kept fixed during differentiation. The smooth zero extensions imply that all their derivatives have support in the same interval. Define $G_{k,m,n}(z, t)$ as the supremum over $r \in [R_-, R_+]$ and the original auxiliary phases of the angular average of the following nonnegative expression: the radial cost $F_{r;(k,0,m,n)}$ in (197) with its last pressure term deleted. This definition involves only $\mathbf{A}$, the actual current velocity derivatives, ν_{NS} , and the inverse-radius arrays; it has no dependence on Π . When auxiliary phases are retained, the velocity derivative arrays mean the corresponding full lifted physical derivatives before evaluation. The chain rule for source evaluation identifies these with derivatives of the evaluated fields at each phase. The force identities and the derivative proof below therefore imply

$$|\partial_r^k \partial_z^m \partial_t^n (\delta g_r)_{\text{phys}}| \leq G_{k,m,n}, \quad |\partial_z^m \partial_t^n \delta P_{\text{step}}| \leq D_{m,n} := L_R G_{0,m,n}.$$

For the first inequality use the negative angularly averaged radial force before pressure, as proved above, and the triangle inequality under that average. For the second, differentiate (182) under

its compact radial integral, commute axial and temporal differentiation with Haar average, and integrate the same bound over the interval of width L_R . The Haar integral of each original constant torus direction derivative is zero by periodicity, which justifies that commutation for the full physical time derivative as well.

The derivative of the pressure bump is already an explicit original- scale coefficient in (162). Define its exact support supremum

$$\mathbf{H}_{k,m,n}(z, t) = q^{-(1+k)/2-mD-n} \sup_{r \in [R_-, R_+]} |f_{0;k,m,n}(r/\sqrt{q}, \eta)|,$$

where $f_{0;k,m,n}$ is the recursion starting with the actual $\hat{\rho}$, rather than another normalized bump. It follows that $|\partial_r^k \partial_z^m \partial_t^n \rho| \leq \mathbf{H}_{k,m,n}$. Set

$$\mathbf{E}_{k,m,n} = \mathbf{G}_{k,m,n} + \sum_{i=0}^m \sum_{j=0}^n \binom{m}{i} \binom{n}{j} \mathbf{H}_{k,m-i,n-j} \mathbf{D}_{i,j}. \quad (192)$$

Since δP_{step} has no radial dependence, all k radial derivatives in its product with ρ act on ρ . The repeated axial and temporal product rule proves $|\partial_r^k \partial_z^m \partial_t^n f_{\text{phys}}| \leq \mathbf{E}_{k,m,n}$ on the full interval.

Write $\mathbf{X}_{k,i,j}(r, z, t) = |\partial_r^k \partial_z^i \partial_t^j \chi_m(r, z, t)|$, retaining the actual cutoff and its moving-scale derivatives. Define

$$\begin{aligned} \Pi_{(k,0,m,n)} &= \begin{cases} L_R \mathbf{E}_{0,m,n} & k = 0, \\ \mathbf{E}_{k-1,m,n} & k \geq 1 \end{cases} \\ &+ L_R \sum_{i=0}^m \sum_{j=0}^n \binom{m}{i} \binom{n}{j} \mathbf{X}_{k,i,j} \mathbf{E}_{0,m-i,n-j}, \quad \Pi_{(k,\ell,m,n)} = 0 \quad (\ell > 0). \end{aligned} \quad (193)$$

This proves the explicit bound $|\partial^\alpha \delta p| \leq \Pi_\alpha$. In fact differentiating (191) yields the exact first-integral derivative

$$\partial_r^k \partial_z^m \partial_t^n \int_0^r f_{\text{phys}}(s, z, t) ds = \begin{cases} \int_0^r \partial_z^m \partial_t^n f_{\text{phys}}(s, z, t) ds & k = 0, \\ \partial_r^{k-1} \partial_z^m \partial_t^n f_{\text{phys}}(r, z, t) & k \geq 1. \end{cases}$$

The absolute values are bounded by the first term in (193). In the cutoff product all radial derivatives act on χ_m , whereas the axial and temporal derivatives distribute with the two displayed binomial factors. Every derivative of its radial integral is bounded by $L_R \mathbf{E}_{0,m-i,n-j}$. This proves its second term, with every cutoff derivative and the actual support width retained. Angular invariance proves the stated zero entries. This construction first computes the pressure-free $\mathbf{G}$, then $\mathbf{D}$, $\mathbf{H}$, $\mathbf{E}$, Π , and finally the full force array below, so the pressure budget is a finite calculation without circularly assuming a pressure bound.

For nonnegative arrays define their exact Leibniz convolution and derivative shift by

$$(X \star Y)_\alpha = \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} X_\beta Y_{\alpha-\beta}, \quad (S_\delta X)_\alpha = X_{\alpha+\delta}. \quad (194)$$

Products with three factors use either parenthesization; multiplying the binomial coefficients gives $\alpha! / (\beta! \gamma! (\alpha - \beta - \gamma)!)$, proving the two parenthesizations agree. For the exact inverse-radius arrays put

$$(\mathbf{R}_j)_\alpha = \begin{cases} (j)_k r^{-j-k} & \ell = m = n = 0, \\ 0 & \text{otherwise,} \end{cases} \quad (j)_0 = 1, \quad (j)_k = j(j+1) \cdots (j+k-1), \quad j = 1, 2. \quad (195)$$

Induction gives $\partial_r^k r^{-j} = (-1)^k (j)_k r^{-j-k}$, which proves that R_j is its exact absolute derivative array. No radial derivative of an inverse-radius coefficient is omitted in the following costs.

For $i = r, \theta, z$ define the common transport cost

$$\begin{aligned} T_i &= S_{e_t} A_i + U_r \star S_{e_r} A_i + U_z \star S_{e_z} A_i + A_r \star S_{e_r} U_i + A_z \star S_{e_z} U_i \\ &\quad + R_1 \star A_\theta \star S_{e_\theta} U_i + A_r \star S_{e_r} A_i + A_z \star S_{e_z} A_i, \end{aligned} \quad (196)$$

and the exact finite upper-cost arrays

$$\begin{aligned} F_r &= T_r + 2R_1 \star U_\theta \star A_\theta + R_1 \star A_\theta \star A_\theta \\ &\quad + \nu_{NS} (S_{2e_r} A_r + R_1 \star S_{e_r} A_r + S_{2e_z} A_r + R_2 \star A_r) + S_{e_r} \Pi, \\ F_\theta &= T_\theta + R_1 \star (U_r \star A_\theta + A_r \star U_\theta + A_r \star A_\theta) \\ &\quad + \nu_{NS} (S_{2e_r} A_\theta + R_1 \star S_{e_r} A_\theta + S_{2e_z} A_\theta + R_2 \star A_\theta) + R_1 \star S_{e_\theta} \Pi, \\ F_z &= T_z + \nu_{NS} (S_{2e_r} A_z + R_1 \star S_{e_r} A_z + S_{2e_z} A_z) + S_{e_z} \Pi. \end{aligned} \quad (197)$$

Every addition and scalar multiplication here is entrywise. For every finite α , these arrays prove the full physical component inequalities

$$|\partial^\alpha \delta \mathcal{R}_i| \leq F_{i,\alpha}, \quad i = r, \theta, z. \quad (198)$$

Indeed the repeated Leibniz rule gives $|\partial^\alpha (fg)| \leq (X \star Y)_\alpha$ whenever the respective derivative absolute values are bounded by X, Y . The three-factor version is its iterated rule, proved by the explicit multinomial coefficient above. Apply these rules term by term to (188)–(190), using (181) for the increment, the actual derivatives for $U, \delta p$, and (195) for each inverse radius. The constant viscosity passes through every derivative. The three resulting sums are precisely (196)–(197), proving the claim at every order.

These are component derivatives in the original orthonormal basis. For an explicit bound on the actual Cartesian vector after angular differentiation, the moving basis gives

$$\left| \partial_r^k \partial_\theta^\ell \partial_z^m \partial_t^n \delta \mathcal{R} \right| \leq \sum_{j=0}^{\ell} \binom{\ell}{j} (F_{r;(k,j,m,n)} + F_{\theta;(k,j,m,n)}) + F_{z;(k,\ell,m,n)}.$$

To prove it, expand each angular derivative by the binomial rule between a scalar component and its basis vector. Every derivative of e_r, e_θ has norm one, whereas the derivatives of e_z of positive order vanish. The triangle inequality gives exactly the displayed bound. Thus the component formulas also provide a bound for the physical vector, with its angular basis differentiation.

There is also an exact recursion for every Cartesian derivative. Let O_θ be the orthogonal matrix with columns (e_r, e_θ, e_z) , and let $J(f_r, f_\theta, f_z) = (-f_\theta, f_r, 0)$. Use $x_C = r \cos \theta, y_C = r \sin \theta$ for the physical Cartesian coordinates, distinct from the original scaled radius $x = r/\sqrt{q}$. Define the operators on component vectors

$$\begin{aligned} \mathcal{X}f &= \cos \theta \partial_r f - \frac{\sin \theta}{r} (\partial_\theta f + Jf), \\ \mathcal{Y}f &= \sin \theta \partial_r f + \frac{\cos \theta}{r} (\partial_\theta f + Jf), \quad \mathcal{Z}f = \partial_z f. \end{aligned} \quad (199)$$

The scalar coordinate rules $\partial_{x_C} = \cos \theta \partial_r - r^{-1} \sin \theta \partial_\theta$ and $\partial_{y_C} = \sin \theta \partial_r + r^{-1} \cos \theta \partial_\theta$, together with $\partial_\theta O_\theta = O_\theta J$, give exactly $\partial_{x_C}(O_\theta f) = O_\theta \mathcal{X}f$, $\partial_{y_C}(O_\theta f) = O_\theta \mathcal{Y}f$, and $\partial_z(O_\theta f) = O_\theta \mathcal{Z}f$. Applying these identities repeatedly proves

$$\partial_{x_C}^i \partial_{y_C}^j \partial_z^k \partial_t^n (O_\theta f) = O_\theta \mathcal{X}^i \mathcal{Y}^j \mathcal{Z}^k \partial_t^n f.$$

The order written is a fixed composition order and is exact because it was obtained from commuting smooth Cartesian derivatives. For a fully explicit majorant recursion, if $\mathbf{C}_r, \mathbf{C}_\theta, \mathbf{C}_z$ majorize the component derivative arrays of f , set $(\mathbf{JC})_r = \mathbf{C}_\theta$, $(\mathbf{JC})_\theta = \mathbf{C}_r$, $(\mathbf{JC})_z = 0$. Let $\mathbf{c}_\alpha = |\partial^\alpha \cos \theta|$, $\mathbf{s}_\alpha = |\partial^\alpha \sin \theta|$, whose entries vanish unless $k = m = n = 0$ and whose remaining entries are the actual derivatives of these elementary functions. Then one $\mathcal{X}$ application is bounded componentwise by

$$\mathbf{c} \star S_{e_r} \mathbf{C}_i + \mathbf{s} \star \mathbf{R}_1 \star (S_{e_\theta} \mathbf{C}_i + (\mathbf{JC})_i),$$

and one $\mathcal{Y}$ application by the same expression with $\mathbf{c}, \mathbf{s}$ interchanged. A $\mathcal{Z}$ or time application is the array shift S_{e_z} or S_{e_t} . These rules follow from the exact product rule (194). Starting with $\mathbf{F}$, iterating them the indicated finite number of times, and taking the sum of the three resulting zero-order component bounds gives a finite bound for each stated Cartesian vector derivative. Thus all Cartesian inverse-radius and angular-basis costs are also explicitly supplied.

All costs above are finite for the actual finite state on each compact interior physical domain. Their dependence on the original scale, target derivatives, inverse-radius factors, viscosity, full current velocity, and reconstructed pressure is explicit. The exact nonlinear defects (186) remain after this operation. No uniform estimate over an increasing family of source stages, no singular-time limit, and no infinite correction gain is asserted by these finite derivative identities and costs.